

SPARSE SAMPLING AND INVERSE CAPACITY IN SAFE FEEDBACK SYSTEMS

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ABSTRACT. We study the instruction names of globally safe Borel feedbacks at a fixed control period τ . Their complexity is measured by the number of patterns observed at selected block times. The order of optimization distinguishes the minimum maximal-pattern rate M_τ from the best lower bound L_τ obtainable with one sampling sequence for all feedbacks. Our first theorem shows that uniform contraction of safe inverse-transfer products over global label lists forces multivalued independence on a positive fraction of the positions in every sufficiently long name set. At the minimum safety-cover cardinality r , this gives $M_\tau = \log r / \tau$. For finite-cylinder domains and branches $P_u \sigma$, exact matrix representations yield contraction criteria based on source weights, dispersion, and gain-loss compensation. For common successors conjugate to σ^p on a mixing one-sided shift of finite type, every prescribed sequence with diverging gaps gives the exact minimum $\log r / \tau$. Countable-language attainment connects the two optimizations under an additional hypothesis. Symbolic and affine examples distinguish these results from invariance entropy and show why open-loop coverage and probability entropy do not determine actual feedback pattern counts.

1. INTRODUCTION

A safe feedback rule chooses a control block from the current state and applies the same rule at every block boundary. Even when safety can be maintained with a small collection of blocks, overlap between their legal domains can leave considerable freedom in this choice. We ask how much complexity must remain in the resulting instruction names after the feedback is chosen to minimize it. Observations are made at selected block times. A name observed at n such times is normalized by $n\tau$, where τ is the fixed control period; the gaps between observations do not enter the denominator.

Invariance entropy counts finite-horizon spanning controls and measures their growth per elapsed time. Here a single state rule must govern the whole trajectory, while observations may be sparse. A small spanning family therefore need not give a feedback with few observed names. The distinction is made precise in Section 9.

For a finite control alphabet and a legal Borel feedback s , write $h_A(s)$ for its counting entropy along an increasing sequence A and $h^*(s)$ for its maximal-pattern rate. The two quantities of interest are

$$L_\tau = \sup_A \inf_s h_A(s), \quad M_\tau = \inf_s h^*(s), \quad L_\tau \leq M_\tau.$$

Both infima include every legal Borel state feedback and, equivalently, every finite Borel safe partition with atom names counted. For M_τ , the positions witnessing a lower bound may depend on the feedback. For L_τ , one sequence must work for the entire class. This distinction is central when different legal choices at the same state have different successors.

Inverse capacities. For each admissible block u , let K_u contain precisely the states from which the complete block stays safe, and let L_u be its inverse-transfer operator relative to a full-support probability. Theorem 3.2 assumes exponential decay, uniform over all histories,

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of products obtained by summing the operators in a global list of $k - 1$ labels at each time. It then gives k -valued independence on a positive fraction of the positions in every sufficiently long feedback name set. In particular, $h^*(s) \geq \log k/\tau$ for every feedback. If $k = r$, where r is the minimum number of safety domains covering the state space, an ordered subcover selector supplies the matching upper bound and $M_\tau = \log r/\tau$.

The proof passes from measures of actual initial-state fibers to a lower bound on the number of label-list boxes needed to cover the names. The finite shattering lemma of Kerr and Li [8, Lemma 3.3], also obtained from the estimates of Huang and Ye [4, Lemma 4.1 and Remark 4.2], then yields independent positions. The control-specific step retains the safety restrictions at every intermediate time in the operator products. The closed-loop map may be discontinuous, and the reference probability need not be invariant.

For branches $P_u\sigma$ with finite-cylinder safety domains, these product norms have exact finite-dimensional representations. Section 4 derives contraction from strict source weights, cross-layer dispersion, and compensation of gain columns by unavoidable losses at the next step. The last criterion applies to an explicit family with no positive common nonstrict left weight in the original representation. The positive-matrix mechanisms are related to the loss estimates of Shi et al. [11] and the Lyapunov-template comparisons of Debauche et al. [1]; the stated safety conditions and all-list estimates are proved here.

Common successors. When all legal blocks have the same successor, the state trajectory is independent of the feedback. The problem becomes one of refining an overlapping safety cover. Theorem 5.1 shows that a finite closed cover of a mixing one-sided shift of finite type has minimum Borel-refinement counting entropy $\log r$ along every prescribed sequence with diverging gaps. Its control consequence is $L_\tau = M_\tau = \log r/\tau$ (Theorem 5.5). The proof uses the Baire property of Borel atoms and concatenation of sufficiently separated cylinders.

Romagnoli [10] computed the gap between an excluded-letter cover and its clopen refinements at consecutive times. Here the refinements may be Borel and the sparse sequence is fixed before they are chosen. In the measure-theoretic setting, Huang, Maass, and Ye [3, Lemma 3.3 and Theorem 3.5] constructed a sequence with a common positive threshold for refinements of a suitable measurable cover. Their invariant-measure result and our exact counting formula concern different notions of entropy. Sequence entropy and independence provide the broader background [2, 5, 6, 8, 9].

The remaining results explain when the two optimizations agree and where the arguments stop. A different-successor example has $L_1 = M_1 < \log r$, and the cyclic-permutation system gives a direct application of the capacity theorem (Section 6). A deletion argument produces one attaining sequence for countably many finite-alphabet languages, leading to a language-quotient minimax theorem (Section 7). Symbolic safety realizations then give a Thue–Morse example with zero invariance entropy and positive fixed-period pattern minimum (Section 8). The comparison with optimization over all periods is made in Section 9. Affine systems in Section 10 give computable values and bounds, while their proofs expose the difference between a covering language and one globally consistent state feedback.

The appendices contain longer counting arguments and examples needed to interpret these bounds: the greedy even-sampling computation, actual even-odd pattern pairs, feedback-consistency obstructions, and the distinction between a common minimum and pointwise common attainment. They also show why exceptional states, transfer peaks, and Shannon estimates cannot replace support counts. The general common-sampling problem for the inverse-capacity criterion remains open; the independent positions supplied by that criterion may vary with the feedback.

2. SAFE FEEDBACKS AND PATTERN COUNTS

2.1. Controlled invariance and invariant partitions. Write $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. All logarithms are natural. Invariance entropy and consecutive-name entropy are measured per physical time unit; sparse rates are defined below. Let X be a metrizable

space, let U be a nonempty control set, and suppose that $f_u := f(\cdot, u) : X \rightarrow X$ is continuous for every $u \in U$. For a finite control word $\omega = (\omega_0, \dots, \omega_{N-1}) \in U^N$, set

$$\varphi(0, x, \omega) = x, \quad \varphi(n+1, x, \omega) = f(\varphi(n, x, \omega), \omega_n) \quad (0 \leq n < N).$$

A nonempty compact set $Q \subset X$ is *controlled invariant* if every $x \in Q$ admits an infinite control whose complete forward trajectory stays in Q .

For $N \in \mathbb{N}$, let $r_{\text{inv}}(N, Q)$ be the least cardinality of a set $S \subset U^N$ such that for every $x \in Q$ some $\omega \in S$ satisfies

$$\varphi(n, x, \omega) \in Q \quad (0 \leq n \leq N).$$

If no finite such set exists, put $r_{\text{inv}}(N, Q) = +\infty$. The discrete-time invariance entropy is

$$h_{\text{inv}}(Q) = \limsup_{N \rightarrow \infty} \frac{1}{N} \log r_{\text{inv}}(N, Q). \quad (2.1)$$

Throughout, the admissible control space is the full sequence space $\mathcal{U} = U^{\mathbb{N}_0}$; equivalently one may use Kawan's two-sided full control space and retain its nonnegative coordinates. Every finite word therefore extends to an admissible infinite control, and finite-horizon dependence identifies the preceding spanning number with Kawan's. Thus (2.1) is Kawan's definition with the logarithm converted from base two to base e ; see [7, Definition 2.2, p. 44].

Fix a sampling period $\tau \in \mathbb{N}$. A *finite invariant cover* of Q with period τ is a triple $\mathcal{C} = (\mathcal{A}, \tau, \nu)$, where $\mathcal{A}$ is a finite cover of Q and $\nu(D) \in U^\tau$ is assigned to each $D \in \mathcal{A}$, such that

$$\varphi(s, x, \nu(D)) \in Q \quad (x \in D, 0 \leq s \leq \tau). \quad (2.2)$$

This is the discrete-time specialization of Kawan's Definition 2.8 [7, p. 79]. An *invariant partition* is an invariant cover whose underlying family is a finite Borel partition of Q into nonempty atoms. It is *clopen* if its atoms are clopen in Q . We write

$$\begin{aligned} \mathfrak{C}_{\text{Bor}}(Q, \tau) &= \{\text{finite Borel invariant partitions of } Q \text{ of period } \tau\}, \\ \mathfrak{C}_{\text{clop}}(Q, \tau) &= \{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q, \tau) : \mathcal{C} \text{ is clopen}\}, \\ \mathfrak{C}_{\text{Bor}}(Q) &= \bigcup_{\tau \in \mathbb{N}} \mathfrak{C}_{\text{Bor}}(Q, \tau). \end{aligned}$$

We use the extended-value convention $\inf \emptyset = +\infty$.

Let $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, \tau, \nu)$ be an invariant partition, with $I_{\mathcal{C}}$ a finite nonempty alphabet. For $a = (a_0, \dots, a_{N-1}) \in I_{\mathcal{C}}^N$, concatenate the assigned blocks in that order. The word a is *admissible* if some $x \in D_{a_0}$ follows this concatenated control and satisfies

$$\varphi(j\tau, x, \nu(D_{a_0}) \cdots \nu(D_{a_{N-1}})) \in D_{a_j} \quad (0 \leq j < N). \quad (2.3)$$

The set of such words is $W_N(\mathcal{C}; Q)$. These are the partition words of Kawan's Definition 2.9 [7, pp. 79–80].

Lemma 2.1 (Safety, suffixes, and right extension). *Let $\mathcal{C}$ be an invariant partition.*

- (i) *A witness for $a \in W_N(\mathcal{C}; Q)$ remains in Q at every physical time from 0 through $N\tau$ under the concatenated blocks.*
- (ii) *Every prefix and every suffix of an admissible word is admissible.*
- (iii) *Every $a \in W_N(\mathcal{C}; Q)$ has a right extension in $W_{N+1}(\mathcal{C}; Q)$. In particular, every finite admissible word has an infinite right extension.*
- (iv) *For $M, N \in \mathbb{N}$,*

$$\#W_{M+N}(\mathcal{C}; Q) \leq \#W_M(\mathcal{C}; Q) \#W_N(\mathcal{C}; Q). \quad (2.4)$$

Proof. On block j , condition (2.2), applied to the state in D_{a_j} , keeps the trajectory in Q throughout the next τ steps. Induction proves (i). Prefixes are immediate. A suffix beginning at block j is witnessed by the state at time $j\tau$, which proves (ii).

At time $N\tau$ the terminal state belongs to a unique atom D_b . Appending b and its assigned block gives an admissible word of length $N+1$, proving (iii) without compactness. Finally,

an admissible word of length $M + N$ is determined injectively by its length- M prefix and length- N suffix. Part (ii) places these in W_M and W_N , respectively, and gives (2.4). $\square$

Define the finite-prefix name system

$$\Sigma_{\mathcal{C}} = \{a \in I_{\mathcal{C}}^{\mathbb{N}_0} : a|_{[0, N-1]} \in W_N(\mathcal{C}; Q) \text{ for every } N \in \mathbb{N}\}. \quad (2.5)$$

We give $I_{\mathcal{C}}$ the discrete topology and $\Sigma_{\mathcal{C}}$ the subspace topology from $I_{\mathcal{C}}^{\mathbb{N}_0}$. Its complement is a union of cylinders determined by forbidden finite prefixes, so $\Sigma_{\mathcal{C}}$ is closed. Lemma 2.1(iii) shows that every word in W_N is the initial word of some element of $\Sigma_{\mathcal{C}}$. Lemma 2.1(ii) shows $\sigma(\Sigma_{\mathcal{C}}) \subset \Sigma_{\mathcal{C}}$; surjectivity need not hold. For Borel atoms, an element of $\Sigma_{\mathcal{C}}$ need not have one common state witness for all its prefixes. Our definitions use only finite projections and do not assume such a witness.

There is a useful concrete description of $\Sigma_{\mathcal{C}}$. The feedback block map and its consecutive atom names also appear in the numerical framework of [12, Section 2]. Define

$$T_{\mathcal{C}}(x) = \varphi(\tau, x, \nu(D_i)) \quad \text{if } x \in D_i,$$

and its atom itinerary $\iota_{\mathcal{C}}(x)$ by $(\iota_{\mathcal{C}}(x))_j = i$ if $T_{\mathcal{C}}^j(x) \in D_i$. Each branch of $T_{\mathcal{C}}$ is continuous, and $T_{\mathcal{C}}$ is Borel measurable; it need not be continuous when the atoms are merely Borel. Every finite admissible word is the prefix of the actual itinerary of its witness. Conversely every actual itinerary has admissible prefixes. It follows that

$$\Sigma_{\mathcal{C}} = \overline{\iota_{\mathcal{C}}(Q)}. \quad (2.6)$$

Indeed, each prefix cylinder meeting $\Sigma_{\mathcal{C}}$ meets $\iota_{\mathcal{C}}(Q)$, which proves density, and all actual itineraries belong to the closed set $\Sigma_{\mathcal{C}}$.

Remark 2.2 (Infinite witnesses). *If the atoms are clopen, $T_{\mathcal{C}}$ and $\iota_{\mathcal{C}}$ are continuous. Compactness of Q makes $\iota_{\mathcal{C}}(Q)$ closed, so every element of $\Sigma_{\mathcal{C}}$ has a common state witness. Equivalently, the sets of witnesses for its finite prefixes are nested nonempty compact sets.*

This conclusion can fail for a Borel partition even in a compact continuous system with one control. Take $Q = X = [0, 1]$, $f(x) = \min\{2x, 1\}$, $D_0 = (0, 1/2)$ and $D_1 = Q \setminus D_0$. Both atoms receive the unique control, so the partition is invariant. For every N , the point $x_N = 2^{-N-1}$ witnesses the word 0^N : for $0 \leq j < N$, $f^j(x_N) = 2^{j-N-1} \in (0, 1/2)$. Thus $0^\infty \in \Sigma_{\mathcal{C}}$. No point witnesses this infinite name: zero is outside D_0 , and every positive point eventually leaves D_0 .

For a nonempty finite $F \subset \mathbb{N}_0$, put

$$P_{\mathcal{C}}(F) = \#\{a|_F : a \in \Sigma_{\mathcal{C}}\}. \quad (2.7)$$

Equivalently, if $m = \max F$, this is the number of projections of $W_{m+1}(\mathcal{C}; Q)$ onto F . Thus a counted pattern always has a finite witness that remains in Q also between the selected block times.

Let

$$\mathcal{S} = \{A = (a_n)_{n \geq 1} : 0 \leq a_1 < a_2 < \dots, a_n \in \mathbb{N}_0\}.$$

Define

$$p_{\mathcal{C}}^*(n) = \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} P_{\mathcal{C}}(F), \quad (2.8)$$

$$h^*(\mathcal{C}; Q) = \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log p_{\mathcal{C}}^*(n), \quad (2.9)$$

$$h_{\text{seq}}^A(\mathcal{C}; Q) = \limsup_{n \rightarrow \infty} \frac{1}{n\tau} \log P_{\mathcal{C}}(\{a_1, \dots, a_n\}). \quad (2.10)$$

The existence of the limit in (2.9) follows from Lemma 7.1 below. Although there are infinitely many n -element subsets of $\mathbb{N}_0$, the maximum in (2.8) exists: its possible values form a nonempty subset of $\{1, \dots, (\#I_{\mathcal{C}})^n\}$.

The entries of A are *block indices*; the corresponding physical observation times are $\tau A = (\tau a_n)_{n \geq 1}$. The normalization in (2.10) is $n\tau$, not the elapsed horizon $a_n\tau$. For fixed τ , $A \mapsto \tau A$ is a bijection between block-index schedules and physical schedules contained in $\tau\mathbb{N}_0$. If sampling periods vary, the same block-index sequence represents different physical schedules.

The ordinary consecutive-name entropy is

$$h(\mathcal{C}; Q) = \lim_{N \rightarrow \infty} \frac{1}{N\tau} \log \#W_N(\mathcal{C}; Q), \quad (2.11)$$

whose limit exists by (2.4) and Fekete's lemma.

2.2. Fixed-period selectors and the two optimizations. Use the control setting of Section 2, now with a finite control set U . Fix $\tau \geq 1$ and write $W = U^\tau$. For $w \in W$, its closed safety domain and endpoint map are

$$K_w = \{x \in Q : \varphi(j, x, w) \in Q \ (0 \leq j \leq \tau)\}, \quad g_w(x) = \varphi(\tau, x, w). \quad (2.12)$$

The domains cover Q by controlled invariance.

The class $\mathcal{S}_\tau$ consists of *all* Borel selectors $s : Q \rightarrow W$ such that $x \in K_{s(x)}$. Its closed-loop block map is $T_s(x) = g_{s(x)}(x)$. All Borel mixtures of legal selectors belong to $\mathcal{S}_\tau$. No continuity or dependence on finitely many coordinates is imposed. A selector is used without an additional clock, register, or feedback memory.

For a finite window $F \subset \mathbb{N}_0$, let

$$\mathbf{P}_s(F) = \left| \{(s(T_s^k x))_{k \in F} : x \in Q\} \right|.$$

For $A = (a_i)_{i \geq 1} \in \mathcal{S}$, define

$$h_A(s) = \limsup_{n \rightarrow \infty} \frac{1}{n\tau} \log \mathbf{P}_s(\{a_1, \dots, a_n\}), \quad h^*(s) = \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log \max_{|F|=n} \mathbf{P}_s(F).$$

The same submultiplicativity used above gives the second limit. The denominator counts sampled blocks of length τ ; elapsed time τa_n defines a different rate. Put

$$L_\tau = \sup_A \inf_{s \in \mathcal{S}_\tau} h_A(s), \quad M_\tau = \inf_{s \in \mathcal{S}_\tau} h^*(s). \quad (2.13)$$

Always $L_\tau \leq M_\tau$. Integer-logarithm quantization makes the pattern infimum a minimum; Appendix F gives the counting argument and its fixed-period consequence.

The counts use actual closed-loop trajectories. If instead one takes the closure of their one-sided name set, the finite patterns are unchanged: a cylinder specifying finitely many coordinates is open, so if it meets that closure, it meets the original name set. Taking a name closure therefore introduces no extra finite-window witnesses.

Lemma 2.3. *The infima in (2.13), and the inner infimum at any fixed A , are unchanged if one allows every finite Borel safe partition with a control block assigned to each atom, and counts atom names rather than control-block names.*

Proof. Merge all atoms assigned the same block. Their finite union is Borel and remains inside that block's safety domain. The resulting selector has exactly the same closed-loop state map. The map from original atom names to block names is coordinatewise and onto the actual block-name patterns, so every block-pattern count is no larger than the original atom-pattern count. Conversely, the nonempty fibers of any legal selector form a legal finite Borel safe partition whose atom and block counts agree. Both directions of the infimum equality follow. $\square$

Example 2.4 (Merging observation labels and merging controls). *Let $Q = \{0, 1\}^{\mathbb{N}_0} \cup \{2^\infty\}$ and let the controls be $0, 1, 2$. A control equal to x_0 sends x to σx ; every other control sends it to an isolated cemetery point. These maps are continuous, and safety forces the unique selector $s(x) = x_0$. Every nonempty n -position window has $2^n + 1$ actual control patterns. Replacing the observed label 2 by 0 leaves exactly 2^n patterns, so both observations have*

rate $\log 2$. The merged observed zero fiber, however, contains both $[0]$ and 2^∞ . The first set requires physical control 0, whereas the latter point requires physical control 2. No one control is safe on their union. Thus a symbolic factor may preserve the entropy without defining a safe merged control. Lemma 2.3 applies because the atoms it merges were already assigned the same physical control block.

3. INVERSE CAPACITIES AND MAXIMAL-PATTERN ENTROPY

3.1. Branch measures, lists, and actual fibers. We use the fixed-period rates of Section 2.2. In particular $p_s^*(n) = \max_{|F|=n} P_s(F)$ and $h^*(s) = \lim_n (n\tau)^{-1} \log p_s^*(n)$.

Fix a period $\tau \geq 1$ and a finite collection W of admissible control blocks, with $q = |W|$. For each $u \in W$ let $K_u \subset Q$ be its closed safety domain, including all intermediate-time safety conditions, and $g_u : K_u \rightarrow Q$ its continuous endpoint map. Assume these domains cover the nonempty compact metric space Q . The feedback class and optimization quantities in this section are relative to this specified block set W , as are its finite atom refinements. For the system optimization of Section 2.2, we take $W = U^\tau$. The legal class $\mathcal{S}_\tau$ consists of all Borel maps $s : Q \rightarrow W$ with $x \in K_{s(x)}$. Define $T_s(x) = g_{s(x)}(x)$. Reusing this same map at each block boundary gives an infinite safe trajectory for every state. No continuity of s or of T_s is imposed.

Let μ be a full-support Borel probability on Q and suppose

$$\nu_u = (g_u)_*(\mu|_{K_u}) \ll \mu, \quad \rho_u = \frac{d\nu_u}{d\mu} \in L^\infty(\mu). \quad (3.1)$$

The positive bounded operator $L_u : L^\infty(\mu) \rightarrow L^\infty(\mu)$ is specified by

$$\int h L_u f d\mu = \int_{K_u} h(g_u x) f(x) d\mu(x) \quad (h, f \text{ bounded Borel}). \quad (3.2)$$

Indeed, the signed pushforward of $f\mu|_{K_u}$ is dominated by $\|f\|_\infty \nu_u$; its density defines $L_u f$ and $|L_u f| \leq \|f\|_\infty \rho_u$. Changing f on a μ -null set changes no pushforward measure, so this is well defined on L^∞ classes. In particular $L_u \mathbf{1} = \rho_u$.

Fix $2 \leq k \leq q$ and put $l = k - 1$. A list $S \subset W$ is a *global* set of l labels, not a state-dependent choice. Write $A_S = \sum_{u \in S} L_u$, and define

$$b_0 = 1, \quad b_n = \max_{|S_0|=\dots=|S_{n-1}|=l} \|A_{S_{n-1}} \cdots A_{S_0} \mathbf{1}\|_\infty. \quad (3.3)$$

Products are ordered in the forward pushforward order. Positivity gives $\|P\|_{\infty \rightarrow \infty} = \|P\mathbf{1}\|_\infty$ for every positive product P . Consequently $b_{n+m} \leq b_n b_m$. If $b_h < 1$ for some $h \geq 1$, choose $b_h < \theta^h < 1$ and set

$$H = \max_{0 \leq j < h} b_j \theta^{-j}. \quad b_n \leq H \theta^n \quad (n \geq 0). \quad (3.4)$$

This follows by writing $n = ah + j$ and using submultiplicativity. The proof length h is not a change of control period.

For a fixed feedback let

$$E_w(s) = \{x : s(T_s^j x) = w_j \ (0 \leq j < n)\}, \quad F_n(s) = \{w \in W^n : E_w(s) \neq \emptyset\}.$$

The $E_w(s)$ are Borel, pairwise disjoint, and cover Q . For $w = w_0 \cdots w_{n-1}$ put $L_w = L_{w_{n-1}} \cdots L_{w_0}$ and $P_F = \sum_{w \in F} L_w$.

Lemma 3.1 (Actual-fiber capacity). *For every legal Borel feedback and every $n \geq 1$,*

$$1 \leq \int P_{F_n(s)} \mathbf{1} d\mu. \quad (3.5)$$

If $F_n(s)$ is covered by t boxes $S_0^a \times \cdots \times S_{n-1}^a$ whose lists have size at most l , then $1 \leq t b_n$.

Proof. Fix a nonempty fiber E_w and let η_j be the pushforward of $\mu|_{E_w}$ after its first j actual blocks. Set $f_0 = \mathbf{1}_{E_w}$. We claim inductively that $\eta_j = f_j \mu$ and $f_{j+1} = L_{w_j} f_j$. Indeed, η_j is concentrated on K_{w_j} , and for every bounded Borel h ,

$$\int h d\eta_{j+1} = \int_{K_{w_j}} h(g_{w_j} x) f_j(x) d\mu(x) = \int h L_{w_j} f_j d\mu.$$

The boundedness of each L_u permits the induction. None of these restrictions removes mass, because all states in E_w follow the specified safe word. Thus $\int L_w \mathbf{1}_{E_w} d\mu = \mu(E_w)$. Positivity gives $L_w \mathbf{1}_{E_w} \leq L_w \mathbf{1}$; summing over the disjoint fibers proves (3.5). The set $F_n(s)$ includes null but nonempty fibers as well.

Complete every nonempty list to size l . Distributivity and positivity give

$$P_{F_n(s)} \mathbf{1} \leq \sum_{a=1}^t A_{S_{n-1}^a} \cdots A_{S_0^a} \mathbf{1}.$$

Repeated boxes and overlaps only enlarge this upper bound. Unsafe words in a box are subject to the original safety gates in L_u . Integrating and using (3.3) yields $1 \leq t b_n$. $\square$

3.2. The finite combinatorial input and its application. For $F \subset W^n$, let $C_l(F)$ be its minimum covering number by l -list boxes. Given a k -element set $V \subset W$, let $C_V(F)$ be the minimum number of boxes of the form

$$\prod_{j=0}^{n-1} (W \setminus \{v_j\}), \quad v_j \in V,$$

needed to cover F . This is a cover, since $k \geq 2$.

We use the following finite covering lemma. For every $k \geq 2$ and $b > 0$ there are $c > 0$ and N such that if $G \subset \{0, 1, \dots, k\}^n$, $n \geq N$, needs at least e^{bn} boxes omitting one nonzero letter per coordinate, then some $J \subset \{0, \dots, n-1\}$ with $|J| \geq cn$ has

$$G|_J \supset \{1, \dots, k\}^J. \quad (3.6)$$

This is Kerr–Li’s finite covering lemma [8, Lemma 3.3]. Huang and Ye recover the same conclusion from their sharper estimates in [4, Lemma 4.1 and Remark 4.2, pp. 1701–1705]. Its constants depend only on b, k ; its use here requires no dynamical assumptions on the closed loop.

For clarity, the entire reduction from list boxes to that lemma is included. Choose a minimum omitted-letter cover for every V and intersect all these covers. The intersections cover F , with at most $\prod_V C_V(F)$ members. In a nonempty intersection each coordinate contains at most $k-1$ letters: if it contained a k -set V_0 , its factor from the V_0 cover would have omitted a letter of V_0 . Hence

$$C_l(F) \leq \prod_{V \in \binom{W}{k}} C_V(F). \quad (3.7)$$

For actual names Lemma 3.1 gives $C_l(F_n(s)) \geq b_n^{-1}$. Under (3.4), for all sufficiently large n this lower bound is at least $e^{\eta n/2}$, where $\eta = -\log \theta > 0$. With $J_0 = \binom{q}{k}$, (3.7) implies that some V satisfies

$$C_V(F_n(s)) \geq e^{\eta n/(2J_0)}.$$

Code the members of V by $1, \dots, k$ and all other letters by 0. Omitted-letter boxes pull back exactly to the boxes defining C_V , so the coded set has precisely that covering number. Applying (3.6) and undoing this coding yields $F_n(s)|_J \supset V^J$. No distinct members of V were merged.

Theorem 3.2 (Product inverse-capacity criterion). *Assume (3.1). If $b_h < 1$ for some $h \geq 1$, then there are $\alpha > 0$ and N , depending only on the branches, μ, k , such that, for every legal*

Borel feedback s and every $n \geq N$, its actual length- n names have a k -valued cube projection on at least αn coordinates. Consequently

$$p_s^*(m) \geq k^m \quad (m \geq 1), \quad h^*(s) \geq \frac{\log k}{\tau}. \quad (3.8)$$

The same lower bound holds for all finite Borel atom refinements. If $k = r \geq 2$ is the minimum number of safety domains covering Q , then

$$M_\tau = \min_s h^*(s) = \frac{\log r}{\tau}, \quad (3.9)$$

both over unrefined selectors and over all finite Borel safe partitions.

Proof. The preceding argument supplies α, N independent of s . Given m , choose n so large that $\alpha n \geq m$ and retain m of the independent positions. All k^m patterns are actual feedback names, proving (3.8) with the prescribed normalization by $m\tau$.

For $k = r$, choose a minimum subcover $K_{u_1}, \dots, K_{u_r}$ and select u_j on $K_{u_j} \setminus \bigcup_{i < j} K_{u_i}$. This is a Borel state function, legal at every point. Each chosen block stays safe at its intermediate times and ends in Q ; applying the same rule at the next state gives infinite safety. It uses r labels and has at most r^m patterns on any m positions. This proves (3.9) and attainment. For a refined partition, projection to its physical control labels is onto the actual coarse patterns and preserves the dynamics. The lower bound therefore passes to every refinement, while the unrefined selector remains an available upper-bound witness in the enlarged class. $\square$

The theorem asserts a uniform *density*, not a common set of independent positions. The sets J and V may depend on both s and n . In particular it proves no equality $L_\tau = M_\tau$ without an additional common-sampling argument.

3.3. Single-step scalar and joint-density criteria. Define

$$\Theta_k = \left\| \max_{|S|=k-1} \sum_{u \in S} \rho_u \right\|_\infty. \quad (3.10)$$

If $\Theta_k < 1$, then $b_1 = \Theta_k < 1$ and Theorem 3.2 applies. If the branch capacities obey $\mu(K_u \cap g_u^{-1}E) \leq c_u \mu(E)$ for every Borel E , then $\rho_u \leq c_u$ almost everywhere, and the sum of the largest $k-1$ of the c_u bounds Θ_k . Thus a scalar-capacity sum smaller than one is a sufficient special case. The order of taking the branch supremum and summing matters: joint densities may pass the test while the individual optimal scalar bounds do not.

The joint-density case also has a direct quantitative proof worth recording. For nonempty $F \subset W^n$, let $d_k(F)$ be the largest $|J|$ for which there are sets $V_j \subset W$, $|V_j| = k$, such that $F|_J \supset \prod_{j \in J} V_j$. The sets V_j may differ with j . Set $d_k(\emptyset) = -1$. Put $\theta = \Theta_k < 1$, $C_0 = \|\sum_u \rho_u\|_\infty$ and $B = C_0 \binom{q}{k}$. For nonnegative f let $\Phi f = \max_{|S|=k-1} \sum_{u \in S} L_u f$ and $\mathcal{T}f = \sum_u L_u f$. Their norms are bounded by θ and C_0 , respectively.

Proposition 3.3 (Weighted shattering estimate). *With $H(n, -1) = 0$, for every $F \subset W^n$,*

$$\|P_F\|_{\infty \rightarrow \infty} \leq H(n, d_k(F)), \quad H(n, d) = \sum_{j=0}^{\min(d, n)} \binom{n}{j} B^j \theta^{n-j}. \quad (3.11)$$

For actual names this gives $d_k(F_n(s)) > \alpha n$, uniformly in every legal feedback, whenever $0 < \alpha < 1/2$ satisfies

$$\log \theta + \mathfrak{h}(\alpha) + \alpha \log(B/\theta) < 0, \quad \mathfrak{h}(t) = -t \log t - (1-t) \log(1-t). \quad (3.12)$$

Proof. Take $H(n, -1) = 0$ and treat the empty word at $n = 0$ separately. Split F at its first letter: $F_a = \{v : av \in F\}$, $U_0 = \bigcup_a F_a$, and $I_S = \bigcap_{a \in S} F_a$ for $|S| = k$. If $d = d_k(F)$, then $d_k(U_0) \leq d$ and $d_k(I_S) \leq d-1$: independence of d tail positions in the latter would also make the first position independent and give $d+1$. For each $v \in U_0$ the set of its permitted first

letters has either at most $k - 1$ members, or contains a k -set S . Applying the positive tail product L_v and summing gives

$$P_F f \leq P_{U_0} \Phi f + \sum_{|S|=k} P_{I_S} \mathcal{T} f.$$

This respects the order $L_{av} = L_v L_a$; no operators are commuted. Induction and Pascal's identity give $\theta H(n - 1, d) + BH(n - 1, d - 1) = H(n, d)$. Positive domination handles signed f and proves (3.11).

By actual-fiber mass, $1 \leq H(n, d_k(F_n(s)))$. The cover property gives $C_0 \geq 1$, and positivity gives $0 < \theta \leq C_0 \leq B$. For $d \leq \alpha n$, therefore, $B/\theta \geq 1$ and $\sum_{j \leq \alpha n} \binom{n}{j} \leq e^{n\mathfrak{h}(\alpha)}$ give $H(n, d) \leq \exp(n[\log \theta + \mathfrak{h}(\alpha) + \alpha \log(B/\theta)]) < 1$, a contradiction. The binomial bound follows by comparing each term to its probability under independent Bernoulli(α) trials; those probabilities sum to at most one. Condition (3.12) is possible because its last two terms tend to zero as α tends to zero. $\square$

3.4. A smooth joint-density example. On $Q = [0, 1]$ take three full inverse branches with

$$\begin{aligned} l_u &= \frac{19u}{60}, & \rho_u(y) &= \frac{11}{30} + \frac{7}{30} \cos(2\pi(y - u/3)), \\ \psi_u(y) &= l_u + \frac{11}{30}y + \frac{7}{60\pi} [\sin(2\pi(y - u/3)) - \sin(-2\pi u/3)]. \end{aligned} \quad (3.13)$$

Here $g_u = \psi_u^{-1}$, for $u = 0, 1, 2$. Their derivatives lie between $2/15$ and $3/5$, so the branches are strictly increasing and expanding in the forward direction. The closed safety intervals are $[0, 22/60]$, $[19/60, 41/60]$, and $[38/60, 1]$. They cover Q and each is needed: the middle interval alone covers the open gap between $22/60$ and $38/60$, while the two endpoints require the first and third intervals. Hence $r = 3$.

With Lebesgue probability, the densities are exactly $\psi'_u = \rho_u$. The three cosine terms sum to zero. The sum of two is the negative of the remaining cosine, whose maximum is one; therefore

$$\Theta_3 = \frac{22}{30} + \frac{7}{30} = \frac{29}{30} < 1.$$

Each branch has optimal scalar bound $3/5$, since its continuous density reaches that value. The largest two scalar bounds sum to $6/5$. The joint criterion consequently gives $M_1 = \log 3$ although that Lebesgue scalar criterion fails.

To place these interval branches in a continuous compact ambient system, take $X = [-1, 2]$. Extend each g_u to the left of its safety interval $[a_u, b_u]$ by $-\min\{a_u - x, 1\}$ and to its right by $1 + \min\{x - b_u, 1\}$. At the endpoints these formulas match 0 and 1. Outside the closed safety interval the image lies outside Q , so this extension changes neither safety domains nor legal dynamics. The same construction applies to full monotone interval branches elsewhere in the paper.

The fixed-reference separation proved here is not, by itself, a separation from every full-support reference probability. The bounded positive-density obstruction is recorded in Appendix D; its extension to arbitrary full-support probabilities remains unresolved for this smooth example.

3.5. Product contraction without a single-step reference test. Let $Q = (\mathbb{Z}/3\mathbb{Z})^{\mathbb{N}_0}$, let P_u add u to each coordinate, and use $g_u = P_u \sigma$ at period one. Put

$$K_0 = [02] \cup [12] \cup [22], \quad K_1 = [00] \cup [20], \quad K_2 = [01] \cup [10] \cup [11] \cup [12] \cup [21].$$

The only overlap is $[12]$. The cylinders $[02], [00], [01]$ require controls 0, 1, 2, respectively, so $r = 3$. On $[12]$, choosing 0 or 2 changes the successor's first letter from 2 to 1. Every Borel mixture on this entire overlap is allowed. Illegal controls go to an isolated fixed cemetery point; the resulting maps on $Q \sqcup \{\dagger\}$ are continuous.

For fair Bernoulli probability, the operators leave first-coordinate functions invariant and have matrices $M_u/3$, where

$$M_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

This follows from splitting the source measure by its first letter, and retaining exactly the allowed two-letter cylinders. For a positive product the full L^∞ norm equals its norm on $\mathbf{1}$, whose entire orbit stays in this invariant space. Thus this finite representation computes the full product norms, not an approximation to arbitrary Borel functions.

Let $A = M_0 + M_1$, $B = M_0 + M_2$, $C = M_1 + M_2$. The vectors after two of these unnormalized factors are

$$\begin{array}{ccc} (0, 3, 5) & (0, 7, 8) & (0, 4, 7) \\ (5, 2, 7) & (8, 1, 9) & (7, 3, 10) \\ (5, 5, 2) & (8, 8, 1) & (7, 7, 3). \end{array}$$

They are coordinatewise dominated by one of the vectors in

$$V = \{(0, 7, 8), (7, 3, 10), (7, 7, 3), (8, 1, 9), (8, 8, 1)\}.$$

For $v = (p, q', r')$, $Av = (0, p + r', p + q' + r')$, $Bv = (p + q' + r', q', p + 2q' + r')$, and $Cv = (p + q' + r', p + q' + r', q')$. For the five vectors of V , their largest resulting coordinate is, respectively, 22, 23, 24, 19, 25. Also $BCB\mathbf{1} = (17, 8, 25)$. Consequently $b_3 = 25/27 < 1$. Submultiplicativity, rather than an extrapolation from finite calculations, controls all lengths. Theorem 3.2 gives $M_1 = \log 3$, and every unrefined feedback has that maximal-pattern rate. An explicit safe selector chooses 0 if $x_1 = 2$, otherwise 1 if $x_1 = 0$ and $x_0 \neq 1$, and otherwise 2.

On the three target first-letter cylinders the densities are $(0, 0, 1)$, $(0, 2/3, 1/3)$, and $(1, 0, 1/3)$. Hence $\Theta_3 = 4/3$. This failure persists under *every* full-support probability ν . If one of the safe branch measures has no bounded ν -density, the single-step criterion already fails its prerequisites. Otherwise $g_0(K_0) \subset [2]$ and $g_1(K_1) \subset [1]$ give disjoint density supports. At most two densities are nonzero at a point, so their largest two sum to their total. Its integral is

$$\sum_{u=0}^2 \nu(K_u) = 1 + \nu([12]) > 1.$$

Thus $\Theta_3(\nu) > 1$ for every such ν , including singular or atomic probabilities. This excludes both original single-step tests, but not all possible changes of presentation or iteration.

There is also a distinction from iterating the pointwise envelope Φ . On nonnegative functions the disjoint supports give $\Phi = \sum_u L_u$. Under fair measure its first-letter matrix is $\frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$, with positive eigenvector $(1, 1, (\sqrt{13} - 1)/2)$ and eigenvalue $(3 + \sqrt{13})/6 > 1$. Hence $\|\Phi^n \mathbf{1}\|_\infty$ grows, while every history of *global* two-label lists decays. Maximization by state and composition cannot be interchanged.

4. FINITE-CYLINDER CRITERIA FOR CONTRACTION

We now make the product condition of Theorem 3.2 checkable from finite-cylinder safety data. The exact representation comes first; three sufficient criteria then describe how a deficit propagates through every sequence of global label lists.

4.1. An exact finite representation. Let D be a finite alphabet of size $d \geq 2$, $Q = D^{\mathbb{N}_0}$, and μ its fair Bernoulli probability. The finite controls have legal maps $g_u = P_u \sigma$, where p_u permutes D and P_u acts in every coordinate. Suppose

$$K_u = \bigcup_{\xi \in R_u} [\xi], \quad R_u \subset D^h, \quad h \geq 2,$$

and the K_u cover Q . Shorter cylinders can be completed to the same length h . The continuous cemetery extension used above applies to these clopen domains. The control period is always one.

For every bounded Borel function the full operator is

$$L_u f(y) = \frac{1}{d} \sum_{a \in D} \mathbf{1}_{R_u}(a, p_u^{-1} y_0, \dots, p_u^{-1} y_{h-2}) f(a P_u^{-1} y). \quad (4.1)$$

Indeed, conditioned on $x_0 = a$ the tail of x still has law μ , and P_u preserves that law. Substitution in (3.2) gives this formula with the exact safety gate.

Index source and target vectors by D^{h-1} . For target η and source ξ , set

$$T_u(\eta, \xi) = \frac{1}{d} \# \left\{ a \in D : \begin{array}{l} (a, p_u^{-1} \eta_0, \dots, p_u^{-1} \eta_{h-2}) \in R_u, \\ (a, p_u^{-1} \eta_0, \dots, p_u^{-1} \eta_{h-3}) = \xi \end{array} \right\}. \quad (4.2)$$

When $h = 2$ the second tuple is just a . Functions of the first $h - 1$ coordinates form an invariant subspace of every L_u , represented by T_u . Starting at $\mathbf{1}$, every list product stays in this space. Every such cylinder has positive measure and a positive product has norm equal to its value on $\mathbf{1}$. Therefore, for every length and every history,

$$\|A_{S_{n-1}} \cdots A_{S_0}\|_{\infty \rightarrow \infty} = \|T_{S_{n-1}} \cdots T_{S_0} \mathbf{1}\|_{\infty}, \quad T_S = \sum_{u \in S} T_u. \quad (4.3)$$

This equality is on the full L^∞ space. It does not claim that an arbitrary Borel function is approximated by a fixed finite space, or that a Borel feedback uses a finite-state rule. The finite indices are proof tools, not new state coordinates or a clock.

For two-letter domains, rows are target letters and columns are source letters, with

$$T_u(j, i) = d^{-1} \mathbf{1}_{R_u}(i, p_u^{-1} j), \quad d(w^\top T_u)_i = \sum_{z: (i, z) \in R_u} w(p_u z). \quad (4.4)$$

The structural conditions below use this source-column identity. A left budget $w^\top T_S \leq \theta w^\top$ is different from changing a reference density, which leads instead to a right inequality $T_S a \leq \theta a$. They cannot be interchanged.

4.2. A strict source-weight criterion. Partition $D = A \sqcup B \sqcup C$ into nonempty sets of sizes a, b, c , so $d = a + b + c$. Use three controls with

$$p_0(C) \subset D \setminus A, \quad p_1(B) \subset A, \quad p_2(A) \subset D \setminus A. \quad (4.5)$$

Choose arbitrary relations $E_0 \subset A \times C$ and $E_1 \subset A \times B$, with $E_0 \cup E_1 \neq \emptyset$, and define

$$\begin{aligned} R_0 &= ((D \setminus A) \times C) \cup E_0, \\ R_1 &= ((D \setminus A) \times B) \cup E_1, \\ R_2 &= (A \times D) \cup (D \times A), \quad K_u = [R_u]. \end{aligned} \quad (4.6)$$

Put $e = \max_{i \in A} |\{z : (i, z) \in E_1\}|$. Require only the following counts and landing conditions:

$$s_- := \max \left\{ \frac{c}{b+c}, \frac{e}{b+c-e} \right\} < s_+ := \frac{\min\{a, c\}}{b}. \quad (4.7)$$

These are direct conditions on the original allowed cylinders.

Theorem 4.1 (Strict source budget). *For every system satisfying (4.5)–(4.7), $r = 3$, all histories of global two-label lists have exponentially decaying inverse capacity, and every legal unrefined Borel feedback has $h^*(s) = M_1 = \log 3$. Every finite Borel atom refinement has $h^* \geq \log 3$. Moreover, no full-support reference probability can directly satisfy the joint-density or scalar single-step criterion for $k = 3$ on these original branches.*

Proof. Take $s = (s_- + s_+)/2$, $\gamma = 1 + s$, and let $w_i = \gamma$ on A , $w_i = 1$ elsewhere. Write $H = \sum_i w_i = d + sa$. Before division by d , the weighted contributions of controls 0, 1, 2 in a source column are

	0	1	2
$i \in A$	$e_0(i) \leq c$	$\gamma e_1(i) \leq \gamma e$	H
$i \notin A$	c	γb	a

Each entry follows from the original domains and (4.5). For example, on A control 2 permits every tail, hence the sum of all weights irrespective of its permutation. Off A , its permitted tails lie in A and land outside A , each with weight one.

Since $sb < a, c$, both c and $\gamma e \leq \gamma b$ are smaller than H . Thus every two-label sum on A is bounded by $H + \max\{c, \gamma e\}$. Define

$$\theta = \max \left\{ \frac{H + \max\{c, \gamma e\}}{d\gamma}, \frac{c + \gamma b}{d}, \frac{c + a}{d}, \frac{\gamma b + a}{d} \right\}.$$

We have $d\gamma - H = s(b + c) > c, \gamma e$ because $s > s_-$. The other three terms are smaller than one by $sb < a, c$ and $b > 0$. Hence $0 < \theta < 1$ and $w^\top T_S \leq \theta w^\top$ for every global two-label list. Repeatedly applying this single inequality gives

$$\|T_{S_{n-1}} \cdots T_{S_0} \mathbf{1}\|_\infty \leq w^\top T_{S_{n-1}} \cdots T_{S_0} \mathbf{1} \leq H\theta^n,$$

since $\min_i w_i = 1$. Equation (4.3) controls the complete capacity for all lengths.

If $x_0 \in A$, control 2 is legal; otherwise its next letter in A, B, C requires, respectively, 2, 1, 0. Thus the domains cover Q . Taking $x_0 \notin A$ and x_1 successively in C, B, A gives cylinders on which each of 0, 1, 2 alone is legal, so $r = 3$. A clopen attaining selector chooses 2 when $x_0 \in A$ or $x_1 \in A$, otherwise 1 when $x_1 \in B$, otherwise 0. It is safe, single-valued, and stays in Q at each step; the same rule at every successor gives all-time safety. Applying Theorem 3.2 proves all pattern conclusions, including the lower bound for arbitrary refinements.

Let $O = [E_0 \cup E_1]$. It is nonempty and clopen, and safety multiplicity is exactly $1 + \mathbf{1}_O$. For a full-support probability ν , either some branch fails the bounded-density prerequisite, or all densities exist. In the latter case the images of controls 0 and 1 lie, respectively, in $[D \setminus A]$ and $[A]$; their supports are disjoint. The largest two densities consequently sum to all three almost everywhere. Their integral is $\sum_u \nu(K_u) = 1 + \nu(O) > 1$. Thus $\Theta_3(\nu) > 1$. The scalar condition implies $\Theta_3 < 1$ and also fails. $\square$

For instance $1 \leq b \leq a \leq b + c$ and $c > b$ imply (4.7) for every possible $e \leq b$: $e/(b + c - e) \leq b/c < 1$, $c/(b + c) < 1$, while $s_+ \geq 1$. Every allowed overlapping relation and permutation landing pattern is covered. Constants can be uniform when these counts have fixed bounds.

For a concrete example take $D = \{0, 1, 2, 3, 4\}$, $A = \{0, 1\}$, $B = \{2\}$, $C = \{3, 4\}$, $E_0 = \{(0, 3), (1, 4)\}$, $E_1 = \{(0, 2)\}$. Let p_0 be the identity, $p_1 = (0 \ 2)$, and specify p_2 by $0 \mapsto 2$, $1 \mapsto 3$, $2 \mapsto 0$, $3 \mapsto 4$, $4 \mapsto 1$. Here $s = 4/3$, $\gamma = 7/3$, $H = 23/3$, $\theta = 13/15$, and $b_n \leq (23/3)(13/15)^n$. The target densities at first letters 0, 1, 2, 3, 4 are, respectively,

$$(0, 4/5, 2/5), \quad (0, 0, 2/5), \quad (0, 0, 1), \quad (4/5, 0, 1), \quad (4/5, 0, 2/5).$$

Thus $\Theta_3 = 9/5$. In [14] the two legal successors have first letters 4 and 1. No common successor is present. The natural $A/B/C$ merging also fails to commute with the listed permutations; this observation does not exclude every possible factor.

4.3. Cross-layer dispersion without a strict left budget. Let U have size $q \geq 2$. Partition $D = \bigsqcup_{t \in U} I_t$ into nonempty source classes, and let $m(i) = t$ on I_t . For each i partition the tail alphabet by

$$D = \bigsqcup_{u \neq m(i)} V_{i,u}, \quad F_i \subset D.$$

Control $u \neq m(i)$ is the primary legal control on $V_{i,u}$, while control $m(i)$ is also legal on F_i . Precisely,

$$R_u = \{(i, z) : u \neq m(i), z \in V_{i,u}\} \cup \{(i, z) : u = m(i), z \in F_i\}. \quad (4.8)$$

The permutations p_u are arbitrary subject to the count below. Assume

$$a_0 = \min_{i, u \neq m(i)} (|V_{i,u}| - |F_i|) > 0, \quad (4.9)$$

$$a_1 = \min_{i,t} \left[d - \sum_{u \neq m(i)} |p_u(V_{i,u}) \cap I_t| \right] > 0. \quad (4.10)$$

The second count includes multiplicities of coincident landings. It says that the primary successor mass cannot be confined to any one source class. Set $\varepsilon = a_0/d$, $\eta = a_1/d$, and $\zeta = 1 - \varepsilon\eta \in (0, 1)$.

Theorem 4.2 (Dispersion criterion). *Under (4.8)–(4.10), $r = q$ and*

$$b_n \leq d\zeta^{\lfloor n/2 \rfloor} \leq \frac{d}{\sqrt{\zeta}} (\sqrt{\zeta})^n \quad (n \geq 0). \quad (4.11)$$

Every legal unrefined Borel feedback has $h^(s) = M_1 = \log q$; all finite Borel atom refinements have $h^* \geq \log q$. There is no $w \in (0, \infty)^d$ and $\vartheta < 1$ such that $w^\top T_S \leq \vartheta w^\top$ for every global $q - 1$ label list. A common nonstrict left budget does exist.*

Proof. Write $S_t = U \setminus \{t\}$ and $T^{(t)} = T_{S_t}$. Its column mass is exactly

$$c_i(t) = \sum_j T^{(t)}(j, i) = \begin{cases} 1, & m(i) = t, \\ 1 - |V_{i,t}|/d + |F_i|/d \leq 1 - \varepsilon, & m(i) \neq t. \end{cases}$$

Thus every $T^{(t)}$ has induced l^1 norm at most one. Let $Q_i(j) = d^{-1} \sum_{u \neq m(i)} \mathbf{1}_{p_u(V_{i,u})}(j)$. It has total mass one, and by (4.10) $\sum_{j \notin I_t} Q_i(j) \geq \eta$ for every i, t .

Fix two global lists S_s, S_t . The source- i column mass of $T^{(t)}T^{(s)}$ is $\sum_j c_j(t)T^{(s)}(j, i)$. If $i \notin I_s$, it is at most $1 - \varepsilon \leq 1 - \varepsilon\eta$. If $i \in I_s$, its first column is exactly Q_i , and at least η of that mass lands outside I_t , where the next list loses at least ε . Its mass is again at most ζ . Hence every pair has l^1 norm at most ζ . Grouping arbitrary histories into pairs proves their l^1 norm is at most $\zeta^{\lfloor n/2 \rfloor}$. Applying the product to the all-one vector and using (4.3) gives (4.11). No restrictions on switching order or on the feedback are used.

For every control u choose i with $m(i) \neq u$. Condition (4.9) gives a $z \in V_{i,u} \setminus F_i$, on whose cylinder only u is legal. This proves $r = q$. The unique primary control on each two-letter state is a clopen selector, legal at every state and at each of its future successors. The capacity theorem now proves the entropy statements.

If a strict common left budget existed, take a coordinate i with minimum w_i and select the already assumed inequality with $t = m(i)$. That column is Q_i , of mass one, and

$$(w^\top T^{(m(i))})_i \geq w_i > \vartheta w_i,$$

a contradiction. This checks an inequality that was assumed for every global list; it does not alter lists by source in an actual product. Finally $\mathbf{1}^\top T^{(t)} \leq \mathbf{1}^\top$ gives the nonstrict budget, with strict contraction after two steps. $\square$

An explicit family is obtained with $q \geq 2$, $m \geq 2$, and $d = (q - 1)m$. Partition tails into $q - 1$ blocks of size m . For each source, biject its $q - 1$ primary labels with these blocks, and let every p_u preserve each tail block. Choose arbitrary F_i of size at most $m - 1$, and any partition into q nonempty source classes whenever such a partition exists. The primary landing mass is uniform, so $a_0 \geq 1$ and $a_1 = d - \max_t |I_t| > 0$. This supplies arbitrary overlapping choices and nontrivial permutations, within the same estimate.

For a small witness use $D = \{0, 1, 2, 3\}$, $U = \{0, 1, 2\}$, $I_0 = \{0, 1\}$, $I_1 = \{2\}$, $I_2 = \{3\}$, and $F_i = \{0\}$ for every source. At sources 0, 1 the primary labels 1, 2 use tails $\{0, 1\}, \{2, 3\}$; at source 2 the labels 0, 2 use those tails; at source 3 the labels 0, 1 do so. Let p_0 be the identity, $p_1 = (0 \ 1)$, $p_2 = (2 \ 3)$. The densities at target first letters are

$$(1, 1/2, 1/4), \quad (1/2, 3/4, 0), \quad (0, 1/4, 3/4), \quad (0, 1/4, 3/4).$$

Thus $\Theta_3 = 3/2$, but $\varepsilon = 1/4$, $\eta = 1/2$ give $b_n \leq 4(7/8)^{\lfloor n/2 \rfloor}$ and $M_1 = \log 3$. On $[00]$ controls 0, 1 have different successors. The preceding minimum-coordinate proof excludes all strict left budgets, not just a numerically tested weight. This example does not inherit the all-reference-probability exclusion of Theorem 4.1; its proved exclusion concerns the specified matrix representation.

4.4. Gain columns and next-step losses. The preceding dispersion theorem retains a common nonstrict column budget. We now describe a class in which every source has a gain list, no positive common nonstrict left budget exists, and all global list histories nevertheless contract. The condition is on original safety sets and their cardinalities, not on a computed product norm.

4.4.1. Safety domains and a contraction condition. Let $q = |U| \geq 3$ and partition $D = \bigsqcup_{t \in U} L_t$ into nonempty source classes. Write $d = |D|$, $\kappa(i) = t$ on L_t , and $M = \min_t |L_t|$. For every source i choose a spare label $m(i) \neq \kappa(i)$. For each $u \in A_i := U \setminus \{\kappa(i), m(i)\}$ choose mutually disjoint tail sets $H_{i,u}$, all of size $a \geq 1$. Put

$$H_i = \bigsqcup_{u \in A_i} H_{i,u}, \quad b = (q-2)a, \quad B_i = D \setminus H_i.$$

Also choose an arbitrary spare tail set F_i of size f . It may overlap any of the primary tail sets. Define the original allowed relation by

$$\begin{aligned} R_u &= \{(i, z) : u = \kappa(i), z \in B_i\} \\ &\cup \{(i, z) : u \in A_i, z \in H_{i,u}\} \\ &\cup \{(i, z) : u = m(i), z \in F_i\}, \quad K_u = [R_u]. \end{aligned} \tag{4.12}$$

Every two-letter state has one primary control and possibly a spare. The permutations p_u may be arbitrary. Use $g_u = P_u \sigma$ with the same continuous cemetery extension and period one. Require

$$M > b, \quad a < f, \quad b + f < d, \tag{4.13}$$

and

$$2d(f-a) + (f-a)^2 < [d - (q-1)a][M - (q-2)a]. \tag{4.14}$$

These conditions contain only cardinalities of the original source classes, small primary tails, and spare tails. The theorem applies to every placement of these tails and every permutation, not only to matrices previously known to be stable.

Set

$$\begin{aligned} \gamma &= (f-a)/d, & C &= 1 + \gamma, & \varepsilon &= 1 - (b+f)/d, & \eta &= (M-b)/d, \\ \zeta &= \max\{C(1-\varepsilon), C^2 - (\varepsilon + \gamma)\eta\}, & \theta &= \sqrt{\zeta}, & H &= dC/\sqrt{\zeta}. \end{aligned} \tag{4.15}$$

Theorem 4.3 (Gain-loss safety criterion). *For every system satisfying (4.12)–(4.14), $r = q$, $0 < \zeta < 1$, and the complete inverse capacities satisfy*

$$b_n \leq dC^{n \bmod 2} \zeta^{\lfloor n/2 \rfloor} \leq H\theta^n \quad (n \geq 0). \tag{4.16}$$

Every source has an original $q-1$ label list whose column mass is $C > 1$. There is no $w \in (0, \infty)^d$ satisfying $w^\top T_S \leq w^\top$ for all original $q-1$ label lists. Nevertheless every unrefined legal Borel feedback has

$$h^*(s) = M_1 = \log q = \log r, \tag{4.17}$$

and every finite Borel atom refinement has $h^ \geq \log q$.*

4.4.2. Complete estimate for every pair of global lists.

Proof. Write $S_t = U \setminus \{t\}$ and $T^{(t)} = T_{S_t}$. The exact source-column masses, counted with edge multiplicities, are

$$c_i(t) = \sum_j T^{(t)}(j, i) = \begin{cases} (b+f)/d = 1 - \varepsilon, & t = \kappa(i), \\ 1 + (f-a)/d = C, & t \in A_i, \\ 1, & t = m(i). \end{cases} \quad (4.18)$$

Indeed, omitting the large primary control leaves all small tails and the spare; omitting a small primary tail removes a edges and adds f spare edges to the primary total d ; omitting the spare leaves exactly the primary partition. Distinct controls landing on the same target are still distinct counted edges.

Fix two whole-space lists S_s, S_t . The source- i column mass of their product is

$$\sum_j (T^{(t)} T^{(s)})(j, i) = \sum_j c_j(t) T^{(s)}(j, i) = c_i(s) + \sum_j [c_j(t) - 1] T^{(s)}(j, i). \quad (4.19)$$

If $s = \kappa(i)$, then $c_i(s) = 1 - \varepsilon$ and every next-column mass is at most C . The product column is at most $C(1 - \varepsilon)$.

If $s \neq \kappa(i)$, the whole large primary tail is retained. Its permutation image misses just b letters. Hence, for every next loss class L_t ,

$$\sum_{j \in L_t} T^{(s)}(j, i) \geq \frac{|p_{\kappa(i)}(B_i) \cap L_t|}{d} \geq \frac{|L_t| - b}{d} \geq \eta. \quad (4.20)$$

Small and spare tails can only increase this mass. On L_t the next list has mass $1 - \varepsilon$, whereas elsewhere its mass is at most C . Thus

$$\begin{aligned} \sum_j c_j(t) T^{(s)}(j, i) &\leq C c_i(s) - [C - (1 - \varepsilon)] \sum_{j \in L_t} T^{(s)}(j, i) \\ &\leq C^2 - (\varepsilon + \gamma)\eta. \end{aligned} \quad (4.21)$$

This includes both the current gain and the worst next gain, rather than subtracting a loss while omitting the cost of spare labels. The lists s, t remain fixed over the state space throughout.

Dividing (4.14) by d^2 yields

$$2\gamma + \gamma^2 < (\varepsilon + \gamma)\eta, \quad \varepsilon + \gamma = [d - (q-1)a]/d > 0.$$

Since $M \leq d/q$ and $M > b$, $0 < \eta < 1$. The last inequality therefore implies $\varepsilon > \gamma + \gamma^2 > \gamma$. It follows that $C(1 - \varepsilon) < 1$; it also gives $C^2 - (\varepsilon + \gamma)\eta < 1$ directly. Both quantities are positive: the first by $b + f > 0$, the second because $\varepsilon + \gamma < 1$, $\eta < 1$, and $C^2 > 1$. Thus $0 < \zeta < 1$ and every list pair has induced l^1 norm at most ζ .

For every history, group consecutive factors into pairs. The possible odd remaining factor has norm at most C , not one. Consequently

$$\|T^{(t_{n-1})} \dots T^{(t_0)}\|_1 \leq C^{n \bmod 2} \zeta^{\lfloor n/2 \rfloor}.$$

Equation (4.3) and $\|\mathbf{1}\|_1 = d$ prove (4.16), including $b_0 = 1 \leq d$.

To exclude *all* common positive nonstrict left budgets, suppose one existed and take a coordinate i of smallest w_i . Since $q \geq 3$, choose $t \in A_i$. The already assumed inequality for this global list would give

$$w_i \geq (w^\top T^{(t)})_i = \sum_j w_j T^{(t)}(j, i) \geq w_i c_i(t) = C w_i > w_i,$$

a contradiction. The contradiction holds for every positive vector w .

All states have a primary legal control. For each u choose $i \in L_u$. As $|B_i| = d - b > f$, a $z \in B_i \setminus F_i$ exists. Only u is legal on $[iz]$. Thus every control is needed in a subcover, giving $r = q$. The actual primary selector is

$$s_0(x) = \begin{cases} \kappa(x_0), & x_1 \in B_{x_0}, \\ u, & x_1 \in H_{x_0, u}, u \in A_{x_0}. \end{cases} \quad (4.22)$$

Its primary domains partition all states, so it is single-valued and clopen. At every successor it again uses the same legal state function; all infinite trajectories, including periodic and constant states, stay safe. The capacity theorem gives the lower bound for every Borel rule, not only this selector. With only q physical labels every unrefined rule has at most q^m patterns on m positions. This proves (4.17) and the refinement statement. $\square$

The unavailable common budget is the *original first-letter, one-step, positive left* budget. Two-step products do have a strict common budget. No claim excludes higher-dimensional lifts, right budgets, arbitrary changes of reference measure, or path-dependent Lyapunov functions. In particular no conclusion about L_1 follows from (4.17).

4.4.3. A family for arbitrary control counts. For any $q \geq 3$ and integer $m \geq q + 1$, take $d = qm$, all $|L_t| = m$, $a = 1$, and $f = 2$. Each source may assign its spare label arbitrarily, choose its $q - 2$ distinct small-tail letters arbitrarily, choose any two spare-tail letters, and use arbitrary permutations. Conditions (4.13) hold. The right side minus the left side of (4.14) is

$$P(m) = qm^2 - (q^2 + q - 1)m + q^2 - 3q + 1.$$

Here $P(q + 1) = q^2 - 2q + 2 > 0$ and $P(m + 1) - P(m) = 2qm - q^2 + 1 > 0$ for $m \geq q + 1$. Thus the entire family satisfies the theorem, with constants depending only on these counts. The condition is uniform over all placements of the allowed tails. It does not assert that every member lacks all protected symbolic subsystems; no such subsystem is a hypothesis of the proof.

4.4.4. A three-control example with gain columns. Let $D = \{0, \dots, 23\}$ and $U = \{0, 1, 2\}$. Take $L_0 = \{0, \dots, 7\}$, $L_1 = \{8, \dots, 15\}$, $L_2 = \{16, \dots, 23\}$. For $i \in L_t$ the large label is t , the small label is $t + 1$, and the spare is $t + 2$ modulo three. The complete tails are specified in the following table; every source in a row has that row's tails.

Source class	Large label	Small tail H_t	Spare tail F_t
L_0	0	$\{0, 1, 2, 3\}$	$\{4, 5, 6, 7, 8\}$
L_1	1	$\{8, 9, 10, 11\}$	$\{0, 1, 2, 3, 4\}$
L_2	2	$\{8, 9, 10, 11\}$	$\{0, 1, 2, 3, 4\}$

The large tail is $D \setminus H_t$. Let p_0 be the identity, $p_1 = (0\ 1)$, $p_2 = (16\ 17)$, fixing all other letters. These permutations preserve each of their actually used tail sets. Thus the exact matrix description, with subscripts modulo three, is

$$T_u(j, i) = \frac{1}{24} [\mathbf{1}_{L_u}(i) \mathbf{1}_{D \setminus H_u}(j) + \mathbf{1}_{L_{u-1}}(i) \mathbf{1}_{H_{u-1}}(j) + \mathbf{1}_{L_{u-2}}(i) \mathbf{1}_{F_{u-2}}(j)].$$

It computes capacity without replacing the actual permutations by common successor maps. Its target densities are

Target letter	ρ_0	ρ_1	ρ_2	Largest two sum
0, 1, 2, 3	1/3	1	1/3	4/3
4	2/3	2/3	2/3	4/3
5, 6, 7	1/3	1/3	2/3	1
8	2/3	0	2/3	4/3
9, 10, 11	2/3	0	1/3	1
12, ..., 23	1/3	1/3	1/3	2/3

In particular $\Theta_3 = 4/3$. Here $d = 24$, $M = 8$, $a = b = 4$, $f = 5$, and the strict count margin is $64 - 49 = 15$. The constants are

$$\begin{aligned} \varepsilon &= 5/8, \quad \gamma = 1/24, \quad C = 25/24, \quad \eta = 1/6, \\ C(1 - \varepsilon) &= 25/64, \quad C^2 - (\varepsilon + \gamma)\eta = 187/192. \end{aligned}$$

The full-history estimate is therefore

$$b_n \leq 24(25/24)^{n \bmod 2} (187/192)^{\lfloor n/2 \rfloor}. \quad (4.23)$$

Source columns have masses $3/8$, $25/24$, or 1 , depending on which of their large, small, or spare labels is omitted. The cylinders $[0, 12]$, $[8, 12]$, $[16, 12]$ require controls $0, 1, 2$. On $[16, 0]$ controls $1, 2$ are legal but have successor first letters $1, 0$. Consequently the full Borel class includes choices genuinely changing the future trajectory.

There is no consecutive 000 trajectory with uniquely legal controls at every step. A unique 0 at a source in L_0 has its next first letter in $\{9, \dots, 23\}$; a unique 0 in L_2 has its next first letter in $\{8, \dots, 11\} \subset L_1$. In L_1 , control 0 is only spare and is never unique. Thus from L_0 there can be at most one further unique 0 , and from L_2 none. This excludes a direct consecutively forced three-symbol full language. It does not exclude sparse bridges, other factors, or every protected coding construction.

The gain–loss argument is elementary positive-matrix reasoning, applied here through exact allowed-tail quotas and actual feedback fibers. Shi et al. [11, Definition 2.5 and Theorems 2.6, 2.8, pp. 8–11 of arXiv:2004.01867v2] already allow losses to compensate growth in matrix products. Their positive-diagonal and ordered-window assumptions do not automatically hold here: for example $T^{(1)}(0, 0) = 0$ in this witness. Our proof uses the mass quota (4.20) for every next global list and does not depend on that theorem. The separation concerns the specified common-left template in the original representation.

5. COMMON SUCCESSORS AND A CLOSED-COVER FORMULA

The inverse-capacity argument permits the independent positions to depend on the feedback. A common successor gives a different route: the orbit is fixed, and only the names assigned on an overlapping cover vary. We first prove the exact formula for Borel refinements of a closed cover, then apply it to safe feedbacks.

Let Q be a nonempty compact metric space and let $T : Q \rightarrow Q$ be a Borel map. For a finite Borel partition α of Q and a finite $F \subset \mathbb{N}_0$, put

$$P_\alpha(F) = \left| \left\{ (\alpha(T^k x))_{k \in F} : x \in Q \right\} \right|,$$

where $\alpha(y)$ denotes the atom containing y and empty atoms are discarded. Thus $P_\alpha(F)$ is the number of nonempty atoms in $\bigvee_{k \in F} T^{-k} \alpha$. For a strictly increasing sequence $A = (a_i)_{i \geq 1}$ in $\mathbb{N}_0$, define

$$h_A(T, \alpha) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log P_\alpha(\{a_1, \dots, a_n\}), \quad (5.1)$$

$$p_\alpha^*(n) = \max_{\substack{F \subset \mathbb{N}_0 \\ |F|=n}} P_\alpha(F), \quad h^*(T, \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} \log p_\alpha^*(n). \quad (5.2)$$

All logarithms are natural. These are combinatorial counting quantities; for Borel atoms they are not Shannon entropies of an invariant measure. The maximum in (5.2) exists because the counts are integers between 1 and $|\alpha|^n$. Splitting a window into two parts gives $p_\alpha^*(n+m) \leq p_\alpha^*(n)p_\alpha^*(m)$. Subadditivity proves the existence of the limit, and $h_A(T, \alpha) \leq h^*(T, \alpha)$. No invertibility or continuity of the observation is used here.

Let $\mathcal{K} = (K_w)_{w \in W}$ be a finite closed cover of Q , and write

$$r = N(\mathcal{K}) = \min\{|I| : I \subset W, \bigcup_{w \in I} K_w = Q\}.$$

The set $\mathcal{B}(\mathcal{K})$ consists of all finite Borel partitions refining $\mathcal{K}$: each atom must be contained in at least one K_w . Equivalently, one may first consider all Borel maps $s : Q \rightarrow W$ with $x \in K_{s(x)}$, and then allow arbitrary finite Borel refinements of their fibers.

Let H be a primitive finite 0–1 matrix, with vertex set V , and let

$$Y_H = \{y \in V^{\mathbb{N}_0} : H_{y_j y_{j+1}} = 1 \text{ for all } j \geq 0\}.$$

Primitivity means that all sufficiently large powers of H have strictly positive entries. Write σ for the one-sided shift.

Theorem 5.1. *Suppose $T : Q \rightarrow Q$ is topologically conjugate to $\sigma^p : Y_H \rightarrow Y_H$ for some primitive H and integer $p \geq 1$. Let $\mathcal{K}$ be any finite closed cover of Q . For every strictly increasing $A = (a_i)$ satisfying*

$$a_{i+1} - a_i \longrightarrow \infty, \quad (5.3)$$

one has

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \min_{\alpha \in \mathcal{B}(\mathcal{K})} h^*(T, \alpha) = \log r. \quad (5.4)$$

An ordered difference partition of any minimum subcover attains both minima, simultaneously for every A satisfying (5.3). Consequently,

$$\max_A \inf_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \log r, \quad (5.5)$$

where the maximum ranges over all strictly increasing infinite sequences and is attained, for example, by $a_i = (i - 1)^2$.

The order of quantifiers is part of the assertion. Every A satisfying (5.3) works for every allowed partition, although the finite initial segment discarded in the proof can depend on that partition. The proof does not interchange an infimum over partitions with a finite-window limit.

5.1. A category and concatenation lemma. We use meagerness relative to Q . A Borel set has the Baire property, so a nonmeager Borel set is comeager in some nonempty open set. Indeed, the sets differing from open sets by meager sets form a sigma-algebra: for complements, the additional difference is contained in the nowhere dense boundary of an open set. A compact metric space is a Baire space. In particular, a nonempty open set cannot be covered by countably many nowhere dense sets, as is also seen by choosing nested nonempty open sets with closures avoiding those sets and diameters tending to zero.

If $F : Q \rightarrow Q$ is continuous and open, preimages of meager sets are meager. For a closed nowhere dense N , the closed set $F^{-1}N$ has empty interior: otherwise the image of a nonempty open subset would be a nonempty open subset of N . Passing to countable unions proves the assertion.

Lemma 5.2. *Under the dynamical assumptions of Theorem 5.1, T is continuous, open, and onto. Given finitely many nonempty cylinder sets $C_1, \dots, C_q$ in symbolic coordinates, there is an integer $b \geq 1$ such that, for every $k \geq 1$, every $t_1, \dots, t_k \in \{1, \dots, q\}$, and all $0 \leq n_1 < \dots < n_k$ with $n_{j+1} - n_j \geq b$,*

$$\bigcap_{j=1}^k T^{-n_j} C_{t_j}$$

is nonempty and open.

Proof. Work on Y_H through the conjugacy. The image under σ of a prefix cylinder of length at least two is the cylinder obtained by removing its first symbol. The image of a one-symbol cylinder is the union of the permitted successor cylinders. Hence σ is open. Every vertex has a predecessor, so every infinite path has a predecessor; thus σ is onto. These properties pass to powers and conjugates.

Choose within each C_j a nonempty prefix cylinder, and extend their defining words, if necessary, to a common length ℓ . Choose N with $H^n > 0$ for every $n \geq N$, and then choose b with $pb - \ell + 1 \geq N$. Place the word for C_{t_j} at coordinate pn_j . Between the end of one placed word and the start of the next there are $p(n_{j+1} - n_j) - \ell + 1$ edges to fill. Primitivity supplies a path of that length between any two endpoint vertices. Predecessors fill the finite segment before the first placed word, and successors provide an infinite continuation after the last. This constructs an actual point in the intersection. The intersection is open because it is a finite intersection of preimages of open sets under continuous maps. $\square$

Lemma 5.3. *Under the same assumptions, let α be a finite Borel partition with q nonmeager atoms. For every A satisfying (5.3) there exists $m \geq 1$ such that*

$$P_\alpha(\{a_1, \dots, a_n\}) \geq q^{n-m+1} \quad (n \geq m). \quad (5.6)$$

In particular $h_A(T, \alpha) \geq \log q$.

Proof. List the nonmeager atoms as $D_1, \dots, D_q$. The Baire property provides a nonempty cylinder C_j for which $C_j \setminus D_j$ is meager. Let b be supplied by Lemma 5.2, and choose m so that $a_{i+1} - a_i \geq b$ for all $i \geq m$.

Fix $n \geq m$ and any target labels $t_m, \dots, t_n$. The set

$$G = \bigcap_{i=m}^n T^{-a_i} C_{t_i}$$

is nonempty and open. Each power T^{a_i} is continuous and open, so

$$\bigcup_{i=m}^n T^{-a_i} (C_{t_i} \setminus D_{t_i})$$

is meager and cannot cover G . A point outside this union realizes all the specified atoms at the sampled times. Distinct target label strings give distinct patterns; restriction from the full window to this tail is onto. This proves (5.6). Dividing its logarithm by n and taking a limit inferior proves the asserted bound. $\square$

5.2. Proof of the exact formula.

Proof of Theorem 5.1. Let $\alpha \in \mathcal{B}(\mathcal{K})$. Assign to each atom D one index $w(D)$ with $D \subset K_{w(D)}$. Let J be the set of indices assigned to the nonmeager atoms. If the closed sets $(K_w)_{w \in J}$ did not cover Q , their complement would be a nonempty open set covered by the finitely many meager atoms of α , which is impossible. Thus $|J| \geq r$, and the number q of nonmeager atoms is at least r . Lemma 5.3 gives

$$h_A(T, \alpha) \geq \log q \geq \log r$$

for every A satisfying (5.3). Closedness of the cover members is used precisely in this passage from category to a full cover.

Choose a minimum subcover $K_{w_1}, \dots, K_{w_r}$, and set

$$E_j = K_{w_j} \setminus \bigcup_{k < j} K_{w_k}, \quad \alpha_0 = \{E_1, \dots, E_r\}. \quad (5.7)$$

Every E_j is Borel and is nonempty, since a minimum subcover has no redundant member. The partition refines $\mathcal{K}$ and has r atoms. Hence $P_{\alpha_0}(F) \leq r^{|F|}$ for all finite F , giving $h_A(T, \alpha_0) \leq h^*(T, \alpha_0) \leq \log r$ for every A . The established lower bound and $h^* \geq h_A$ yield (5.4) and simultaneous attainment. For arbitrary A , the partition α_0 bounds the inner infimum in (5.5) by $\log r$. Any A satisfying (5.3) gives the reverse bound and attains the maximum. $\square$

5.3. An excluded-letter cover. For any finite cover $\mathcal{K}$ define its combinatorial sequence entropy by

$$h_A(T, \mathcal{K}) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log N \left(\bigvee_{i=1}^n T^{-a_i} \mathcal{K} \right).$$

For a partition this is (5.1). For overlapping sets, covering each finite orbit window need not correspond to one fixed refining partition used at every time.

Corollary 5.4. *Let $q \geq 3$, $Q = \{0, \dots, q-1\}^{\mathbb{N}_0}$, $T = \sigma$, and*

$$K_j = \{x : x_0 \neq j\} \quad (0 \leq j < q), \quad \mathcal{K} = (K_j)_{j=0}^{q-1}.$$

For every increasing infinite A ,

$$h_A(\sigma, \mathcal{K}) = \log \frac{q}{q-1}. \quad (5.8)$$

For every such A with diverging successive gaps,

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(\sigma, \alpha) = \log 2 > h_A(\sigma, \mathcal{K}). \quad (5.9)$$

Proof. The cover has no single member equal to Q , and any two distinct members cover Q . Thus $r = 2$, and (5.9) follows from Theorem 5.1 once (5.8) is proved.

For a window F of size n , the full shift has all q^n possible input vectors. Each member of the joined cover specifies one excluded letter at each coordinate, and covers $(q-1)^n$ of these vectors. Consequently the minimum number $c(n)$ of such boxes is at least $(q/(q-1))^n$. For completeness, the following elementary probabilistic covering bound gives the matching exponent. Choose R excluded-letter vectors independently and uniformly from the q^n possibilities. A fixed input vector is left uncovered with probability at most

$$\left(1 - \left(\frac{q-1}{q} \right)^n \right)^R \leq \exp \left(-R \left(\frac{q-1}{q} \right)^n \right).$$

Taking $R = \lceil (q/(q-1))^n (n \log q + 1) \rceil$ makes the sum of these probabilities over all input vectors smaller than one. Some choice therefore covers every vector. We have proved

$$\left(\frac{q}{q-1} \right)^n \leq c(n) \leq \left\lceil \left(\frac{q}{q-1} \right)^n (n \log q + 1) \right\rceil.$$

The count depends only on n , not on the positions of the sampled coordinates, so its exponential rate proves (5.8). $\square$

The excluded-letter cover and its covering exponent are already present in Romagnoli [10]; the calculation above is included to identify the sparse-time quantity without an external dependency. The consequence of Theorem 5.1 is the exact minimum over *all Borel* refinements for a sequence fixed in advance. It does not assert that every Borel refinement has consecutive-time entropy at least $\log 2$, or that every sampling sequence gives that lower bound.

5.4. Application to safe feedbacks. For a fixed period, the common-successor hypothesis identifies every feedback orbit with the same dynamical system. The preceding cover formula therefore applies to the control-label fibers.

Theorem 5.5. *In the setting of Section 2.2, assume that there is a map $T : Q \rightarrow Q$ such that*

$$g_w(x) = T(x) \quad \text{for every } w \in W \text{ and every } x \in K_w, \quad (5.10)$$

and that T is conjugate to σ^p on a mixing one-sided shift of finite type. With $r = N((K_w)_{w \in W})$, for every A satisfying (5.3) one has

$$\min_{s \in \mathcal{S}_\tau} h_A(s) = \min_{s \in \mathcal{S}_\tau} h^*(s) = L_\tau = M_\tau = \frac{\log r}{\tau}. \quad (5.11)$$

The maximum defining L_τ is attained by every such A , and an ordered minimum-subcover selector attains all the indicated minima. The same minimum and minimax values hold when all finite Borel safe partitions, including every finite Borel refinement, are allowed. Every refinement satisfies the stated lower bound; no upper bound for each individual refinement is asserted.

Proof. For every legal selector, (5.10) gives $T_s = T$ pointwise. Its fibers form a finite Borel refinement of $(K_w)_{w \in W}$. Theorem 5.1, with normalization by τ , gives the lower bound at every A with diverging gaps. If $w_1, \dots, w_r$ form a minimum subcover, define $s_0(x) = w_j$ on the sets E_j of (5.7). This is Borel and legal at every point. Each selected block stays in Q at its intermediate times by (2.12), and the endpoint again lies in Q , so recursion gives an infinite safe closed-loop orbit for every initial state. The selector uses only r blocks and thus gives the upper bound for every window and every sampling sequence. This proves attainment, (5.11), and the outer maximum. Lemma 2.3 proves the last assertion. $\square$

The standard safe-partition framework is described by Kawan [7, Chapter 2]. His minimization over control periods in invariance entropy is a different optimization. Here τ is fixed, intermediate safety is retained, and the exact value comes from the additional common-successor and finite-type hypotheses.

5.4.1. *An overlapping example at period two.* Take

$$Q = \{0, 1, 2\}^{\mathbb{N}_0}, \quad X = Q \sqcup \{\dagger\}, \quad U = \{0, 1, 2\}, \quad \tau = 2,$$

where the isolated point $\dagger$ lies outside Q . Define

$$f_u(x) = \begin{cases} \sigma x, & x \in Q, \ x_0 \neq u, \\ \dagger, & x \in Q, \ x_0 = u, \end{cases} \quad f_u(\dagger) = \dagger.$$

Each f_u is continuous since the cases are clopen. For $w = (a, b)$,

$$K_{(a,b)} = \{x : x_0 \neq a, \ x_1 \neq b\}, \quad g_{(a,b)}|_{K_{(a,b)}} = \sigma^2.$$

All nine control blocks are available to Borel feedbacks, and each state permits four of them. A safety domain covers four of the nine two-symbol input blocks, so two domains cannot cover Q . The three diagonal blocks $(0, 0)$, $(1, 1)$, $(2, 2)$ do cover Q , since some symbol differs from both x_0 and x_1 . Hence $r = 3$, and

$$L_2 = M_2 = \frac{\log 3}{2}.$$

An attaining selector is

$$c(x) = \min(\{0, 1, 2\} \setminus \{x_0, x_1\}), \quad s_0(x) = (c(x), c(x)).$$

It is clopen and legal everywhere. Its actual block-boundary states are $\sigma^{2k}x$. To prescribe any sequence of diagonal labels, use successive input blocks 11, 00, and 01 for the desired labels 0, 1, and 2, respectively. Therefore its actual name set is the full shift on these three blocks. The lower bound, however, holds for every legal Borel rule using any of the nine blocks.

This example does not preassign a nonconstant common control-letter observation to every feedback. More precisely, suppose a map $\rho : U^2 \rightarrow V'$ and an observation $F : Q \rightarrow V'$ satisfy $\rho(s(x)) = F(x)$ for every legal Borel selector and every x . Any two blocks (a, b) and (a', b') are colegal at a state beginning with ij , where $i \notin \{a, a'\}$ and $j \notin \{b, b'\}$. A legal baseline selector may be changed at this single state to use either block, while remaining Borel and legal. Hence their ρ values agree. All pairs of blocks are colegal somewhere, so ρ is constant. This rules out this particular forced-letter explanation; it makes no claim about every possible dynamical factor.

6. DIFFERENT SUCCESSORS: A BOUNDARY EXAMPLE AND AN APPLICATION

The common-successor assumption cannot be removed solely because each branch is continuous and open. We give a system with individual prefix cylinders as safety domains and compute its full-class value.

Let $Q = \{0, 1\}^{\mathbb{N}_0}$, let $\mathcal{C} = \{0, 10, 11\}$ be a set of finite words, and take $U = \mathcal{C} \times \{0, 1\}$ and $\tau = 1$. For $e \in \{0, 1\}$ let P_e add e modulo two to every coordinate. On $X = Q \sqcup \{\dagger\}$ define

$$f_{(c,e)}(x) = \begin{cases} P_e \sigma x, & x \in [c], \\ \dagger, & x \in Q \setminus [c], \end{cases} \quad f_{(c,e)}(\dagger) = \dagger. \quad (6.1)$$

The cylinders $[0], [10], [11]$ are disjoint and cover Q . Each state has exactly two legal controls with different successors. The global maps $f_{(c,e)}$ are continuous, and their legal restrictions are open. The safety domains are $K_{(c,e)} = [c]$, and $r = 3$.

Proposition 6.1. *For the system (6.1), over all legal Borel feedbacks and arbitrary Borel mixtures,*

$$\min_s h_{A_0}(s) = \min_s h^*(s) = \max_A \inf_s h_A(s) = \log 2 < \log r, \quad A_0 = (0, 1, 2, \dots). \quad (6.2)$$

There is one clopen legal feedback s_f satisfying $h_A(s_f) = \log 2$ for every increasing infinite A . The same full-class values hold if arbitrary finite Borel atom refinements are allowed.

Proof. Let $c(x)$ be the unique member of $\mathcal{C}$ with $x \in [c(x)]$, and set

$$s_f(x) = (c(x), x_1), \quad T_f x = P_{x_1} \sigma x.$$

This is a clopen selector, legal at every state. Its four nonempty initial control fibers, for labels $(0, 0), (0, 1), (10, 0), (11, 1)$, are $[00], [01], [10], [11]$, respectively. For all $n \geq 1$, direct induction gives

$$T_f^n x = P_{x_n} \sigma^n x, \quad s_f(T_f^n x) = (0, x_{n+1} \oplus x_n), \quad (6.3)$$

where $\oplus$ is addition modulo two. In the induction step the chosen flip is $x_{n+1} \oplus x_n$; it cancels the previous flip and leaves $P_{x_{n+1}} \sigma^{n+1} x$. The first coordinate is always zero after time zero, which also proves all-time safety.

For every finite nonempty window F , with $n = |F|$, one has

$$P_{s_f}(F) = \begin{cases} 2^n, & 0 \notin F, \\ 2^{n+1}, & 0 \in F. \end{cases} \quad (6.4)$$

The four initial labels and two later labels give the upper bounds. To prove the lower bounds, prescribe any of the four initial labels if $0 \in F$, thereby fixing (x_0, x_1) ; if $0 \notin F$, set $x_0 = x_1 = 0$. At positive sampled times prescribe the desired second components d_k , and put $d_k = 0$ at the other positive times. Recursively set $x_{k+1} = x_k \oplus d_k$ for every $k \geq 1$. This constructs a full state in Q whose actual trajectory under the same selector has the required sampled controls by (6.3). Thus all counted patterns are realized by true infinite closed-loop trajectories. Equation (6.4) implies $h_A(s_f) = h^*(s_f) = \log 2$ for every A .

For the opposite bound fix any legal Borel selector s . If its first n actual controls from x are (c_k, e_k) , let $E_0 = 0$ and $E_k = e_0 \oplus \dots \oplus e_{k-1}$. Since shifts commute with flips,

$$T_s^k x = P_{E_k} \sigma^k x.$$

Let $b(c)$ be the first bit of the word c . Safety at time k gives

$$x_k = b(c_k) \oplus E_k \quad (0 \leq k < n). \quad (6.5)$$

Thus one actual length- n control word determines at most one input prefix of length n . Every binary input prefix has an extension in Q , and the feedback is defined and safe on every such extension. Distinct prefixes must yield distinct actual control words. Hence

$$P_s(\{0, \dots, n-1\}) \geq 2^n$$

for every n and every legal s . This proves $h_{A_0}(s) \geq \log 2$ and $h^*(s) \geq \log 2$. Combining these inequalities with s_f proves (6.2), including attainment. Lemma 2.3 handles atom refinements. $\square$

The obstruction has a precise orbit interpretation. In this example $T_f(Q) = [0]$: for the reverse inclusion, given $y \in [0]$, take $x = 0y$. Therefore neither $[10]$ nor $[11]$ is visited at a positive time. The initial fibers with those labels are nonmeager, but their labels cannot be freely concatenated in the future. The map T_f is itself continuous and open, as follows by restricting to the four clopen two-symbol cylinders. It is not onto Q .

Even preservation of meager preimages does not repair this problem. For any selector s and any n , the value of $T_s^n x$ is either $\sigma^n x$ or $P_1 \sigma^n x$. Hence, for a meager set N ,

$$(T_s^n)^{-1}N \subset (\sigma^n)^{-1}N \cup (P_1 \sigma^n)^{-1}N$$

is meager. What fails is the nonempty open intersection used in Lemma 5.2 for the *selected* dynamics. Also, the same s_f gives

$$\inf_s h_A(s) \leq \log 2 = \log r - \log(3/2) \quad \text{for every } A.$$

The failure of the cover formula is therefore a uniform numerical gap over all samplings, rather than a failure at one finite window. It does not refute $L_\tau = M_\tau$: equality holds in this very example. Nor does the proof determine the inner infimum for every other fixed sampling sequence.

6.1. A cyclic-permutation application. Let $q \geq 3$, $G = \mathbb{Z}/q\mathbb{Z}$, and $Q = G^{\mathbb{N}_0}$. Write P_u for coordinatewise addition of u . On $X = Q \sqcup \{\dagger\}$ use the continuous controls

$$f_u(x) = P_u \sigma x \quad \text{if } x_0 \neq u, \quad f_u(x) = \dagger \quad \text{otherwise,} \quad f_u(\dagger) = \dagger.$$

Their safety domains are the excluded-letter cylinders $K_u = \{x : x_0 \neq u\}$, so the minimum cover cardinality is two.

Corollary 6.2. *For this system, at period one and over all legal Borel feedbacks and finite atom refinements,*

$$M_1 = \min_s h^*(s) = \log 2, \quad \log \frac{q}{q-1} \leq L_1 \leq \log 2, \quad h_{\text{inv}}(Q) = \log \frac{q}{q-1}.$$

There is a clopen feedback with $P_s(F) = 2^{|F|}$ for every finite window F .

Proof. Use the uniform Bernoulli probability on Q . The safe inverse operator is

$$(L_u f)(y) = \frac{1}{q} \sum_{a \neq u} f(a P_{-u} y), \quad L_u \mathbf{1} = \frac{q-1}{q} \mathbf{1}.$$

Products of any single-label lists therefore have norm $((q-1)/q)^n$. Theorem 3.2, with $k = r = 2$, gives $M_1 = \log 2$.

For a real control word $w = (w_0, \dots, w_{n-1})$, put $E_k = \sum_{j < k} w_j$ and $v_k = w_k - E_k$, with arithmetic in G . The actual orbit satisfies $T_s^k x = P_{E_k} \sigma^k x$, so its initial fiber lies in

$$K(w) = \{x : x_k \neq v_k \ (0 \leq k < n)\}.$$

This is also the complete open-loop safety domain of w . Each such domain contains exactly $(q-1)^n$ input prefixes. Every input prefix extends to a state with an infinite safe feedback trajectory; hence the actual length- n word count is at least $(q/(q-1))^n$. Consecutive sampling gives the stated lower bound for L_1 .

Define $s_0(x) = 1$ if $x_0 = 0$, and $s_0(x) = 0$ otherwise. This is clopen and safe on all of Q . Given any infinite binary sequence d , let $E_k = \sum_{j < k} d_j$ and let $b_k = 0$ when $d_k = 1$, $b_k = 1$ when $d_k = 0$. The state with coordinates $x_k = b_k - E_k$ has current first coordinate b_k at time k , by induction. Its real feedback name is therefore d . This proves the finite-window equality and the upper bound for L_1 with the same feedback for every sampling sequence.

The map $w \mapsto v$ is triangular and bijective: recover $w_k = v_k + E_k$, $E_{k+1} = 2E_k + v_k$, starting with $E_0 = 0$. This requires no inverse of 2, and includes even q . Thus the open-loop spanning number is the least number $c_q(n)$ of boxes $\prod_{k < n} (G \setminus \{v_k\})$ covering G^n . Writing $p = ((q-1)/q)^n$, the counting lower bound and the elementary random-cover upper bound give

$$p^{-1} \leq c_q(n) \leq \lceil p^{-1}(n \log q + 1) \rceil.$$

Indeed, $t = \lceil p^{-1}(n \log q + 1) \rceil$ independent uniform boxes leave some input uncovered with probability at most $q^n(1-p)^t \leq q^n e^{-pt} \leq e^{-1} < 1$. A covering choice exists, and removal of repetitions can only decrease its size. Taking normalized logarithms proves the invariance-entropy formula. Lemma 2.3 supplies the refinement statement. $\square$

The safe spanning number is a lower bound for the actual feedback word count, not an upper bound. For example, when $q = 3$, the three safe words 00, 12, 21 cover all two-symbol inputs: their excluded vectors are 00, 11, 22. The clopen rule $s(x) = \min(G \setminus \{x_0, x_1\})$ repeatedly selects the first control of a covering word and is safe everywhere. Starting from $x = (1, 1, 0, 0, \dots)$, however, it selects 0 and then 2, giving the real word 02 outside that cover. The subsequent state is 2^∞ and all subsequent controls are 0, so the witness has an infinite safe continuation. In general its real prefix fibers obey

$$H_s(uw) = s^{-1}(u) \cap (P_u \sigma)^{-1} H_s(w), \quad H_s(w) \subset K(w).$$

These compatibility equations are absent from a separately chosen spanning family. The lower and upper bounds for L_1 remain distinct; neither their equality nor a strict minimax gap is asserted for this cyclic system.

7. COUNTABLE LANGUAGES AND COMMON SAMPLING

A common sampling sequence can also be obtained from countability of the possible name languages. The argument below needs neither shift invariance nor surjectivity: deleting an initial set of coordinates allows successive nearly maximizing windows to be placed farther out.

7.1. Simultaneous attainment for pattern systems. A *finite-alphabet pattern system* is a nonempty set $\Omega \subset E^{\mathbb{N}_0}$ with E finite and nonempty. No topology or shift invariance is imposed. For finite $F \subset \mathbb{N}_0$, write

$$P_\Omega(F) = \#\{x|_F : x \in \Omega\}, \quad p_\Omega^*(n) = \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} P_\Omega(F).$$

With a fixed scale $\tau > 0$, define

$$h^*(\Omega; \tau) = \lim_{n \rightarrow \infty} \frac{1}{n\tau} \log p_\Omega^*(n),$$

$$h_{\text{seq}}^A(\Omega; \tau) = \limsup_{n \rightarrow \infty} \frac{1}{n\tau} \log P_\Omega(\{a_1, \dots, a_n\}).$$

The maximum exists: its possible values form a nonempty subset of the finite integer set $\{1, \dots, (\#E)^n\}$.

We use $P_\Omega(\emptyset) = 1$ when needed. Taking the product-topology closure of Ω changes none of its finite projections: a cylinder meeting $\overline{\Omega}$ already meets Ω . Thus dispensing with closedness is only a matter of formulation. The absence of shift invariance, in contrast, matters to the placement of the coordinate sets.

Lemma 7.1 (Pattern submultiplicativity and deletion). *Let $\Omega \subset E^{\mathbb{N}_0}$ be a pattern system and $q = \#E$. For $m, n \in \mathbb{N}$ and finite $B \subset F \subset \mathbb{N}_0$,*

$$p_\Omega^*(m+n) \leq p_\Omega^*(m) p_\Omega^*(n), \tag{7.1}$$

$$P_\Omega(F) \leq q^{\#(F \setminus B)} P_\Omega(B). \tag{7.2}$$

If $B \subset G$, then $P_\Omega(B) \leq P_\Omega(G)$. Consequently,

$$h^*(\Omega; \tau) = \inf_{n \geq 1} \frac{1}{n\tau} \log p_\Omega^*(n), \quad \log p_\Omega^*(n) \geq n\tau h^*(\Omega; \tau). \quad (7.3)$$

Proof. Split any $(m+n)$ -element set into disjoint subsets of sizes m and n . Restriction sends a pattern on the union injectively to the pair of its two restrictions, proving (7.1). Restriction from F to B has fibers of cardinality at most $q^{\#(F \setminus B)}$, which proves (7.2). If $B \subset G$, every pattern on B extends to a pattern on G using the same element of Ω , giving monotonicity. Fekete's lemma applied to $\log p_\Omega^*(n)$ proves (7.3). $\square$

For comparison, let $\Omega = \{0^\infty, 10^\infty\}$. This set is a closed one-sided subshift, but $P_\Omega(\{0\}) = 2$ and $P_\Omega(\{m\}) = 1$ for $m \geq 1$. One cannot simply translate a maximizing set. Deleting a finite initial interval has a different advantage: the number of lost coordinates is bounded independently of the size of the maximizing set. Inequality (7.2) turns this bound into a controlled information loss.

Theorem 7.2 (Simultaneous attainment). *Fix $\tau > 0$. For every countable family $(\Omega_j)_{j \geq 1}$ of finite-alphabet pattern systems, there is one $A \in \mathcal{S}$ such that*

$$h_{\text{seq}}^A(\Omega_j; \tau) = h^*(\Omega_j; \tau) \quad (j \geq 1). \quad (7.4)$$

Thus the maximal-pattern entropy rate of each system is attained along the same sequence. No closedness, compactness, shift invariance, shift surjectivity, or backward extension is required.

Proof. Let E_j be the alphabet of Ω_j , put $q_j = \#E_j$ and $h_j^* = h^*(\Omega_j; \tau)$. For every A and n ,

$$P_{\Omega_j}(\{a_1, \dots, a_n\}) \leq p_{\Omega_j}^*(n),$$

so (7.3) gives $h_{\text{seq}}^A(\Omega_j; \tau) \leq h_j^*$.

Choose an enumeration of stages

$$(j_k)_{k \geq 1} = (1; 1, 2; 1, 2, 3; 1, 2, 3, 4; \dots),$$

where semicolons only separate finite groups. Every positive integer occurs infinitely often. Choose $\eta_k > 0$ with $\eta_k \rightarrow 0$. We recursively construct nonempty finite sets B_k , each lying strictly to the right of all previous ones. Put

$$G_{k-1} = \bigcup_{i < k} B_i, \quad r_{k-1} = \#G_{k-1}, \quad M_{k-1} = \max G_{k-1},$$

with $r_0 = 0$ and $M_0 = -1$.

If $h_{j_k}^* > 0$, choose n_k so large that

$$n_k > M_{k-1} + 1, \quad \frac{(M_{k-1} + 1) \log q_{j_k}}{n_k \tau} < \eta_k, \quad \frac{r_{k-1}}{n_k} < \eta_k. \quad (7.5)$$

The quantities on the numerators are fixed before n_k is chosen, so these requirements can be satisfied even if the previously selected coordinates are extremely far apart. Notice that $M_{k-1} + 1$ bounds the number of *possible* old positions, whereas r_{k-1} counts the positions actually selected. The former controls deletion; the latter controls dilution by historical observations. No uniform bound on q_j or on the coordinate gaps is required. Choose an n_k -element F_k attaining $p_{\Omega_{j_k}}^*(n_k)$ and set

$$B_k = F_k \cap \{M_{k-1} + 1, M_{k-1} + 2, \dots\}.$$

At most $M_{k-1} + 1$ coordinates are deleted, and the first condition in (7.5) makes B_k nonempty. If $h_{j_k}^* = 0$, set $B_k = \{M_{k-1} + 1\}$.

Let A be the increasing enumeration of $\bigcup_{k \geq 1} B_k$. Fix j with $h_j^* > 0$ and restrict to the infinitely many stages with $j_k = j$. Write $b_k = \#B_k$ and $R_k = r_{k-1} + b_k = \#G_k$; then G_k

consists of the first R_k entries of A . By deletion, (7.3), and monotonicity,

$$\begin{aligned} \log P_{\Omega_j}(G_k) &\geq \log P_{\Omega_j}(B_k) \\ &\geq \log p_{\Omega_j}^*(n_k) - (M_{k-1} + 1) \log q_j \\ &\geq n_k \tau h_j^* - (M_{k-1} + 1) \log q_j. \end{aligned} \quad (7.6)$$

Also $R_k \leq r_{k-1} + n_k$. At all sufficiently late such stages the last line of (7.6) is positive. Dividing by $R_k \tau$ and using (7.5) yields the explicit endpoint estimate

$$\frac{1}{R_k \tau} \log P_{\Omega_j}(G_k) > \frac{h_j^* - \eta_k}{1 + \eta_k}. \quad (7.7)$$

For a fixed j , the indices k with $j_k = j$ tend to infinity, hence $\eta_k \rightarrow 0$ along those stages. Since every B_i is nonempty, $R_k \geq k \rightarrow \infty$. The limsup along these endpoints is at least h_j^* , proving the lower bound. If $h_j^* = 0$, the universal upper bound and nonnegativity give equality for every A . This proves (7.4) for all j . $\square$

Remark 7.3 (What is attained). *The theorem asserts equality of limsup rates. It does not assert a common subsequence of endpoints on which every system is simultaneously close to its maximal rate. The good endpoints used for different j can be different. Nor does it assert convergence of the normalized counts. For example, take $\tau = 1$, let $B \subset \mathbb{N}_0$ and its complement both be infinite, and let Ω_B consist of all binary sequences supported on B . For any schedule, the two normalized logarithmic counts for Ω_B and $\Omega_{\mathbb{N}_0 \setminus B}$ add to $\log 2$ at every n , although both maximal-pattern rates are $\log 2$. They cannot both converge to $\log 2$. The two limsups, however, can both equal $\log 2$.*

Exact finite-level maximization is also a stronger and generally false requirement. Put

$$\Omega = \{0x : x \in \{0, 1\}^{\mathbb{N}_0}\} \cup \{10^\infty, 20^\infty\} \subset \{0, 1, 2\}^{\mathbb{N}_0}.$$

The unique maximizing singleton is $\{0\}$, with three patterns. For a three-element set containing zero there are six patterns, whereas any three positive coordinates realize eight patterns. No increasing sequence maximizes the counts at both levels $n = 1$ and $n = 3$.

Corollary 7.4 (Fixed-language attainment). *For every finite-alphabet pattern system Ω and every $\tau > 0$,*

$$h^*(\Omega; \tau) = \max_{A \in \mathcal{A}} h_{\text{seq}}^A(\Omega; \tau). \quad (7.8)$$

The same identity holds for every finite Borel invariant partition $\mathcal{C}$ after putting $\Omega = \Sigma_{\mathcal{C}}$.

7.2. A language-quotient minimax theorem. Fix a compact controlled-invariant set Q and one period $\tau \in \mathbb{N}$. For two invariant partitions $\mathcal{C}$ and $\mathcal{D}$ of this period, write $\mathcal{C} \sim_{\text{lang}} \mathcal{D}$ if there is a bijection $\pi : I_{\mathcal{C}} \rightarrow I_{\mathcal{D}}$ such that, for every $N \in \mathbb{N}$,

$$\pi^N(W_N(\mathcal{C}; Q)) = W_N(\mathcal{D}; Q), \quad (7.9)$$

where π^N acts coordinatewise. This is *language equivalence*. Identity, inverse, and composition of the alphabet bijections show that $\sim_{\text{lang}}$ is an equivalence relation. If $F \subset \mathbb{N}_0$ is nonempty and finite and $m = \max F$, then (7.9) at length $m + 1$, followed by projection onto F , gives a bijection between the two F -pattern sets. Thus language-equivalent partitions have identical finite-coordinate pattern counts and identical sequence and maximal-pattern complexities.

For a nonempty class $\mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$, set

$$H_{\text{pat}}^*(Q; \mathfrak{C}) = \inf_{\mathcal{C} \in \mathfrak{C}} h^*(\mathcal{C}; Q), \quad (7.10)$$

$$H_{\text{seq}}^A(Q; \mathfrak{C}) = \inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^A(\mathcal{C}; Q). \quad (7.11)$$

Every individual rate is nonnegative and finite, and because the class is nonempty both infima are finite; no uniform bound on the numbers of atoms is needed.

The elementary minimax inequality gives only

$$\sup_A \inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^A(\mathcal{C}; Q) \leq \inf_{\mathcal{C} \in \mathfrak{C}} \sup_A h_{\text{seq}}^A(\mathcal{C}; Q) = H_{\text{pat}}^*(Q; \mathfrak{C}).$$

Separate maximizing schedules for separate partitions do not give the opposite inequality. The next theorem obtains it by constructing one sequence that attains every individual rate.

Theorem 7.5 (Minimax over countably many language types). *Let $\emptyset \neq \mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$ and suppose that the quotient $\mathfrak{C}/\sim_{\text{lang}}$ is finite or countable. There is one $A \in \mathcal{S}$ such that*

$$h_{\text{seq}}^A(\mathcal{C}; Q) = h^*(\mathcal{C}; Q) \quad (\mathcal{C} \in \mathfrak{C}). \quad (7.12)$$

Consequently,

$$\max_{B \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^B(\mathcal{C}; Q) = \inf_{\mathcal{C} \in \mathfrak{C}} h^*(\mathcal{C}; Q). \quad (7.13)$$

The outer maximum is attained. The pattern infimum and the inner infimum at the common attaining schedule are minima, by Corollary F.2.

Proof. Choose one representative $\mathcal{C}_j$ of each language-equivalence class. Enumerate the representatives, repeating them periodically when the quotient is finite, and apply Theorem 7.2 to the resulting countable family $\Sigma_{\mathcal{C}_j}$, using the common scale τ . The resulting A gives equality for each representative. Language equivalence preserves all finite pattern counts, so it gives (7.12) for every $\mathcal{C} \in \mathfrak{C}$. Taking the infimum over $\mathcal{C}$ gives equality at this A .

For any $B \in \mathcal{S}$, Corollary 7.4 gives $h_{\text{seq}}^B(\mathcal{C}; Q) \leq h^*(\mathcal{C}; Q)$ for every $\mathcal{C}$. Taking first the infimum over $\mathcal{C}$ and then the supremum over B proves the reverse minimax inequality and hence (7.13). If the pattern infimum is zero, the pointwise upper bound and nonnegativity in fact give $\inf_{\mathcal{C} \in \mathfrak{C}} h_{\text{seq}}^B(\mathcal{C}; Q) = 0$ for every $B \in \mathcal{S}$, without any countability assumption. $\square$

The hypothesis concerns the symbolic behavior generated on Q , not the names of controls used to present a strategy. It identifies alphabet renumberings and strategies with the same admissible language after such a renumbering. It does not merge distinct atoms merely because their assigned controls agree. It is therefore closer to an intrinsic statement about the chosen partition class, although it remains relative to that class and to the fixed period.

Proposition 7.6 (Countability of clopen feedbacks). *Assume that Q is compact metrizable and that the control alphabet U is countable. Then $\mathfrak{C}_{\text{clon}}(Q, \tau)$ is countable. Hence, if this class is nonempty, Theorem 7.5 applies to it. In particular this holds when U is a finite digital alphabet.*

Proof. A compact metrizable space has a countable base. Every clopen subset is compact and hence is a finite union of basic open sets contained in it. Thus Q has only countably many clopen subsets and only countably many finite clopen partitions. Since τ is finite, U^τ is countable. A finite partition admits only countably many block assignments. The invariant assignments form a subcollection. $\square$

Theorem 7.5 fixes τ . It does not produce one common physical-time schedule for partitions of different periods and proves no minimax identity over $\mathfrak{C}_{\text{Bor}}(Q)$.

The countability hypothesis has concrete limitations. Appendix G develops the elementary obstruction of Example G.1 into autonomous constructions. In particular, Proposition G.4 uses four distinct controls and has $L_1 = M_1 = \log 2$, although no sequence attains every feedback's own maximal-pattern rate. It therefore separates pointwise common attainment from a common full-class minimum.

8. SYMBOLIC SAFETY REALIZATIONS

Symbolic systems allow the optimized feedback counts to be computed without choosing a reference probability. The realization below forces the physical control labels to reproduce the state symbols; the Thue–Morse example then separates sparse complexity from invariance entropy.

8.1. Realizing a one-sided subshift. Let E be a finite nonempty discrete alphabet and let $\emptyset \neq Y \subset E^{\mathbb{N}_0}$ be closed with $\sigma(Y) \subset Y$. Write

$$\mathcal{L}_N(Y) = \{y_0 \cdots y_{N-1} : y \in Y\}, \quad P_Y(F) = \#\{y|_F : y \in Y\}.$$

The coordinate complexities are

$$h_{\text{seq,coord}}^A(Y) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log P_Y(\{a_1, \dots, a_n\}),$$

$$h_{\text{pat,coord}}^*(Y) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} P_Y(F).$$

Adjoin an isolated cemetery point and give $X = Y \sqcup \{\dagger\}$ the topological disjoint-union topology. Put $U = E$ and define

$$f(y, u) = \begin{cases} \sigma y, & y_0 = u, \\ \dagger, & y_0 \neq u, \end{cases} \quad f(\dagger, u) = \dagger. \quad (8.1)$$

Theorem 8.1 (Symbolic-to-control realization). *For the matching-control system (8.1):*

- (i) X is compact metrizable, $f : X \times U \rightarrow X$ is continuous, and Y is a compact controlled-invariant set.
- (ii) For every $N \in \mathbb{N}$,

$$r_{\text{inv}}(N, Y) = \#\mathcal{L}_N(Y), \quad (8.2)$$

and therefore

$$h_{\text{inv}}(Y) = h_{\text{top}}(Y, \sigma) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \#\mathcal{L}_N(Y). \quad (8.3)$$

- (iii) At period one identify U^1 with U . If $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, 1, \nu) \in \mathfrak{C}_{\text{Bor}}(Y, 1)$, set $\lambda_{\mathcal{C}}(i) = \nu(D_i) \in E$. Then every atom satisfies

$$D_i \subset [\lambda_{\mathcal{C}}(i)]_Y := \{y \in Y : y_0 = \lambda_{\mathcal{C}}(i)\}. \quad (8.4)$$

Moreover, the one-block label map

$$\Lambda_{\mathcal{C}} : \Sigma_{\mathcal{C}} \longrightarrow Y, \quad (a_j)_{j \geq 0} \longmapsto (\lambda_{\mathcal{C}}(a_j))_{j \geq 0}, \quad (8.5)$$

is a continuous, onto, shift-commuting map.

- (iv) For every $A \in \mathcal{S}$,

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = h_{\text{seq,coord}}^A(Y), \quad (8.6)$$

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h^*(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h^*(\mathcal{C}; Y) = h_{\text{pat,coord}}^*(Y), \quad (8.7)$$

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h(\mathcal{C}; Y) = h_{\text{top}}(Y, \sigma). \quad (8.8)$$

Every Borel and clopen infimum in (8.6)–(8.8) is attained by the same first-coordinate cylinder partition.

Proof. The space X is the union of the two clopen compact sets Y and $\{\dagger\}$, hence is compact metrizable. For fixed $u \in E$, the sets $[u]_Y$, $Y \setminus [u]_Y$, and $\{\dagger\}$ are clopen; on them f_u is, respectively, the shift or a constant. Thus f_u is continuous. Since U is finite discrete, f is jointly continuous. Choosing $u = y_0$ sends y to $\sigma y \in Y$, proving controlled invariance.

A direct induction gives

$$\varphi(n, y, \omega) \in Y \quad (0 \leq n \leq N) \iff \omega_j = y_j \quad (0 \leq j < N). \quad (8.9)$$

Indeed, after j matches the state is $\sigma^j y$, whereas a mismatch enters $\dagger$ permanently. Hence a length- N control word is safe from at least one state of Y exactly when it belongs to $\mathcal{L}_N(Y)$, and then it safely covers precisely its nonempty prefix cylinder. Distinct language words give disjoint cylinders. More explicitly:

Lower bound. For each $w \in \mathcal{L}_N(Y)$ choose $y^w \in Y$ with prefix w . If $S \subset U^N$ spans Y , some member of S is safe from y^w . By (8.9) that member must equal w . Thus $\mathcal{L}_N(Y) \subset S$, and $r_{\text{inv}}(N, Y) \geq \#\mathcal{L}_N(Y)$.

Upper bound. Take $S = \mathcal{L}_N(Y)$. For any $y \in Y$, its prefix $y_0 \cdots y_{N-1}$ belongs to S and is safe through time N by (8.9). Hence $r_{\text{inv}}(N, Y) \leq \#\mathcal{L}_N(Y)$. This proves (8.2). The language counts are submultiplicative, so their normalized logarithms converge.

For completeness, this limit is the usual topological entropy even when $\sigma|_Y$ is not onto. The join of the first-coordinate cylinder cover through times $0, \dots, N-1$ has exactly $\#\mathcal{L}_N(Y)$ nonempty atoms. Conversely, given a finite open cover of compact Y , the partition into cylinders of some fixed length m refines that cover; its N -fold dynamical join is then refined by the length- $(N+m-1)$ cylinders. Taking exponential rates proves (8.3).

Let $\mathcal{C} = (\{D_i : i \in I_{\mathcal{C}}\}, 1, \nu)$ be a Borel invariant partition. If D_i contained y with $y_0 \neq \lambda_{\mathcal{C}}(i)$, its assigned control would send y to $\dagger \notin Y$, contradicting safety. This proves (8.4).

Take $a = (a_j)_{j \geq 0} \in \Sigma_{\mathcal{C}}$ and put $z_j = \lambda_{\mathcal{C}}(a_j)$. For each N , choose a witness $y^{(N)}$ for the first N symbols of a . Induction using matching controls and (8.4) gives $y_j^{(N)} = \lambda_{\mathcal{C}}(a_j) = z_j$ for $0 \leq j < N$. Hence $y^{(N)} \rightarrow z$ in the product topology; since Y is closed, $z \in Y$. Thus $\Lambda_{\mathcal{C}}$ is well defined, and it is continuous: the preimage of a cylinder in $E^{\mathbb{N}_0}$ is a finite union of cylinders in $I_{\mathcal{C}}^{\mathbb{N}_0}$, restricted to $\Sigma_{\mathcal{C}}$. It is shift commuting because $(\Lambda_{\mathcal{C}}(\sigma a))_j = \lambda_{\mathcal{C}}(a_{j+1}) = (\sigma \Lambda_{\mathcal{C}}(a))_j$. Given $y \in Y$, let a_j be the unique index with $\sigma^j y \in D_{a_j}$. Its assigned label is y_j , and y witnesses every finite prefix of this atom itinerary. Thus (8.5) is onto. Here each prefix witness $y^{(N)}$ may be different. Its first N control labels nevertheless equal those of z , so convergence to $z \in Y$ follows. This does not prove that z follows the original atom name a ; its orbit may meet different atoms with the same labels. Thus neither shift surjectivity nor a common atom-name witness is used.

For every finite F , the coordinate projection of $\Lambda_{\mathcal{C}}$ is onto, so

$$P_{\mathcal{C}}(F) \geq P_Y(F). \quad (8.10)$$

It follows that every Borel, and hence every clopen, invariant partition has sequence, maximal-pattern, and consecutive-name entropy at least the corresponding coordinate quantity.

Let $E_Y = \{e \in E : [e]_Y \neq \emptyset\}$ and define

$$\mathcal{C}_0 = (\{D_e = [e]_Y : e \in E_Y\}, 1, \nu_0), \quad \nu_0(D_e) = e.$$

This is a clopen invariant partition. Under the symbol identification, $W_N(\mathcal{C}_0; Y) = \mathcal{L}_N(Y)$ and $\Sigma_{\mathcal{C}_0} = Y$; the latter follows from closedness of Y . Equality holds in (8.10) for every F . Thus $\mathcal{C}_0$ attains all the lower bounds, proving (8.6)–(8.8). $\square$

8.2. A Thue–Morse separation. Let $\vartheta(0) = 01$, $\vartheta(1) = 10$, and let

$$\mathbf{t} = \lim_{r \rightarrow \infty} \vartheta^r(0) = 0110100110010110 \cdots$$

Equivalently, $t_n = s_2(n) \pmod{2}$, where $s_2(n)$ is the number of ones in the binary expansion of n . Let

$$Y = \overline{\{\sigma^m \mathbf{t} : m \geq 0\}} \subset \{0, 1\}^{\mathbb{N}_0}. \quad (8.11)$$

This is the one-sided Thue–Morse orbit closure. The following direct computation supplies both rates used below.

Proposition 8.2 (Exact symbolic rates). *For every $N \geq 1$,*

$$\#\mathcal{L}_N(Y) \leq 8N. \quad (8.12)$$

Hence $h_{\text{top}}(Y) = 0$. On the single schedule

$$A_{\text{TM}} = (a_i)_{i \geq 1}, \quad a_i = 4^{i-1} - 1, \quad (8.13)$$

one has, for every $k \geq 1$,

$$P_Y(\{a_1, \dots, a_k\}) = 2^k. \quad (8.14)$$

The chosen coordinate set contains zero. Since a cylinder is clopen, its patterns in the orbit closure are exactly the patterns obtained along shifts of $\mathbf{t}$; this calculation uses actual shifts of the fixed point. Consequently

$$h_{\text{seq,coord}}^{A_{\text{TM}}}(Y) = h_{\text{pat,coord}}^*(Y) = \log 2. \quad (8.15)$$

Proof. A finite word is in $\mathcal{L}_N(Y)$ exactly when it occurs as a factor of $\mathbf{t}$: one direction follows from the orbit, and the other from convergence in a prefix cylinder. Choose $\ell = \lceil \log_2 N \rceil$. The identity $\mathbf{t} = \vartheta^\ell(\mathbf{t})$ decomposes $\mathbf{t}$ into aligned supertiles of length 2^ℓ . Since $N \leq 2^\ell$, every length- N factor is contained in $\vartheta^\ell(ab)$ for an adjacent two-letter factor ab . There are at most four choices for ab and at most 2^ℓ starting offsets in the first supertile. Therefore $\#\mathcal{L}_N(Y) \leq 4 \cdot 2^\ell \leq 8N$, proving zero topological entropy.

Fix $k \geq 1$ and $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{k-1}) \in \{0, 1\}^k$. Put

$$r = \#\{0 \leq i < k : \varepsilon_i = 0\}, \quad q = \sum_{\substack{0 \leq i < k \\ \varepsilon_i = 0}} 4^i + (r \bmod 2)4^k.$$

Only even binary positions occur in this sum. The binary digit sum of q is $r + (r \bmod 2)$, which is even, so $t_q = 0$. If $\varepsilon_i = 1$, bit $2i$ of q is zero, and adding $4^i = 2^{2i}$ changes the digit-sum parity. If $\varepsilon_i = 0$, bit $2i$ is one and bit $2i + 1$ is zero; adding 2^{2i} replaces these two bits by zero and one without a further carry, so the parity is unchanged. In both cases

$$t_{q+4^i} = \varepsilon_i \quad (0 \leq i < k). \quad (8.16)$$

The extra bit at position $2k$ only corrects the initial parity of q . For $i < k$, a carry stops at the empty odd position $2i + 1$, which is strictly below $2k$. The parity correction therefore interferes with none of these tests.

Set $n = q + 1$. Since $a_{i+1} = 4^i - 1$,

$$(\sigma^n \mathbf{t})_{a_{i+1}} = t_{q+4^i} = \varepsilon_i.$$

Thus every binary k -pattern occurs on the first k members of A_{TM} , proving (8.14). The binary alphabet gives the reverse upper bound $p_Y^*(k) \leq 2^k$, and (8.15) follows. $\square$

Corollary 8.3 (Zero invariance entropy, full binary sparse-time complexity). *Apply the matching-control construction (8.1) to the Thue–Morse subshift (8.11). Then*

$$h_{\text{inv}}(Y) = 0 < \log 2 = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h^*(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h^*(\mathcal{C}; Y). \quad (8.17)$$

Moreover, the explicit schedule A_{TM} satisfies

$$\max_{A \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Y, 1)} h_{\text{seq}}^{A_{\text{TM}}}(\mathcal{C}; Y) = \log 2, \quad (8.18)$$

$$\max_{A \in \mathcal{S}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^A(\mathcal{C}; Y) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{clon}}(Y, 1)} h_{\text{seq}}^{A_{\text{TM}}}(\mathcal{C}; Y) = \log 2. \quad (8.19)$$

Proof. Combine Theorem 8.1 with Proposition 8.2. For any schedule the binary coordinate complexity is at most $\log 2$, while A_{TM} attains $\log 2$. Equations (8.6)–(8.7) give the Borel and clopen infima. $\square$

The separation persists after minimization over all Borel invariant partitions at period one. It therefore distinguishes the fixed-period pattern infimum from invariance entropy.

9. COMPARISON WITH CLASSICAL AND INVARIANCE ENTROPY

The fixed-period normalization is responsible for the separation just obtained. We recall the relevant entropy notions and show how allowing all control periods recovers invariance entropy.

9.1. Classical entropy and sequence entropy. For a finite open cover $\mathcal{V}$ of a compact space, let $N(\mathcal{V})$ be the minimum cardinality of a subcover. The join of finitely many covers consists of all intersections of one member from each cover, with empty intersections removed. For a continuous map $T : Z \rightarrow Z$ and a finite open cover $\mathcal{V}$, ordinary cover entropy is

$$h_{\text{top}}(T, \mathcal{V}) = \lim_{n \rightarrow \infty} \frac{1}{n} \log N \left(\bigvee_{i=0}^{n-1} T^{-i} \mathcal{V} \right).$$

Submultiplicativity of cover numbers gives this limit. Taking the supremum over finite open covers defines $h_{\text{top}}(T)$. For $A = (a_i)_{i \geq 1}$ with $0 \leq a_1 < a_2 < \dots$, set

$$\begin{aligned} h_{\text{top}}^A(T, \mathcal{V}) &= \limsup_{n \rightarrow \infty} \frac{1}{n} \log N \left(\bigvee_{i=1}^n T^{-a_i} \mathcal{V} \right), \\ p_{Z, \mathcal{V}}^*(n) &= \max_{\substack{F \subset \mathbb{N}_0 \\ \#F=n}} N \left(\bigvee_{i \in F} T^{-i} \mathcal{V} \right), \\ h_{\text{top}}^*(T, \mathcal{V}) &= \lim_{n \rightarrow \infty} \frac{1}{n} \log p_{Z, \mathcal{V}}^*(n). \end{aligned}$$

The sequence in the last line is subadditive before division by n ; the arbitrary sequence in the first line need not have that property. This is why a limit and a limsup appear in different definitions.

Huang and Ye's Theorem 2.1(1) states, under their convention that T is a homeomorphism of a compact metric space, that

$$h_{\text{top}}^*(T, \mathcal{V}) = \max_A h_{\text{top}}^A(T, \mathcal{V}). \quad (9.1)$$

Its proof selects long nearly maximizing time sets and separates them by translations. Surjectivity preserves the cover number under a common translation. The previous sets contribute a vanishing fraction of the selected observations. For a non-surjective map the translation step need not preserve cover numbers; our proof uses deletion instead.

If $\mathcal{V}$ is a clopen partition, the join count is exactly the number of names seen on the selected coordinates. An arbitrary open cover may have overlaps, so its minimum subcover count must not be replaced by the number of nonempty intersections. Similarly, the measurable version of (9.1) uses Shannon entropy $H_\mu(\alpha) = -\sum_{D \in \alpha} \mu(D) \log \mu(D)$, not the logarithm of a support cardinality. Our language counts use the latter quantity. Theorem 7.2 is a counting statement, not a measurable common-sequence theorem.

For a finite-alphabet subshift Y , the first-coordinate cylinders give one particular clopen partition. Its coordinate sequence rate is a cover-level quantity. Unlike ordinary topological entropy, sequence entropy need not be computed by a generating partition: replacing one-coordinate observations by fixed-length windows can change the normalized sparse rate. We therefore keep the subscript ‘‘coord’’ in the realization theorem and do not silently identify this rate with the supremum over all open covers.

Kawan's Chapter 2 supplies the other interface. Definition 2.2 defines invariance entropy from finite-horizon spanning controls. Definitions 2.8 and 2.9 introduce invariant covers and their admissible word entropies. Proposition 2.21 proves word-count submultiplicativity; Proposition 2.22 bounds invariance entropy by invariant-cover entropy. Lemma 2.3 disjointizes a cover without increasing its name entropy. It does *not* turn arbitrary nonmeasurable sets into Borel sets. Borel measurability in Theorem 2.3 comes from the safe sets associated with finitely many spanning controls and their successive differences. The theorem takes an infimum over sampling periods as well as partitions. We provide the discrete-time safe-set argument in Section 9, where its application to pattern rates is also made explicit.

9.2. Comparison with Kawan's invariance entropy. For every invariant partition, consecutive coordinates are one permissible choice in the maximal-pattern count. Therefore

$$h(\mathcal{C}; Q) \leq h^*(\mathcal{C}; Q) \leq \frac{\log \#I_{\mathcal{C}}}{\tau}. \quad (9.2)$$

Kawan's Theorem 2.3, with natural logarithms, gives

$$h_{\text{inv}}(Q) = \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q)} h(\mathcal{C}; Q). \quad (9.3)$$

Its hypotheses are satisfied by the full-control discrete-time systems used here: X is metrizable, each controlled map is continuous, and Q is nonempty, compact and controlled invariant. The period in this formula is allowed to vary.

For clarity, the relevant argument can be given directly. Spanning families concatenate, so $r_{\text{inv}}(m+n, Q) \leq r_{\text{inv}}(m, Q)r_{\text{inv}}(n, Q)$ when the right-hand side is finite. Moreover $r_{\text{inv}}(n, Q)$ is nondecreasing. If any finite-horizon spanning number is finite, then $r_{\text{inv}}(1, Q)$ is finite, all of them are finite by concatenation, and Fekete's lemma gives a limit in (2.1). If none is finite, that limit equals $+\infty$.

Let $S_N = \{\omega^1, \dots, \omega^m\}$ be a minimal finite N -step spanning family. Its safe sets are

$$K_i = Q \cap \bigcap_{s=1}^N \{x \in X : \varphi(s, x, \omega^i) \in Q\}.$$

They are closed in Q , since finite control compositions are continuous and Q is closed. They cover Q . Set $D_1 = K_1$ and $D_i = K_i \setminus \bigcup_{j < i} K_j$, discard empty sets, and assign ω^i to D_i . This gives a Borel invariant partition $\mathcal{C}_N$ of period N with at most m atoms. This is the measurable construction underlying Kawan's Theorem 2.3; Lemma 2.3 alone only provides disjointization. The safe sets are already closed before that lemma is applied.

For the other inequality in (9.3), all admissible k -block words of any $\mathcal{C}$ form a spanning family after their assigned controls are concatenated. Indeed, from each initial state the feedback rule selects one such admissible word. Consequently $r_{\text{inv}}(k\tau, Q) \leq \#W_k(\mathcal{C}; Q)$, and taking limits gives $h_{\text{inv}}(Q) \leq h(\mathcal{C}; Q)$. The reverse inequality follows from $h(\mathcal{C}_N; Q) \leq N^{-1} \log r_{\text{inv}}(N, Q)$ and $N \rightarrow \infty$. If no finite spanning family exists, no finite invariant partition exists, and both sides are $+\infty$.

Thus, for every nonempty fixed-period subclass $\mathfrak{C} \subset \mathfrak{C}_{\text{Bor}}(Q, \tau)$, the general comparison is

$$h_{\text{inv}}(Q) \leq \inf_{\mathcal{C} \in \mathfrak{C}} h(\mathcal{C}; Q) \leq \inf_{\mathcal{C} \in \mathfrak{C}} h^*(\mathcal{C}; Q). \quad (9.4)$$

The minimax theorem concerns the last quantity. The Thue–Morse realization proves that it can strictly exceed $h_{\text{inv}}(Q)$.

9.3. Why the unrestricted all-period infimum collapses. There is a natural candidate for an all-period Borel quantity,

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) := \inf_{\tau \in \mathbb{N}} \inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q, \tau)} h^*(\mathcal{C}; Q).$$

The following proposition records the consequence of Kawan's spanning construction for this all-period quantity.

Proposition 9.1 (All-period collapse). *In the full-control discrete-time setting of Section 2, every nonempty compact controlled-invariant set Q satisfies*

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) = h_{\text{inv}}(Q). \quad (9.5)$$

Proof. The lower bound follows from (9.4). For the upper bound use the same $\mathcal{C}_N$ just constructed. Since its alphabet has at most $r_{\text{inv}}(N, Q)$ symbols, every pattern on k coordinates has at most $r_{\text{inv}}(N, Q)^k$ possibilities. Hence

$$H_{\text{pat,all}}^{*,\text{Bor}}(Q) \leq h^*(\mathcal{C}_N; Q) \leq \frac{\log r_{\text{inv}}(N, Q)}{N}.$$

Let $N \rightarrow \infty$. The infinite-entropy case was already covered by nonexistence of finite invariant partitions. This proves (9.5). $\square$

For the matching-control Thue–Morse example, the collapse is particularly visible. At period N , the length- N cylinder partition with its matching control blocks is clopen, has at most $8N$ atoms and has pattern rate at most $N^{-1} \log(8N)$. Thus even the all-period clopen infimum is zero in this example, while the period-one Borel infimum is $\log 2$. Changing the sampling period changes which information is packed into one observed symbol and the cost assigned to that symbol.

9.4. Countability, trajectories, and physical time. The countability hypothesis in Theorem 7.5 permits one representative of each name language to be assigned infinitely many construction stages. Compact metrizability gives countably many clopen sets, but generally uncountably many Borel languages. Example G.1 shows why pointwise attainment alone does not justify minimax exchange; Appendix G gives autonomous versions of the pointwise obstruction. Theorem 8.1 handles its realization class by a different argument, exhibiting a least language under label projection. Thus the countable-language and common-successor theorems resolve specified classes, while the product-capacity criterion supplies the maximal-pattern bound for more general feedbacks.

The patterns count admissible language projections. Every finite pattern has a finite trajectory witness which stays safe between observations. An infinite point in the completed Borel name system may have only separate finite-prefix witnesses, as Remark 2.2 shows. This distinction cannot be removed by compactness alone when the atoms are not closed. The factor in Theorem 8.1 is a factor of compact *name systems*; it is not a continuous factor defined by an arbitrary Borel feedback map on the original state space.

For one fixed τ , A denotes block indices and τA denotes physical observation times. The normalization $n\tau$ charges one block length per observation and disregards the lengths of the gaps. A common block-index sequence for different periods would represent different physical schedules. Theorem 7.2 could be applied to countably many languages with different positive normalizing constants by rescaling each equality, but that would not solve the distinct physical-time minimax problem. Formula (9.5) is an infimum identity and makes no common-schedule assertion.

Finally, if one safe control block works for every point of Q , the one-atom invariant partition makes all three fixed-period infima zero for any class containing it. On a connected Q , every finite clopen partition has one atom. The full clopen class is therefore either empty or has zero pattern infimum, while Borel strategies can remain nontrivial. These observations explain both the usefulness and the limits of the clopen sufficient condition. The nontrivial separation studied here records the richness of safe symbolic instructions at a specified sampling period, despite a zero asymptotic count per elapsed time.

10. AFFINE EXAMPLES AND FEEDBACK CONSISTENCY

The integer family has a feedback with a finite absorbing core, whereas protected inverse branches force independent choices at one common sampling sequence. A noninteger four-branch system illustrates the gap between these mechanisms: the capacity theorem determines its maximal-pattern minimum, but its common-sampling value is bounded rather than computed. Each upper bound below is realized by a specified feedback; each lower bound counts actual initial-state fibers.

10.1. Integer affine branches and an absorbing feedback. Let $m \geq 2$, $D \geq m$ be integers, $R = D/(m-1)$, $Q = [0, R]$, and controls $u = 0, \dots, D$ act legally by $mx - u$. On $X = [-1, R+1]$ define the continuous ambient maps

$$f_u(x) = \max\{-1, \min\{R+1, mx - u\}\}.$$

For $x \in Q$, this is in Q exactly when $mx - u$ is in Q , and then truncation has no effect. The safety intervals $K_u = [u/m, (u + R)/m]$ cover Q . Fix any period $\tau \geq 1$.

Proposition 10.1. *For this system, over all legal Borel block feedbacks and finite Borel atom refinements,*

$$\min_s h_{(0,1,\dots)}(s) = L_\tau = M_\tau = \log m.$$

There is an actual Borel selector s_τ with $h_A(s_\tau) = h^(s_\tau) = \log m$ for every infinite increasing A , although its absorption time has no uniform bound.*

Proof. For any actual length- n block word, concatenate its controls into $w_0 \cdots w_{n\tau-1}$. Safety gives

$$x_{n\tau} = m^{n\tau}x - \sum_{j=0}^{n\tau-1} w_j m^{n\tau-1-j} \in [0, R].$$

Its initial-state fiber is contained in an interval of length $R/m^{n\tau}$, whether or not that fiber itself is an interval. These containing intervals cover Q , so there are at least $m^{n\tau}$ actual consecutive block words. This gives the full-class lower bound at the common sequence $(0, 1, \dots)$.

Define the one-step rule $v(x) = \min\{D, \lfloor mx \rfloor\}$ and $T(x) = mx - v(x)$. If $x < D/m$, the next state is in $[0, 1)$; if $x \geq D/m$, it is $mx - D \in [0, R]$. The core $[0, 1)$ is invariant and uses only $\Gamma = \{0, \dots, m-1\}$. Both 0 and R are fixed; D/m uses D and goes to zero. For $x < R$, while D is used the state is $R - m^j(R - x)$. Since $R - x > 0$, eventually it is below D/m , then one transitional control $v < D$ sends it into the core. Only R uses D forever. The absorption times are unbounded as x tends to R .

From the current state define $s_\tau(x) = (v(x), v(Tx), \dots, v(T^{\tau-1}x))$. It is Borel, each intermediate step is safe, and its actual endpoint map is T^τ . Put $\mathcal{B} = \Gamma^\tau$, $b = m^\tau$, and $d_0 = (D, \dots, D)$. All its real block names are contained in

$$\bigcup_{j \geq 0} d_0^j \mathcal{C}_\tau \mathcal{B}^{\mathbb{N}_0} \cup \{d_0^\infty\}, \quad |\mathcal{C}_\tau| \leq D \frac{m^\tau - 1}{m - 1} =: C_\tau.$$

The transitional block has some initial D 's, one control below D , and then core controls; this proves the stated count. For any window of n block positions, an unobserved transition leaves just one of $n + 1$ cuts between constant initial labels and core labels. A sampled transition has n possible locations and at most C_τ labels. Thus

$$P_{s_\tau}(F) \leq [n + 1 + nC_\tau]m^{\tau n}, \quad |F| = n.$$

There is no factor depending on the last observation time or on the unbounded absorption time.

For the matching lower count prescribe any n core blocks at the positions of F , fill all other one-step digits with zero, and let $x = \sum_{j \geq 0} e_j m^{-j-1} < 1$. There are only finitely many nonzero digits. Every tail is smaller than one, so $\lfloor mT^j x \rfloor = e_j$ and these are real controls of the same rule. Hence $P_{s_\tau}(F) \geq m^{\tau n}$ for every window. The all-window bounds and the full-class consecutive lower bound prove the proposition, with minima actually attained. Control-label projection treats arbitrary atom refinements. $\square$

When $m \geq 3$ and $D = m$, the minimum one-step safe subcover has $r = m + 1$, although $L_1 = M_1 = \log m$. Indeed, m intervals of length R/m could cover an interval of length R only by abutting without positive-length overlap. Their left endpoints are multiples of $1/m$, so this would force their consecutive control differences to equal the noninteger $R = m/(m - 1)$. All $m + 1$ intervals are therefore needed. There is a common state factor $x \mapsto e^{2\pi i x}$, with induced map $z \mapsto z^m$; it is preserved by every legal branch. The result cannot be presented as eliminating common-factor explanations.

10.2. Protected inverse branches and common sampling. On $Q = [0, 1]$, at period one, use

$$g_a(x) = \frac{5}{4}x, \quad g_b(x) = \frac{20}{3}x - \frac{16}{3}, \quad g_c(x) = 20x - 19, \quad g_d(x) = 100x.$$

Their safety domains are $[0, 4/5]$, $[4/5, 19/20]$, $[19/20, 1]$, and $[0, 1/100]$, respectively. Continuous ambient extensions are obtained by truncating these affine maps to $[-1, 2]$, with no change on legal paths. The minimum subcover is three and the largest two inverse slopes sum to $4/5 + 3/20 = 19/20 < 1$; hence $M_1 = \log 3$ by the scalar special case of the capacity theorem.

Proposition 10.2. *For $A_2 = (0, 2, 4, \dots)$, every legal Borel feedback and every finite Borel atom refinement has at least 3^n actual patterns on its first n observations. Consequently*

$$\min_s h_{A_2}(s) = L_1 = M_1 = \log 3.$$

Proof. The full inverse branches are $\psi_a(y) = 4y/5$, $\psi_b(y) = 4/5 + 3y/20$, $\psi_c(y) = 19/20 + y/20$. Let $J = [1/2, 999/1000]$ and $\theta_u = \psi_u\psi_b$. They satisfy

$$\theta_a(y) = 16/25 + 3y/25, \quad \theta_b(y) = 23/25 + 9y/400, \quad \theta_c(y) = 99/100 + 3y/400.$$

Their complete images and the intermediate image are

image	exact endpoints	only legal control
$\psi_b(J)$	$[7/8, 18997/20000]$	b
$\theta_a(J)$	$[7/10, 18997/25000]$	a
$\theta_b(J)$	$[149/160, 376991/400000]$	b
$\theta_c(J)$	$[159/160, 398997/400000]$	c .

Each $\theta_u(J)$ lies in J . Every displayed closed interval is strictly inside the appropriate uniquely legal region; in particular $\theta_a(J)$ avoids the extra control d .

Given any $u_0 \cdots u_{n-1} \in \{a, b, c\}^n$, set $x_{2n} = 1/2$ and recursively $x_{2j+1} = \psi_b(x_{2j+2})$, $x_{2j} = \theta_{u_j}(x_{2j+2})$. The table shows that at every even state only u_j is legal and at every odd state only b is legal. Thus *every* feedback must execute the actual period-one prefix $u_0, b, \dots, u_{n-1}, b$ from the same constructed initial state. The endpoint remains in Q and the globally legal rule continues safely forever. There is no added phase or period-two control. All three-symbol sampled words are therefore actual patterns. Refinement only increases their count.

The Borel selector using a below $4/5$, b from $4/5$ to $19/20$, and c at and above $19/20$, is legal at all endpoints and uses only three labels. Its upper count is 3^n on every window. This matches the lower bound, including attainment, and $L_1 \leq M_1$ completes the equalities. The normalization is $n\tau = n$, not $2n$. $\square$

The closed contracting branches $\theta_a, \theta_b, \theta_c$ have disjoint first-level images in J . Their nested images give a compact set homeomorphic to the three-symbol full shift, and every T_s^2 on that set is the same shift under this coding. To verify this, a word's image has diameter at most $(3/25)^n \text{diam } J$; nested images intersect in one point, different first differing letters give disjoint images, and the inverse identities used above give the shift relation. The union of this set with its ψ_b image is protected under every legal feedback. These sets can overlap: the all- b state $16/17$ belongs to both and uniquely uses b . Hence the proof does not introduce an implicit disjoint phase register. This is precisely a protected-subsystem mechanism, even if the rest of the state space has feedback-dependent successors.

10.3. A critical boundary of the single-step density condition. For $3/2 < \lambda \leq 2$, on $Q = [0, 1]$ use

$$g_a(x) = \lambda x, \quad g_b(x) = \lambda x - (\lambda - 1)/2, \quad g_c(x) = 3x - 2.$$

The safety intervals are

$$K_a = [0, 1/\lambda], \quad K_b = [(\lambda - 1)/(2\lambda), (\lambda + 1)/(2\lambda)], \quad K_c = [2/3, 1].$$

They cover Q and $r = 3$. Both a and c are forced into every subcover by endpoints 0, 1. Since $1/\lambda < 2/3$, they leave an open gap, so b is also needed. The inverse densities are constants $1/\lambda, 1/\lambda, 1/3$; therefore $\Theta_2 = 1/\lambda < 1$ and $\Theta_3 = 2/\lambda$.

Proposition 10.3. *For every $3/2 < \lambda \leq 2$, this full Borel class has $M_1 = \log 2 < \log 3$. Thus the condition $\Theta_3 < 1$ cannot be uniformly replaced by $\Theta_3 \leq 1$ or by $\Theta_3 < 1 + \eta$ for any $\eta > 0$, while retaining the conclusion $M_1 = \log 3$.*

Proof. The $k = 2$ single-step criterion gives every feedback $h^* \geq \log 2$. Put $t = (\lambda - 1)/(2\lambda)$ and define

$$s_\lambda(x) = \begin{cases} a, & 0 \leq x < t, \\ b, & t \leq x < 2/3, \\ c, & 2/3 \leq x \leq 1. \end{cases}$$

It is Borel and legal: $t < 1/\lambda$ and $(\lambda+1)/(2\lambda) \geq 3/4 > 2/3$. For $x < t$, $g_a(x) < (\lambda-1)/2 \leq 1/2$. For $t \leq x \leq 1/2$, $g_b(x) = \lambda x - (\lambda-1)/2 \in [0, 1/2]$. Hence $J = [0, 1/2]$ is invariant. Before entering J only b, c occur; once in J only a, b occur. This remains true for states which never enter and at all thresholds, including the fixed points $1/2$ and 1 .

At any n selected times a real name has some cut $0 \leq j \leq n$ between the two groups, each of size two. Thus its pattern count is at most $(n+1)2^n$ for *every* window. This actual feedback gives $h^* \leq \log 2$ and completes the equality. At $\lambda = 2$, $\Theta_3 = 1$; as λ increases to 2 from below, Θ_3 decreases to 1 from above. These values give both claimed failures of uniform relaxation. The example does not make $\Theta_3 < 1$ a necessary condition for every system, and does not determine L_1 for this family. $\square$

10.4. A noninteger four-branch system. Let $Q = [0, 6]$ and use, at period one,

$$g_{A0}(x) = \frac{3}{2}x, \quad g_{A1}(x) = \frac{3}{2}x - 3, \quad g_{B0}(x) = \frac{5}{3}x, \quad g_{B1}(x) = \frac{5}{3}x - 4.$$

The original safety intervals are $[0, 4]$, $[2, 6]$, $[0, 18/5]$, $[12/5, 6]$. Truncation to the ambient interval $[-1, 7]$ retains exactly these safe transitions. All legal Borel feedbacks, arbitrary mixtures, and finite atom refinements are allowed. Two A domains cover Q and no single one does, so $r = 2$. The largest scalar inverse capacity is $2/3 < 1$. Thus $M_1 = \log 2$. Every actual length- N control prefix has its fiber inside an interval of length at most $6(3/2)^{-N}$; these intervals cover Q . Its number $D_N(s)$ is therefore at least $(3/2)^N$. Appendix B retains the weights and actual even-odd compatibility in a refinement of this volume argument. This gives

$$\log(3/2) \leq L_1 \leq \log 2. \quad (10.1)$$

For $A_2 = (0, 2, \dots)$ let $\ell_2 = \inf_s h_{A_2}(s)$, and let ℓ_2^A restrict to all two- A Borel feedbacks, without restricting their partitions. The unrefined two- A rules have the exact description

$$s_E(x) = A0 \text{ on } [0, 2) \cup E, \quad s_E(x) = A1 \text{ elsewhere}, \quad E \subset [2, 4] \text{ Borel}.$$

Indeed the two overlap domains are exactly $[2, 4]$. For any feedback, an actual $2n$ prefix is determined by its even and odd sampled patterns. Odd patterns occur among the even patterns from states $T_s x \in Q$. Hence $D_{2n}(s) \leq N_n(s)^2$ and the volume lower bound gives $N_n(s) \geq (3/2)^n$. Consequently $\ell_2 \geq \log(3/2)$.

Proposition 10.4 (Density-one samplings). *If $A = (a_i)$ satisfies $a_n/n \rightarrow 1$, then $\inf_s h_A(s) = \log(3/2)$ over the full class. The greedy feedback $s_g = s_\emptyset$ is an actual minimizer for every such A .*

Proof. Set $\beta = 3/2$. On the invariant core $[0, 3)$, scale $x = 3t$. The greedy dynamics becomes $B(t) = \beta t - \lfloor \beta t \rfloor$ on $[0, 1)$. Every nonempty length- n digit cylinder is a left-closed right-open interval whose B^n image is $[0, l)$ for some $0 < l \leq 1$. This is proved by splitting the current image at $1/\beta$: it has a single child if $\beta l \leq 1$, and otherwise a full-image child and a nonfull-image child. Equality creates no extra singleton at the excluded right endpoint.

Let F_n be the number of full-image cylinders, with $F_0 = 1$. The sum of image lengths is β^n , since each source length is l/β^n and the sources partition $[0, 1)$. Thus $F_n \leq \beta^n$. Each nonfull cylinder has a last full-image ancestor, and each ancestor has at most one descendant at a given later level without another full-image step. The total core count C_n consequently satisfies

$$C_n \leq \sum_{j=0}^n F_j \leq 3\beta^n.$$

For $x \in [3, 6)$ the real greedy control remains $A1$ until entering the core; while waiting its state is $6 - \beta^j(6 - x)$, so entry is finite. State 6 uses $A1$ forever. Therefore the full-space count satisfies

$$D_n(s_g) \leq 1 + \sum_{j=1}^n C_j \leq 10\beta^n.$$

This includes all thresholds and the exceptional endpoint.

Let $N = a_n + 1$, so $N - n = o(n)$. Each sampled word has at most 4^{N-n} completions to a consecutive word, giving

$$P_s(\{a_1, \dots, a_n\}) \geq \frac{D_N(s)}{4^{N-n}} \geq \frac{\beta^N}{4^{N-n}}.$$

For the greedy rule the same count is at most $D_N(s_g) \leq 10\beta^N$. After division by n the two rates tend to $\log \beta$. Refinements preserve the lower bound and the unrefined rule gives the matching upper bound. $\square$

Let $\rho > 1$ be the unique solution

$$\rho^6 = \rho^5 + \rho^4 + \rho + 1, \quad \log \rho = 0.55600938749 \dots \quad (10.2)$$

Appendix A proves the exact real-feedback value $h_{A_2}(s_g) = \log \rho$, with full upper and lower counts. Combining these estimates gives

$$\log(3/2) \leq \ell_2 \leq \ell_2^A \leq \log \rho. \quad (10.3)$$

Neither greedy optimality in the full class nor greedy optimality in the two- A class is established. These fixed- A_2 questions are distinct from (10.1). A lower entropy abstract open-loop language is constructed in Appendix C; its full Borel feedback implementation remains unresolved.

10.5. Expansion extensions of the integer family. The next statement retains a matching real feedback while allowing other feedbacks to change the dynamics inside its absorbing core. The global expansion condition preserves the lower bound from Proposition 10.1; the original feedback still attains it.

Proposition 10.5. *Start with the integer family of Proposition 10.1. Add finitely many continuous branches $g_v : Q \rightarrow \mathbb{R}$ satisfying*

$$|g_v(x) - g_v(y)| \geq m|x - y| \quad (x, y \in Q). \quad (10.4)$$

A branch is legal precisely when its value is in Q . For every fixed period τ , over the entire enlarged Borel class,

$$\min_s h_{(0,1,\dots)}(s) = L_\tau = M_\tau = \log m.$$

If among the added branches is $g_e(x) = kmx$, with an integer $k \geq 2$ and $R \geq k$, there is no nonconstant continuous map $\pi : Q \rightarrow Z$, to any compact metric space, and no map $H : Z \rightarrow Z$ such that $\pi(T_s x) = H(\pi(x))$ for all legal period- τ feedbacks and all states. Continuity of H is not required.

Proof. Use the continuous ambient extension $f_v = c_X \circ g_v \circ c_Q$ on $X = [-1, R + 1]$, where c denotes interval truncation. A value outside Q remains outside Q after truncation to X . The original branches still cover Q . Two states in the same real length- n block fiber follow the same $n\tau$ legal controls, so (10.4) gives $m^{n\tau}|x - y| \leq R$. Their fiber, however fragmented, has diameter at most $Rm^{-n\tau}$. Its containing interval, together with the other containing intervals, covers Q . There must be at least $m^{n\tau}$ real consecutive words. The original actual feedback s_τ is still available and retains every all-window count proved in Proposition 10.1. These matching bounds prove the equalities and treat refinements by projection.

Suppose the asserted common continuous factor exists. Any legal block at a given state can be chosen there by changing a globally safe Borel selector at that singleton. Thus all legal blocks at that state have the same projected endpoint. At $x = y/m^\tau$, for $1 \leq y \leq R$, the legal blocks 0^τ and $0^{\tau-1}1$ end at $y, y - 1$. All intermediate states are in Q . Hence $\pi(y) = \pi(y - 1)$ throughout $[1, R]$. The restriction to $[0, 1]$ induces a continuous $F : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}$, and repeated subtraction of integers gives $\pi(y) = F(y \bmod 1)$ on Q . At the same initial state the blocks 0^τ and $0^{\tau-1}e$ end at y, ky , provided $0 \leq y \leq R/k$. Since $R \geq k$, this implies $F(t) = F(kt \bmod 1)$ for all t on the circle. Iterating gives $F(j/k^n) = F(0)$ for every $0 \leq j < k^n$. Their union is dense, so continuity makes F , and then π , constant, a contradiction. $\square$

For example, $m = 2, D = 3, Q = [0, 3]$ admits both $g_e = 4x$ and $g_v = 2x + x^2$ as additional branches. At $x = 1/8$ the legal successors under $0, e, v$ are $1/4, 1/2, 17/64$, all in the original core. A rule choosing e on $(0, 1/4)$ and otherwise the original greedy control is globally Borel and safe. These are genuine feedback-dependent core dynamics, and the proposition excludes every nonconstant continuous common factor. It does not exclude Borel factors: for integer affine branches and e , rationality of the state is a nonconstant Borel common factor.

Condition (10.4) cannot be replaced by expansion on each monotonicity interval. In the $m = 2, D = 3$ system add $g_b(x) = 2x$ on $[0, 3/2]$ and $g_b(x) = 6 - 2x$ on $[3/2, 3]$. Both slopes have absolute value two, but this single branch maps all of Q into Q . The constant real block b^τ has one instruction name; both optimized rates are zero. Folding hides the spatial branching from the control-label alphabet.

10.6. An alternating open-loop language with no Borel realization. Let $R > 0$, $1 < a, b < 2$, and $\log a / \log b \notin \mathbb{Q}$. On $[0, R]$ take four legal affine branches

$$g_{Ai}(x) = ax - i(a - 1)R, \quad g_{Bi}(x) = bx - i(b - 1)R \quad (i = 0, 1).$$

Each type's two full inverse intervals cover Q . Let Λ_{alt} be the two-phase language whose A, B types alternate strictly and whose digits are otherwise arbitrary. Let

$$\widehat{\Lambda} = \Lambda_{\text{alt}} \cup \{wc^\infty : w \text{ is a finite alternating-type word, } c \in U\}.$$

The empty prefix is allowed. This compact shift-invariant language includes all constant endpoint tails and their finite alternating prehistories. The inverse contractions $\psi_{Ci}(y) = (y + i(\lambda_C - 1)R)/\lambda_C$, with $\lambda_A = a, \lambda_B = b$, define a continuous coding $\pi(d) = \lim_n \psi_{d_0} \cdots \psi_{d_{n-1}}(0)$. Every state has a coding in Λ_{alt} : in each prescribed type choose a safe digit and repeat. The contraction error is at most $R \min(a, b)^{-n}$. This proves full open-loop coverage, not feedback consistency.

Proposition 10.6. *No globally safe Borel period-one feedback has all its real names in $\widehat{\Lambda}$. This remains true with arbitrary Borel mixtures and finite atom refinements. Nevertheless for every n -position window F with $n \geq 1$,*

$$2^{n+1} \leq P_{\widehat{\Lambda}}(F) \leq 8(n + 1)2^n.$$

Proof. The alternating part has two distinct type patterns and 2^n independent digits. For a constant-tail member there are at most $n + 1$ cuts on F , two prefix phases, 2^j prefix digits, and four constant tails. Summing by cuts bounds the count by $8(n + 1)2^n$ (and also covers the infinite alternating case).

For nonrealizability, let C be the union of all finite affine preimages of $\{0, R\}$. This set is countable, and its complement is preserved by every legal branch: a successor in C would put the original state in C . A constant safe tail under any one branch forces its tail state to be the corresponding endpoint, by expansion about 0 or R . Thus any hypothetical feedback has strictly alternating types forever on $Q \setminus C$. Choose

$$0 < \eta < \min\{R/a, R/b, (a-1)R/a, (b-1)R/b\}.$$

On $(0, \eta)$ only zero-digit controls are legal. For $0 < x < \eta/(ab)$ outside C , the true first two steps are x, ax, abx if the first type is A , and x, bx, abx otherwise. The third type is the first type. Put $p = \log a, q = \log b, h = p + q$ and encode the feedback type at $\eta e^t, t < 0$, by a Borel $c(t) \in \{0, 1\}$, with A represented by one. Outside a countable exceptional set,

$$c(t+h) = c(t), \quad c(t) = 1 \Rightarrow c(t+p) = 0, \quad c(t) = 0 \Rightarrow c(t+q) = 1 \quad (t < -h).$$

Translating each negative interval of length h to $[-h, 0)$ gives a measurable circle function, with all countably many translated exceptions still null. Apply the last two implications at $t = r - h, -h \leq r < 0$. For its set E of type one, circle rotations satisfy $R_p E \subset E^c$ and $R_q E^c \subset E$ modulo null sets. Since $q = -p$ modulo h , these imply $R_p E = E^c$, measure $1/2$, and $R_{2p} E = E$ modulo null sets. The number $2p/h$ is irrational. For $G = 2\mathbf{1}_E - 1$, invariance under that rotation and Fourier coefficients give $\widehat{G}(j) = e^{2\pi i j(2p/h)} \widehat{G}(j)$. Every nonzero coefficient vanishes; the mean vanishes as well. Completeness of the trigonometric system gives $G = 0$ almost everywhere, contradicting $|G| = 1$. This proves impossibility. Refinements project to the same physical control rule and cannot avoid the obstruction. $\square$

The noninteger system in Section 10.4 has $a = 3/2, b = 5/3, R = 6$. Its log ratio is irrational by unique prime factorization: $(3/2)^j = (5/3)^k$ would imply $3^{j+k} = 2^j 5^k$. None of its nonempty finite affine composites has an integer slope. The proposition is therefore a substantive implementation boundary for this system; it is not a proof of a strict optimized entropy gap.

11. FURTHER QUESTIONS

The results give two routes to unavoidable sparse complexity. Inverse capacities force independent coordinates separately for each feedback; common successors permit one prescribed sequence to work for all Borel choices. Countability of the name languages supplies another sufficient condition for common attainment.

The main open question is whether the product-capacity hypotheses can force $L_\tau = M_\tau$ without either additional condition, or whether a strict gap occurs. The four-branch affine system provides a concrete test: its M_1 is known, while its exact L_1 , the two fixed-even-sampling infima, and greedy optimality remain open within the proved bounds. The ten-block covering language in Appendix C has the required projection and a smaller even rate, but a globally consistent Borel implementation is still missing. For the smooth example, the individual scalar capacities are already below one. The remaining question is whether a single full-support probability can give simultaneous constants c_u with $\max_{u \neq v} (c_u + c_v) < 1$. Appendix D excludes this pair-sum condition for densities bounded above and away from zero; it leaves other full-support probabilities open.

APPENDIX A. EXACT EVEN-SAMPLING ENTROPY OF THE REAL GREEDY FEEDBACK

We prove $h_{A_2}(s_g) = \log \rho$ for the system of Section 10.4. An upper language bound is followed by inverse paths that realize a family with the same exponential rate. On $C_0 = [0, 3)$ the actual map $R = T_{s_g}^2$ is

$$R(x) = \begin{cases} 9x/4, & 0 \leq x < 4/3, \\ 9x/4 - 3, & 4/3 \leq x < 2, \\ 9x/4 - 9/2, & 2 \leq x < 3. \end{cases} \quad (\text{A.1})$$

The middle state really uses $A0$ or $A1$ according to the original threshold, so these are actual two-step trajectories of a period-one rule. Let $0, 1$ denote its even labels $A0, A1$.

If a core even name begins 11 , its second state is in $[2, 9/4)$, its third in $[0, 9/16)$, and the next two images are in $[0, 81/64)$ and $[0, 729/256)$. The first of these is below $4/3$. If the last image is at least 2 , its next image is smaller than $1953/1024 < 2$. Consequently every core even name avoids

$$111, \quad 1101, \quad 110011. \quad (\text{A.2})$$

All interval upper endpoints in these calculations are excluded by the core or its actual branch convention; $4/3$ and 2 both map to zero under R .

Every word avoiding (A.2) has a unique left-to-right parsing into the prefix-code blocks $a = 0, b = 10, c = 1100$, with no adjacent cc , followed by one of $\emptyset, 1, 11, 110$. The first two exclusions bound one-runs and force two zeros after a full double-one run; the third prevents two complete c blocks. Let H_m count complete block strings of bit length m with no cc , with $H_m = 0$ for $m < 0$. If X and Z count strings ending in a/b and in c , respectively, their formal generating series satisfy

$$X = (z + z^2)H, \quad Z = z^4(1 + X), \quad H = 1 + X + Z.$$

Thus

$$H(z) = \frac{1 + z^4}{1 - z - z^2 - z^5 - z^6}, \quad H_m = H_{m-1} + H_{m-2} + H_{m-5} + H_{m-6} \quad (m \geq 5), \quad (\text{A.3})$$

with $H_0, \dots, H_4 = 1, 1, 2, 3, 6$ and $H_5 = 10$. The equation for ρ in (10.2) is equivalent to $\rho^{-1} + \rho^{-2} + \rho^{-5} + \rho^{-6} = 1$, whose left side is strictly decreasing. Its unique root obeys $5/3 < \rho < 7/4$. The initial counts are at most ρ^m , and the recurrence proves $H_m \leq \rho^m$ at all lengths. The incomplete-tail count is at most $H_m + H_{m-1} + H_{m-2} + H_{m-3} \leq D\rho^m$, where $D = 1 + \rho^{-1} + \rho^{-2} + \rho^{-3}$.

For a real full-space even word, either all labels are 1 , or its first 0 occurs at some position j . At that moment the state is below 2 and lies in the core; its remaining word is governed by (A.2). This includes arbitrarily long waiting outside the core and the fixed endpoint 6 . Therefore

$$N_n(s_g) \leq \sum_{m=0}^n D\rho^m \leq \frac{D\rho}{\rho - 1} \rho^n. \quad (\text{A.4})$$

This is a real feedback upper bound for every length, not an equality of its language with the larger forbidden-block language.

Inverse paths furnishing the matching lower bound. Put $\eta = 4/9$, $L = 162/65$, $d_0 = 729/256$, and define

$$P_0 = [0, 16/27] \cup [72/65, 4/3), \quad P_1 = [L, 3), \\ C = P_0 \cup P_1, \quad D_0 = P_0 \cup [2, 550/243] \cup [L, d_0).$$

These are subsets of the original interval used only to prove realizability, not feedback states or phase variables. The following inverse paths realize the indicated even blocks:

Even block	Target y	Source state	Real one-step controls
$a = 0$	$y \in C$	ηy	00
$b = 10$	$y \in C$	$2 + \eta^2 y$	1000
$b = 10$	$y \in P_0$	$70/27 + \eta^2 y$	1001
$c = 1100$	$y \in D_0$	$26/9 + \eta^4 y$	10100000

The left b rule has its source in D_0 for every $y \in C$, and in $C \cap D_0$ when $y \in P_1$. For a target in P_0 , the right b rule supplies a source in C .

Here are all intermediate checks. For a , the source ηy is in P_0 , the intermediate state is $2y/3 < 2$, and the endpoint is y ; both controls are zero. For left b , the states after its source are $8y/27, \eta y, 2y/3, y$. If $y \in P_0$, the source is in $[2, 550/243)$; if $y \in P_1$, it is in $[L, 70/27)$.

Each source is above 2 and outside $I = [12/5, L)$, while the three subsequent control states are below 2. This gives 1000 with the stated source memberships.

For right b the states are $8/9 + 8y/27$, $4/3 + \eta y$, $2 + 2y/3$, y . The source belongs to $[70/27, 694/243) \subset P_1$. The first two subsequent control states are below 2. If $y \in [0, 16/27]$, the last control state is at most $194/81 < 12/5$; if $y \in [72/65, 4/3)$, it is at least $178/65 > L$ and below $26/9$. Thus it is above 2 and outside I , proving 1001 at all included endpoints.

For c , its source is in $[26/9, 3) \subset P_1$, since $26/9 + \eta^4 d_0 = 3$ and d_0 is excluded from D_0 . Its successive states are

$$\begin{aligned} 4/3 + (3/2)\eta^4 y &< 3/2, & 2 + \eta^3 y &< 9/4, & (3/2)\eta^3 y &< 3/8, & \eta^2 y &< 9/16, \\ (3/2)\eta^2 y &< 27/32, & \eta y &< 81/64, & (3/2)\eta y &< 243/128 < 2, & y. \end{aligned}$$

Only its source and its second subsequent state are above 2; both avoid I . This proves 10100000 with every intermediate state safe. We have obtained true inverse relations

$$C \xrightarrow{a,b} C, \quad C \xrightarrow{c} D_0, \quad D_0 \xrightarrow{a,b} C,$$

where each indicated target set is covered in its entirety by the specified inverse rule.

For any complete block string without cc , start with proof label C and update to D_0 after c , and to C after a, b . End at $0 \in C \cap D_0$ and build the path backwards using the table: for b with source label D_0 use its left rule; with source label C use the left rule for a target in P_1 and the right rule for a target in P_0 . The checked memberships justify every preceding target requirement. Forward execution by the same greedy state rule now realizes the prescribed even word, ends at zero, and continues safely forever. Different block strings give different words because the blocks are a prefix code. Thus $N_m(s_g) \geq H_m$.

Let $c_* = \min_{0 \leq j \leq 5} H_j / \rho^j > 0$. The full recurrence for $m \geq 6$ proves $H_m \geq c_* \rho^m$. This and (A.4) yield

$$h_{A_2}(s_g) = \log \rho. \tag{A.5}$$

All inverse paths above avoid I . Therefore the same lower construction applies when an arbitrary Borel subset $E \subset [12/5, 162/65)$ is changed to control $B1$, with all other states greedy. Such a feedback is globally legal and Borel, but has $h_{A_2} \geq \log \rho$; this entire short-reset template cannot lower the greedy value. This is not a lower bound for the full four-control class or for arbitrary two- A Borel rules.

APPENDIX B. ARITHMETIC SAMPLING AND FINITE-BLOCK OPTIMIZATION

For a fixed globally safe Borel rule and arithmetic sampling $(0, p, 2p, \dots)$, a length- $n + m$ sampled pattern is determined by its first n entries and the length- m pattern starting from $T_s^m x \in Q$. Consequently $N_{n+m}(s) \leq N_n(s)N_m(s)$. The logarithmic rate is a limit and equals $\inf_n n^{-1} \log N_n(s)$ (with division by τ when the control period is τ). This uses forward safety, not surjectivity or unrestricted concatenation.

For the full four-control class at A_2 , let K_n be the least actual count $N_n(s)$ among unrefined legal Borel feedbacks. It exists as a least member of a nonempty subset of $\{1, \dots, 4^n\}$. Merging refinements cannot increase counts, and unrefined rules are themselves allowed, so

$$\ell_2 = \inf_s \inf_n \frac{\log N_n(s)}{n} = \inf_n \frac{\log K_n}{n}. \tag{B.1}$$

The equality exchanges two infima and yields an asymptotic upper bound from any globally implemented finite-block feedback. The minimizing feedback may depend on n , so it gives no asymptotic minimizer. Applying the formula requires actual feedback fibers, rather than an open-loop word table.

B.1. Weighted actual pattern pairs. The volume argument in Section 10.4 has a more informative form that retains the compatibility of even and odd patterns. Throughout this

subsection the feedback is an unrefined legal Borel rule for the original four-branch system. Set $a = 3/2$, $b = 5/3$, and $r_1 = a/b = 9/10$. Let

$$V_n = \{(s(T_s^{2j}x))_{0 \leq j < n} : x \in Q\}, \quad N_n = |V_n|,$$

and let I_n consist of the pairs (α, γ) of even and odd patterns from the *same* initial state. Since $T_s x \in Q$, one has $I_n \subset V_n \times V_n$; surjectivity of T_s is unnecessary. Let $B(\alpha)$ count the $B0, B1$ labels of α , and define

$$W_n = \sum_{\alpha \in V_n} r_1^{B(\alpha)}, \quad Z_n = \sum_{(\alpha, \gamma) \in I_n} r_1^{B(\alpha) + B(\gamma)}, \quad \kappa_n = Z_n / W_n^2.$$

In particular, $0 < \kappa_n \leq 1$ and $W_n \leq N_n$.

Proposition B.1. *With the preceding definitions, put*

$$\theta_n = -\frac{1}{n} \log \frac{W_n}{N_n}, \quad d_n = -\frac{1}{2n} \log \kappa_n.$$

Then $\theta_n, d_n \geq 0$ and

$$\frac{\log N_n}{n} \geq \log a + \theta_n + d_n. \quad (\text{B.2})$$

More precisely, if Λ_w is the slope of the complete affine word w and

$$\Omega_m = \sum_{w \text{ actual}, |w|=m} \Lambda_w^{-1}, \quad e_n = \frac{\log \Omega_{2n}}{2n},$$

then $\Omega_m \geq 1$, $e_n \geq 0$, and

$$\frac{\log N_n}{n} - \log a = \theta_n + d_n + e_n. \quad (\text{B.3})$$

Proof. For each actual complete word w , its Borel initial fiber E_w has diameter at most $6/\Lambda_w$, because two states following the same word separate by the factor Λ_w and their final states lie in $[0, 6]$. All these fibers partition Q . Consequently

$$1 \leq \sum_{w \text{ actual}, |w|=2n} \Lambda_w^{-1} = a^{-2n} Z_n.$$

The equality uses the bijection between complete words and their actual interlaced pairs. Nonempty zero-measure fibers are included. Substitution of $Z_n = \kappa_n W_n^2$ proves (B.2); the same substitution with $\Omega_{2n} = a^{-2n} Z_n$ proves (B.3).

Each original inverse branch carries Q into its safety domain. Thus the complete inverse interval $J_w = G_w^{-1}(Q)$ is contained in Q and has length $6/\Lambda_w$; every intermediate open-loop state is safe. The real fiber need only satisfy $E_w \subset J_w$. The intervals J_w for actual words cover Q , and, writing $\nu_m = \sum_w \mathbf{1}_{J_w}$, one obtains

$$6(\Omega_m - 1) = \int_Q (\nu_m - 1) dx = \sum_w (|J_w| - |E_w|) \geq 0.$$

Here $|E_w|$ denotes Lebesgue measure; no interval or convexity assumption is made on a feedback fiber. This also proves the claimed nonnegativity at every length. $\square$

The three terms distinguish fast-branch weighting, absent actual pattern pairs, and excess complete inverse coverage. They are support-counting quantities, rather than empirical frequencies or Shannon entropies. The estimates transfer to arbitrary atom refinements by projecting to their physical control labels; the equalities themselves refer to those merged labels. Splitting an actual trajectory at time $2n$ also gives

$$N_{n+m} \leq N_n N_m, \quad W_{n+m} \leq W_n W_m, \quad Z_{n+m} \leq Z_n Z_m.$$

The weights multiply under this split, and the suffix is realized from a state in Q . The subadditivity argument at the beginning of this appendix therefore gives finite limits

$$h = \lim_n \frac{\log N_n}{n}, \quad p = \lim_n \frac{\log W_n}{n}, \quad t = \lim_n \frac{\log Z_n}{2n}, \quad \log a \leq t \leq p \leq h,$$

and the exact limiting decomposition $h - \log a = (h - p) + (p - t) + (t - \log a)$.

These formulas do not supply a uniformly positive correction. Even the uniformly strict one-step inequality $\Omega_1 \geq 6/5$ does not iterate: controls at 0 and 6 are different and each inverse slope is at least $3/5$, which proves that inequality, but the greedy feedback has $\Omega_1 = 4/3$ and

$$1 \leq \Omega_m(s_g) = a^{-m} D_m(s_g) \leq 10.$$

The last bound is the all-initial-state word estimate in Section 10.4. Hence $e_n(s_g) \rightarrow 0$. The complete inverse intervals overcount the true fibers at one step without producing an exponential overcount at later steps.

Example B.2 (Full support from a countable modification). *There is a globally safe two- A Borel feedback for which $N_n = 2^n$, $I_n = V_n \times V_n$, and $\theta_n = d_n = 0$ at every length. It agrees with the greedy feedback on the full infinite trajectories of all but countably many initial states.*

Proof. For a purely periodic binary sequence d , set

$$\pi(d) = 3 \sum_{j \geq 0} d_j a^{-(j+1)}.$$

This address is injective on all purely periodic sequences. Indeed, if two different sequences share a common period L and the same address, subtracting their periodic-sum formulas gives a nonzero integer polynomial P with leading coefficient ± 1 and $P(3/2) = 0$. If its degree is m , multiplication by 2^m leaves an odd leading term $\pm 3^m$ and only even remaining terms, a contradiction. Degree zero is immediate.

Let P_i be the addresses of the periodic sequences with first digit i , and $P = P_0 \cup P_1$. These sets are countable Borel sets, and $P_0 \cap P_1 = \emptyset$. Define $s_{\text{per}} = Ai$ on P_i and $s_{\text{per}} = s_g$ on $Q \setminus P$. The identity

$$a\pi(d) - 3d_0 = \pi(\sigma d) \in P \subset Q$$

proves safety on P , including 0 and 6; safety elsewhere is that of s_g . This one state rule is therefore safe on all infinite trajectories. Injectivity and induction show that its actual name from $\pi(d)$ is precisely d .

Any prescribed binary pattern on a nonempty finite window extends to a purely periodic sequence: choose a period larger than its largest coordinate and fill the other digits arbitrarily. The empty window has one pattern. Thus $P_{s_{\text{per}}}(F) = 2^{|F|}$ for every finite F . Applying this to interlaced even and odd patterns gives $V_n = \{A0, A1\}^n$, $I_n = V_n^2$, $W_n = 2^n$, $Z_n = 4^n$, and the asserted zero corrections.

Finally $R = \bigcup_{j \geq 0} T_g^{-j} P$ is countable: each point has at most two one-step greedy preimages. Outside R the greedy trajectory never enters P , so induction gives identical states and controls under the two rules at every time. Nevertheless, their consecutive-time counting entropies are respectively $\log(3/2)$, by Proposition 10.4, and $\log 2$. Discarding countably many initial states would therefore change the counting question. No realization of all infinite nonperiodic binary names is needed or claimed. $\square$

This example excludes an unconditional positive lower bound for $\theta_n + d_n$ over the whole feedback class. Its own pattern count is already large, so it does not exclude an improved estimate restricted to feedbacks with small N_n , and it supplies no low-entropy feedback or strict full-class minimax gap.

APPENDIX C. ITINERARY CLOSURES AND FEEDBACK CONSISTENCY

We construct a closed interval-covering language with smaller even entropy than the greedy feedback. The construction also identifies the additional consistency condition needed for a feedback realization. For an arbitrary two- A rule s_E in Section 10.4, write its real binary name as $B_E(x)_j = \mathbf{1}_{\{s_E(T_E^j x) = A1\}}$ and put

$$Y_E = \text{cl } B_E(Q), \quad \pi(d) = 2 \sum_{j \geq 0} d_j (2/3)^j, \quad P_2(d) = (d_0, d_2, d_4, \dots).$$

Every coordinate of B_E is Borel, and the original state rule gives $\sigma B_E(x) = B_E(T_E x)$. Thus Y_E is nonempty, compact, and $\sigma Y_E \subset Y_E$, without continuity of the closed loop. The true affine recurrence gives

$$x = 2 \sum_{j=0}^{n-1} B_E(x)_j (2/3)^j + (2/3)^n T_E^n x.$$

The remainder is at most $6(2/3)^n$, so $\pi(B_E(x)) = x$. Continuity of π and the range $[0, 6]$ imply $\pi(Y_E) = Q$.

A cylinder fixing the first n even digits is open. If it meets the closure Y_E , it meets $B_E(Q)$; therefore closure adds no finite even patterns. Since $\sigma P_2 = P_2 \sigma^2$, $P_2(Y_E)$ is a one-sided subshift with exactly these real prefixes. Its prefix entropy is its topological entropy, by the same finite-cylinder refinement argument used in Theorem 8.1. Consequently

$$h_{\text{top}}(P_2(Y_E)) = h_{A_2}(s_E). \quad (\text{C.1})$$

No factor $1/2$ is inserted, because the rate is per sampled symbol at control period one.

C.1. A closed shift-invariant interval-covering language. Let

$$W_{10} = \{(a, b, a, c, d, e, f, g, h, g) : a, \dots, h \in \{0, 1\}\},$$

let Y_0 be arbitrary infinite concatenations of these ten-digit blocks, and set $Y = \bigcup_{i=0}^9 \sigma^i Y_0$. The equality conditions $d_{10j} = d_{10j+2}$ and $d_{10j+7} = d_{10j+9}$ make Y_0 closed. Its compact shift images have a finite closed union; adding or removing one full block gives $\sigma^{10} Y_0 = Y_0$ and $\sigma Y = Y$.

Proposition C.1. *The language Y satisfies*

$$\pi(Y) = Q, \quad h_{\text{top}}(P_2(Y)) = \frac{4}{5} \log 2 < \log \rho.$$

It therefore disproves the assertion that closedness, $\sigma Y \subset Y$, and full interval projection alone force the greedy even entropy. It does not provide an implemented Borel feedback or improve ℓ_2^A .

Proof. Put $r_0 = 2/3$, $\vartheta = r_0^{10} = 1024/59049$, $\Delta = 6\vartheta = 2048/19683$. The block projection $t_w = 2 \sum_{j=0}^9 w_j r_0^j$ is a subset sum of the eight weights

$$26/9, \quad 4/3, \quad 16/27, \quad 32/81, \quad 64/243, \quad 128/729, \quad 3328/19683, \quad 512/6561.$$

In increasing order their numerators over denominator 19683 are

$$1536, 3328, 3456, 5184, 7776, 11664, 26244, 56862.$$

The corresponding values of $19683(\Delta + \sum_{j < i} \kappa_j - \kappa_i)$ are

$$512, 256, 3456, 5184, 7776, 11664, 8748, 4374,$$

all positive. Thus each new weight is at most the earlier total plus Δ . Inductively, the intervals $[t, t + \Delta]$ over all subset sums fill the single interval $[0, \Delta + \sum_i \kappa_i] = [0, 6]$: adding a weight unites the preceding interval with its overlapping translate. Hence

$$Q = \bigcup_{w \in W_{10}} (t_w + \vartheta Q).$$

Given $x_0 \in Q$, choose a containing interval and write $x_0 = t_{w_0} + \vartheta x_1$, $x_1 \in Q$; repeat forever. Then $x_0 = \sum_{j < n} \vartheta^j t_{w_j} + \vartheta^n x_n$, with remainder at most $6\vartheta^n$. The infinite block concatenation belongs to Y_0 and projects to x_0 . This proves genuine infinite coverage, including all endpoints, not just finite-depth coverage. The reverse inclusion is automatic for binary sequences.

The even and odd block projections of Y_0 are (a, a, d, f, h) and (b, c, e, g, g) : each has exactly four free bits among five output positions. For a shift of a block concatenation, an even prefix consists of complete five-position outputs and at most two partial boundary blocks. There is a constant C , independent of the length, with prefix count at most $10 \cdot 2^{4n/5+C}$. On lengths $n = 5m$, $P_2(Y_0)$ has 2^{4m} distinct prefixes. Since $P_2(Y)$ is a subshift, its entropy limit is exactly $(4/5) \log 2$. The strict comparison with ρ can be made without decimals: integer arithmetic gives $\rho > 1743/1000 > 2^{4/5}$ by substituting $1743/1000$ in the decreasing reciprocal equation for ρ , and checking $1743^5 > 16 \cdot 1000^5$. $\square$

The coding constructed in this proof is open loop. Its successive tails are safe states, since $\pi(\sigma d) = (3/2)\pi(d) - 3d_0$, but different codings of the same state may require incompatible first controls. The exact unresolved implementation question is whether there is a Borel $B : Q \rightarrow Y$ with, for every x ,

$$\pi(B(x)) = x, \quad B((3/2)x - 3B(x)_0) = \sigma B(x). \quad (\text{C.2})$$

A Borel right inverse of π alone does not meet this requirement. If both equations were proved, the real rule would have $h_{A_2} \leq (4/5) \log 2$ and would disprove greedy optimality. Neither that implementation nor its global impossibility is asserted.

For the phase calculation below, a sequence z belongs to $\sigma^i Y_0$, $0 \leq i < 10$, exactly when

$$z_r = z_{r+2} \quad \text{for every } r \geq 0 \text{ with } r+i \equiv 0 \text{ or } 7 \pmod{10}.$$

The missing coordinates before zero can be supplied independently, since the prescribed copying pairs in each ten-block are disjoint.

Example C.2 (A safe periodic seed need not extend). *The word $w = 1010001101$ belongs to W_{10} . Let $\omega = w^\infty$ and $p_j = \pi(\sigma^j \omega)$, with indices modulo ten. Direct evaluation gives*

$$(p_0, \dots, p_9) = \frac{1}{5275} (17358, 10212, 15318, 7152, 10728, 16092, 24138, 20382, 14748, 22122).$$

The points are distinct and lie in Q . On them prescribe the control A_{w_j} , prescribe A_0 at 0, A_1 at 6, and use the greedy rule elsewhere. This is a globally safe Borel rule: $g_{A_{w_j}}(p_j) = p_{j+1}$ follows from the projection identity, and the other states use already safe controls. Its names on this finite invariant seed all lie in Y .

There is, however, no implementation of Y extending these prescribed names. The point $x_ = 2p_1/3 = 6808/5275 < 2$ has, among the two A -controls, only A_0 legal, and its next state is p_1 . Consistency would force its name to be $z = 0\sigma\omega$. The periodic sequence $e = \sigma\omega = (0100011011)^\infty$ satisfies $e_r = e_{r+2}$ exactly at residues 0, 2, 6, 9 modulo ten. The phase criterion above puts e in exactly phases 1, 8. If $z \in Y$ its phase must consequently be 0 or 7, because deleting its first digit adds one to the phase. Both phases require $z_0 = z_2$, whereas these digits are 0 and 1. Thus $z \notin Y$. This excludes extension of this particular seed, not every Borel implementation of Y .*

APPENDIX D. A BOUNDED-DENSITY OBSTRUCTION TO SIMULTANEOUS SCALAR CAPACITIES

The smooth branches in Section 3.4 cannot all be scalarized by an equivalent probability with a density bounded both above and away from zero. The proof treats every such Borel density, without inserting it at a null periodic point.

Proposition D.1. *There are no Borel a and positive c_0, c_1, c_2 with $0 < \text{ess inf } a \leq \text{ess sup } a < \infty$,*

$$\rho_u(y)a(\psi_u(y)) \leq c_u a(y) \quad \text{almost everywhere,} \quad \max_{u \neq v} (c_u + c_v) < 1.$$

Here all three branches use the same density.

Proof. Choose $0 < m \leq M < \infty$ bounding a almost everywhere, and remove the null exceptions for these bounds and the three inequalities. Also remove their preimages under every finite composition of the ψ_u . These are null because every composition has a Lipschitz inverse on its image: its derivative is bounded below by $(2/15)^n$. The remaining Borel set G has full measure and is preserved by every inverse branch. For $y_0 \in G$ and $y_{j+1} = \psi_{u_j}(y_j)$, multiplication gives

$$\left(\prod_{j < n} \rho_{u_j}(y_j) \right) a(y_n) \leq \left(\prod_{j < n} c_{u_j} \right) a(y_0). \quad (\text{D.1})$$

Since $\psi_0(0) = 0$, $\psi_0(y) \leq 3y/5$, and $\rho_0(0) = 3/5$, every $\eta > 0$ has a small invariant interval on which $\rho_0 \geq 3/5 - \eta$. Choose a point of G in its interior, iterate, and take n th roots in $m(3/5 - \eta)^n \leq M c_0^n$. Letting n grow and then η decrease proves $c_0 \geq 3/5$.

For a second constraint use the positive-length closed intervals

$$\begin{aligned} I_0 &= [9/10, 23/25], & I_1 &= [3/10, 33/100], \\ I_2 &= [9/20, 47/100], & I_3 &= [53/100, 14/25], \\ I_4 &= [19/25, 79/100], & I_5 &= I_0. \end{aligned}$$

The inverse word is $(0, 1, 1, 2, 2)$. The following bounds are rational containing intervals for the *complete* images:

Input and branch	Containing interval	Next interval
$\psi_0(I_0)$	$[3081/10000, 3195/10000]$	int I_1
$\psi_1(I_1)$	$[4510/10000, 4692/10000]$	int I_2
$\psi_1(I_2)$	$[5385/10000, 5494/10000]$	int I_3
$\psi_2(I_3)$	$[7672/10000, 7836/10000]$	int I_4
$\psi_2(I_4)$	$[9002/10000, 9170/10000]$	int I_0

For an explicit verification of every endpoint bound, use $314159/100000 < \pi < 314160/100000$ and $173205/100000 < \sqrt{3} < 173206/100000$. The latter bounds follow by squaring; the former follow from Machin's identity with the alternating arctangent sums through degrees 11 and 3 at $1/5$ and $1/239$. At the two control-zero endpoints use $z = 2\pi(y - 1)$, and at the others use $z = 2\pi(y - u/3)$. In every case $|z| < 1$. Substitute the rational interval for z in

$$S_7(z) = z - z^3/6 + z^5/120 - z^7/5040, \quad |\sin z - S_7(z)| \leq 1/362880.$$

The constant sine terms are $0, -\sqrt{3}/2, \sqrt{3}/2$, and $7/(60\pi)$ is bounded by the reciprocal rational interval. Interval addition and multiplication at each endpoint lie strictly inside the displayed containing bounds. Monotonicity of ψ_u then proves the full-image inclusions, including endpoints.

On each input interval the distance to its corresponding cosine center $(1, 1/3, 2/3)$ is at most $41/300$. Thus

$$\rho_{u_j}(y) \geq \frac{3}{5} - \frac{7}{60}(41\pi/150)^2 > \frac{1214099}{2362500} > \frac{1}{2}.$$

The composite $\psi_2\psi_2\psi_1\psi_1\psi_0$ preserves I_0 . Starting at any $y \in G \cap I_0$, repeat its five-step word n times in (D.1). This yields $m(1/32)^n \leq M(c_0c_1^2c_2^2)^n$, and hence $c_0c_1^2c_2^2 \geq 1/32$. But the pairwise sums and $3/5 \leq c_0 < 1$ give $c_1, c_2 < 1 - c_0$. The function $t(1 - t)^4$ decreases on $[3/5, 1)$, so

$$c_0c_1^2c_2^2 < \frac{3}{5}(2/5)^4 = \frac{48}{3125} < \frac{1}{32},$$

a contradiction. All iterations occur in the same full-measure set; no regularity of a was assumed. $\square$

For $d\mu_a = a dy / \int a$, the density of the branch measure relative to μ_a is exactly

$$\rho_u^a(y) = \rho_u(y)a(\psi_u(y))/a(y),$$

by the change of variables $x = \psi_u(y)$. The proposition excludes this bounded positive density class. It does not exclude arbitrary full-support singular probabilities or densities without uniform bounds; that question remains open for this smooth example. It does not alter its already proved value $M_1 = \log 3$.

There is an exact reduction of that remaining question, without a density assumption. If a full-support probability μ satisfied $\mu(\psi_u(E)) \leq c_u \mu(E)$ for all Borel sets and the same target pair sums, then $q_0 = \max c_u < 1$ would force it to be nonatomic. Indeed, an atom of mass $m > 0$ can be followed along any successive legal controls, since the safety intervals cover Q . Applying the set inequality to the successor singleton raises its mass to at least m/q_0 at each step, eventually exceeding one. Repetition of states does not avoid that contradiction.

Its distribution function $h(x) = \mu([0, x])$ would therefore be a continuous strictly increasing homeomorphism of $[0, 1]$, sending μ to Lebesgue probability. Applying the inequalities to $(x, y]$ shows that every $h\psi_u h^{-1}$ is c_u -Lipschitz. Conversely, any one increasing homeomorphism with these simultaneous Lipschitz bounds defines a full-support nonatomic measure $\mu = (h^{-1})_* \text{Leb}$ with the required inequalities on every Borel set. To see the last assertion, cover a Borel set in the h coordinate by an open set of arbitrarily close measure. Decompose that open set into countably many intervals; each image has length at most c_u times its length. Subadditivity and outer regularity give the full Borel estimate, not merely an interval estimate. This equivalence leaves existence of h undecided and is not an extension of the preceding exclusion.

APPENDIX E. A REAL RETURN CERTIFICATE AND ITS SCOPE

The affine system below is the four-branch system of Section 10.4. A sufficient certificate is stated with its true fibers; extracting it for every Borel feedback is an unresolved step.

Proposition E.1. *Suppose one legal feedback has distinct real control words w^0, w^1 of the same length k , and a nonempty open interval $I \subset (0, 6)$, such that its real prefix fiber E_{w^i} is comeager in the complete inverse interval $J_i = G_{w^i}^{-1}(I)$, and $J_i \subset I$. For the single predetermined sequence $A = (j!)_{j \geq 1}$ this feedback has $h_A(s) \geq \log 2$.*

Proof. Each inverse $\psi_i : I \rightarrow J_i \subset I$ is an affine homeomorphism, and its bad subset $J_i \setminus E_{w^i}$ is meager. For a prescribed finite sequence of the two words, compose their complete inverse branches backwards from I . The resulting initial interval is nonempty and open; the inverse images of every bad set along the finite affine maps are meager. A point outside their finite union executes precisely the prescribed sequence by the same feedback. It remains safe thereafter by global legality.

Choose an offset $0 \leq t < k$ where the two words differ. For all $j \geq k$, $j!$ is divisible by k . Prescribe independently the word in block $j!/k$ of a sufficiently long concatenation, fill all other blocks arbitrarily, and start observation at the actual state $T_s^t x$. Its control at time $j!$ is the offset- t letter of that chosen block. Thus the first n factorial samples have at least $2^{n-O(k)}$ patterns. The ignored head is fixed and the normalized limsup gives the claim. The word length k is a proof device; the feedback period is still one. $\square$

For any expanding affine word $G_w(x) = \lambda_w x - t_w$ and $I = (\alpha, \beta)$, elementary endpoint inequalities give

$$G_w^{-1}(I) \subset I \iff t_w/(\lambda_w - 1) \in [\alpha, \beta].$$

The interval must be the complete inverse of I ; a locally feasible subinterval is insufficient. The proposition would give $L_1 = M_1 = \log 2$ if its certificate were extracted for every legal Borel rule, even with k depending on the rule. Such an extraction is not proved here. Compactness of the state interval does not make the true return graph finite, and openness of each individual affine branch does not imply Borel closed-loop stitching.

A common covered target need not admit the returns in Proposition E.1. To see the distinction on real trajectories, use the globally safe greedy rule s_g , whose interval $[0, 3]$ is forward invariant. The target $I = (2, 5/2)$ has complete one-step inverse intervals

$$J_0 = (4/3, 5/3), \quad J_1 = (10/3, 11/3),$$

and the rule really uses $A0$ throughout J_0 and $A1$ throughout J_1 . Starting in I , however, no trajectory can ever reach J_1 , since it remains in $[0, 3]$. Thus this covered target has no return to its second source at any length. The same rule does have a return certificate on the different target $I_0 = (0, 3)$: the words $A0, A0, A0$ and $A1, A0, A0$ have complete inverse intervals $(0, 8/9)$ and $(2, 26/9)$, both inside I_0 . Their intermediate states lie successively in $(0, 4/3)$, $(0, 2)$, and $(0, 3)$, where the indicated controls are actually chosen. All points on these inverse intervals follow the stated words and continue safely thereafter. The failed return for the first target therefore concerns the choice of certificate, rather than this feedback's sampled entropy or the existence of every possible return certificate.

APPENDIX F. ATTAINMENT OF THE FIXED-PERIOD INFIMUM

Integer-logarithm quantization of maximal-pattern entropy is a known combinatorial fact. For one-sided subshifts it is stated in Huang and Ye [4, Theorem 5.5]. We record the finite-set argument needed here, since it also settles attainment of the feedback infimum. This argument does not address the common-sampling problem.

Proposition F.1. *Let $\emptyset \neq \Omega \subset E^{\mathbb{N}_0}$, where $|E| = q < \infty$, and fix $\tau > 0$. There is an integer $m \in \{1, \dots, q\}$ such that $h^*(\Omega; \tau) = \log m/\tau$.*

Proof. For $H \subset E^n$, let $d_k(H)$ be the largest size of a coordinate set J for which $H|_J$ contains $\prod_{j \in J} V_j$, with $V_j \subset E$ and $|V_j| = k$. Put $d_k(\emptyset) = -1$. The elementary finite bound is

$$|H| \leq B_k(n, d) := \sum_{i=0}^{\min(d, n)} \binom{n}{i} (k-1)^{n-i} \binom{q}{k}^i \quad \text{if } d_k(H) \leq d, \quad 2 \leq k \leq q.$$

Set $B_k(n, -1) = 0$. To prove the bound, write H_e for the $(n-1)$ -letter prefixes ending in e , $V = \bigcup_e H_e$, and $V_S = \bigcap_{e \in S} H_e$ for $|S| = k$. Projection gives $d_k(V) \leq d$, whereas $d_k(V_S) \leq d-1$: a shattered set in V_S can be extended by the last coordinate. A prefix appearing in t of the sets H_e contributes $t \leq (k-1) + \binom{t}{k}$. Hence

$$|H| \leq (k-1)|V| + \sum_{|S|=k} |V_S| \leq (k-1)B_k(n-1, d) + \binom{q}{k} B_k(n-1, d-1) = B_k(n, d).$$

The empty class and the nonempty class at $n = 0$ start the induction.

Let $D_k = \sup_F d_k(\Omega|_F)$, over finite $F \subset \mathbb{N}_0$, and let m be the largest k with $D_k = \infty$. Since Ω is nonempty, $D_1 = \infty$. Restricting shattered sets gives $p_\Omega^*(n) \geq m^n$ for every n . If $m = q$, the alphabet bound gives equality. Otherwise $D_{m+1} = d < \infty$, so the preceding bound gives, for a constant C ,

$$m^n \leq p_\Omega^*(n) \leq \sum_{i=0}^{\min(d, n)} \binom{n}{i} m^{n-i} \binom{q}{m+1}^i \leq C n^d m^n \quad (n \geq 1).$$

Taking logarithms and dividing by $n\tau$ proves the assertion. $\square$

Corollary F.2. *For every nonempty class of finite Borel invariant partitions at one fixed period τ , the infimum of $h^*(\mathcal{C}; Q)$ is a minimum. At the common attaining schedule of Theorem 7.5, the inner sequence-entropy infimum is also a minimum. In particular, for a finite control alphabet the quantity M_τ in (2.13) is attained without a countability assumption.*

Proof. Apply Proposition F.1 to each Σ_C . The corresponding integers form a nonempty subset of $\mathbb{N}$, which has a least member. At the schedule of Theorem 7.5, all individual sequence rates equal their pattern rates. For the full finite-control class, Lemma 2.3 identifies the infimum with that over legal selectors, whose nonempty fibers have at most $|U|^\tau$ labels. $\square$

No attainment assertion for an arbitrary fixed sequence A follows from this corollary. The integer quantization applies to maximal-pattern entropy, and fixed-sequence entropy need not have that form.

APPENDIX G. THE OBSTRUCTION TO POINTWISE COMMON ATTAINMENT

Theorem 7.5 attains the rate of every member of a countable language family. For uncountable families, pointwise common attainment can fail even when the full feedback class has an exact common minimum. We begin with the elementary pattern obstruction, then build genuine autonomous examples.

G.1. The combinatorial obstruction.

Example G.1 (A purely combinatorial uncountable obstruction). *For every infinite $B \subset \mathbb{N}_0$, let*

$$\Omega_B = \{x \in \{0, 1\}^{\mathbb{N}_0} : \text{supp}(x) \subset B\}.$$

At scale $\tau = 1$,

$$P_{\Omega_B}(F) = 2^{\#(F \cap B)}, \quad h^*(\Omega_B; 1) = \log 2.$$

For $A = (a_i) \in \mathcal{S}$,

$$h_{\text{seq}}^A(\Omega_B; 1) = \log 2 \cdot \limsup_{n \rightarrow \infty} \frac{\#(B \cap \{a_1, \dots, a_n\})}{n}.$$

If $\mathbb{N}_0 \setminus \{a_i : i \geq 1\}$ is infinite, choose an infinite B in that complement. Otherwise choose $B = \{a_{k^2} : k \geq 1\}$. In either case the displayed upper density is zero. Therefore

$$\sup_{A \in \mathcal{S}} \inf_{\substack{B \subset \mathbb{N}_0 \\ B \text{ infinite}}} h_{\text{seq}}^A(\Omega_B; 1) = 0 < \log 2 = \inf_{\substack{B \subset \mathbb{N}_0 \\ B \text{ infinite}}} h^*(\Omega_B; 1). \quad (\text{G.1})$$

This proves only that minimax exchange is not a formal consequence of pointwise attainment for arbitrary uncountable pattern families. Every invariant-partition name system satisfies $\sigma(\Sigma_C) \subset \Sigma_C$ by suffix closure. In contrast, $\sigma(\Omega_B) \subset \Omega_B$ forces $k-1 \in B$ whenever $k \in B \setminus \{0\}$; an infinite such B must equal $\mathbb{N}_0$. Hence the proper sparse sets producing (G.1) cannot occur as these autonomous control name systems. The equation is not a deterministic control counterexample and yields no failure statement for the unrestricted Borel invariant-partition class.

G.2. A continuous family of sparse hereditary subshifts. Write $\Theta = \{0, 1\}^{\mathbb{N}}$, with its fair Bernoulli probability ν . For a finite binary word v , including the empty word, set

$$j(v) = 2^{|v|} + \text{val}_2(v), \quad b_v = 4^{j(v)}, \quad B_\theta = \{b_{\theta|_\ell} : \ell \geq 0\}.$$

In $K = \{0, 1\}^{\mathbb{N}_0}$ define

$$Y_\theta = \text{cl} \bigcup_{t \in \mathbb{Z}} \{y : \text{supp}(y) \subset (B_\theta + t) \cap \mathbb{N}_0\}.$$

Lemma G.2. *Each Y_θ is a nonempty hereditary subshift with $\sigma Y_\theta = Y_\theta$ and zero ordinary entropy. The map $\theta \mapsto Y_\theta$ is Hausdorff continuous and injective. Every Y_θ has coordinate maximal-pattern entropy $\log 2$. For each fixed sampling sequence $A = (a_i)$,*

$$\frac{1}{n} \log P_{Y_\theta}(\{a_1, \dots, a_n\}) \longrightarrow 0 \quad \text{for } \nu\text{-almost every } \theta.$$

The exceptional set may depend on A .

Proof. The positive differences of $D = \{4^j : j \geq 1\}$ have unique representations. Indeed, the 4-adic valuation of $4^j - 4^i$ with $j > i$ is i , which determines i and then j . A finite pattern appears in Y_θ exactly when its set of one-positions is contained in a translate of B_θ ; cylinders are clopen and prescribed zero-positions impose no further restriction. If there are at least two one-positions, their difference fixes the candidate nodes and the translation. The remaining one-positions either give incompatible nodes, or the appearance parameters form the cylinder determined by the deepest node. Patterns with at most one one appear for every parameter. Thus each finite prefix language is locally constant in θ , proving Hausdorff continuity. The pattern 11 on $\{0, b_v - b_\emptyset\}$ detects whether θ extends a nonempty word v , proving injectivity.

Setting coordinates to zero preserves the defining supports and then their closure. Shifting replaces t by $t-1$, while prepending zero replaces it by $t+1$; compactness supplies predecessors of limit points. This proves heredity and surjectivity. An interval of length N meets any translate of D in at most $L_N = 2 + \lceil \log_4(N+1) \rceil$ points. Consequently

$$|\mathcal{L}_N(Y_\theta)| \leq (L_N + 1)N^{L_N} \quad (N \geq 2),$$

which gives zero ordinary entropy. On any n positions in B_θ all binary patterns occur, so $p_{Y_\theta}^*(n) = 2^n$.

Fix $F_n = \{a_1, \dots, a_n\}$ and put $M_\theta(F_n) = \sup_t |F_n \cap (B_\theta + t)|$. A translation having at least two intersections is determined by a pair in F_n and a difference in D , so there are at most $\binom{n}{2}$ candidates, independent of θ . Each candidate involves at most n nodes. A branch containing at least k of these nodes contains one at depth at least $k-1$. The parameter-cylinder probabilities therefore give

$$\nu\{M_\theta(F_n) \geq k\} \leq n^3 2^{-(k-1)} \quad (k \geq 2).$$

The events are finite unions of parameter cylinders. With $k_n = \lceil 6 \log_2(n+1) \rceil + 2$, the probabilities are summable. Borel–Cantelli gives $M_\theta(F_n) < k_n$ eventually almost surely. Every occurring pattern then has at most k_n one-positions, and

$$P_{Y_\theta}(F_n) \leq (k_n + 1)n^{k_n} \quad (n \geq 2).$$

Its normalized logarithm tends to zero. $\square$

G.3. Compactness of a designated feedback family is insufficient.

Proposition G.3. *There is a compact continuous control system with a compact designated family of clopen period-one feedbacks $\{\mathcal{C}_\theta\}$ whose actual name languages are Y_θ and for which*

$$\sup_A \inf_\theta h_{\text{seq}}^A(\mathcal{C}_\theta; Q) = 0 < \log 2 = \min_\theta h^*(\mathcal{C}_\theta; Q).$$

The full Borel and clopen classes in the same system have both optimization values zero. The control alphabet in this construction is uncountable.

Proof. Define $r_\theta : K \rightarrow Y_\theta$ by retaining the longest allowed prefix of x and replacing the rest by zeros; if every prefix is allowed, set $r_\theta x = x$. Heredity makes this a retraction. Both one-letter prefixes are allowed, so $(r_\theta x)_0 = x_0$. Every fixed output prefix depends on finitely many prefix tests, whose parameter sets are clopen by Lemma G.2. Thus $(\theta, x) \mapsto r_\theta x$ is jointly continuous.

Take $X = K \sqcup \{\dagger\}$, $Q = K$ and $U = \Theta \times \{0, 1\}$, with $\dagger$ isolated, and define

$$f(x, (\theta, s)) = \begin{cases} \sigma r_\theta x, & x_0 = s, \\ \dagger, & x_0 \neq s, \end{cases} \quad f(\dagger, u) = \dagger.$$

The matching regions are clopen, so f is jointly continuous. Use the two fixed clopen atoms $\{x : x_0 = s\}$ and assign (θ, s) to them. Their actual names are exactly Y_θ : the first retraction preserves the initial symbol and thereafter the state shifts inside Y_θ . The assignments form a compact family in U^2 . Lemma G.2 gives the displayed strict gap.

For the full class, let θ^0, θ^1 be the constant parameters and use $(\theta^0, 0)$ on the zero atom and $(\theta^1, 1)$ on the one atom. The positive difference sets of B_{θ^0} and B_{θ^1} are disjoint. Hence retracting a point of one language into the other leaves at most one one. The languages have no adjacent ones. An initial zero trajectory shifts in Y_{θ^0} until its first one, after which it becomes zero. An initial one trajectory enters Y_{θ^1} with next symbol zero; the next retraction leaves at most one further one. All actual and completed names have at most two ones. Thus $p^*(n) \leq 1 + n + \binom{n}{2}$ for this legal clopen feedback. Nonnegativity gives zero for both full-class infima and for every fixed-sequence inner infimum. $\square$

G.4. Four controls and a positive full-class minimum.

Proposition G.4. *There is a compact system with four distinct continuous control maps and a controlled-invariant compact set Q such that*

$$h_{\text{inv}}(Q) = 0, \quad L_1 = M_1 = \log 2,$$

yet no sampling sequence attains $h^(\mathcal{C}; Q)$ for every legal period-one Borel invariant partition $\mathcal{C}$. This already fails for a family of four-label selectors with Hausdorff continuous, compact name languages and rate $\log 4$.*

Proof. Let Z be the Thue–Morse subshift of Section 8.2, and let

$$\mathcal{E} = \{(\theta, y) : y \in Y_\theta\}, \quad Q = Z \times \mathcal{E} \times \{0, 1\}, \quad X = Q \sqcup \{\dagger\}, \quad U = \{0, 1\}^2.$$

Hausdorff continuity makes $\mathcal{E}$ compact. For $x = (z, \theta, y, r)$ define

$$f(x, (a, b)) = \begin{cases} (\sigma z, \theta, \sigma y, b), & a = z_0, \\ \dagger, & a \neq z_0, \end{cases} \quad f(\dagger, u) = \dagger.$$

These maps are continuous on clopen matching regions. At each state exactly the two controls with $a = z_0$ are safe. Their successors differ in the register coordinate b , so the four control maps are distinct and Q is controlled invariant. The register is a state coordinate of this system; the feedback has no additional memory or time variable.

For every legal Borel partition, the first control label of each atom must equal z_0 . The first state coordinate always shifts, independently of every Borel mixture and atom refinement. Label projection therefore gives $P_{\mathcal{C}}(F) \geq P_Z(F)$ for each finite F . The two-atom clopen feedback $(z_0, 0)$ attains equality for every F . It follows that

$$\inf_{\mathcal{C} \in \mathfrak{C}_{\text{Bor}}(Q, 1)} h_{\text{seq}}^A(\mathcal{C}; Q) = h_{\text{seq, coord}}^A(Z) \quad \text{for every } A, \quad M_1 = \log 2.$$

The schedule in Proposition 8.2 gives $L_1 = \log 2$. A control word of length N is safe exactly when its first components match the initial Z word. Thus $r_{\text{inv}}(N, Q) = |\mathcal{L}_N(Z)| \leq 8N$ and $h_{\text{inv}}(Q) = 0$.

For $\eta \in \Theta$, use the Borel selector

$$s_\eta(z, \theta, y, r) = (z_0, \mathbf{1}_{\{\theta=\eta\}} y_0).$$

Its four fibers are nonempty, Borel, and assigned different controls. Its actual name set is $Z \times Y_\eta$: the slice $\theta = \eta$ realizes the full product, and the other slices add only the zero second sequence. Hence

$$P_{s_\eta}(F) = P_Z(F) P_{Y_\eta}(F).$$

The binary-addition argument in Proposition 8.2 also gives all patterns on any finite subset of $\{4^j : j \geq 0\}$: extend the prescribed digits to the largest exponent and apply the same parity correction. Thus both factors realize 2^n patterns on any n positions in B_η . It follows that $p_{s_\eta}^*(n) = 4^n$ and $h^*(s_\eta) = \log 4$. Their ordinary word entropy is zero by the word bounds for Z and Y_η , and their name languages vary Hausdorff continuously with η .

Fix A . By Lemma G.2, the normalized logarithm of the second pattern factor tends to zero almost surely. Therefore

$$h_A(s_\eta) = h_{\text{seq, coord}}^A(Z) \leq \log 2 < \log 4 \quad \text{for } \nu\text{-almost every } \eta.$$

This disproves pointwise common attainment. In the designated family one also has $\max_A \inf_\eta h_A(s_\eta) = \log 2 < \min_\eta h^*(s_\eta) = \log 4$. For the full Borel class the lower-rate selector $(z_0, 0)$ is available, and the exact equality $L_1 = M_1$ remains valid. $\square$

In this finite-control example the compactness assertion concerns the name languages. It does not assert compactness of the Borel selector class. In particular, the strict gap for the designated family is not a counterexample to the full-class minimax identity.

G.5. Finite switching and a change of sampling alignment.

Lemma G.5. *Let $s_1, \dots, s_m$ be legal feedbacks at the same fixed period τ , let $\lambda : Q \rightarrow \{1, \dots, m\}$ be Borel, and set $s(x) = s_{\lambda(x)}(x)$. Suppose there is one integer K such that, for every $x \in Q$, the index sequence $\lambda(T_s^k x)$ changes at most K times along the entire infinite trajectory. Then*

$$h^*(s) \leq \max_j h^*(s_j).$$

The hypothesis concerns the feedback index, not changes of the control labels produced by a single feedback.

Proof. Put $H = \max_j h^*(s_j)$. For every $\eta > 0$ there is $C_\eta \geq 0$ with $\log p_j^*(k) \leq k\tau(H + \eta) + C_\eta$ for all j, k ; take $p_j^*(0) = 1$. This follows from the finitely many maximal-pattern limits and a bound for their finitely many remaining small lengths.

Fix $F = \{a_1 < \dots < a_n\}$. An actual trajectory has at most $K + 1$ constant-index stages. The stages meeting F divide its ordered indices into $r \leq \min(K + 1, n)$ consecutive groups. For a group $a_b, \dots, a_c$ assigned index j , the trajectory starting at $T_s^{a_b} x$ follows s_j through a_c . Its group pattern therefore has at most $p_j^*(c - b + 1)$ possibilities, using the translated window $\{a_i - a_b : b \leq i \leq c\}$. This window depends on the group, not on the unobserved time at which its stage began. Unobserved stages may be omitted, while stages on either side are kept as separate groups when needed. Taking a Cartesian product gives an upper bound without presuming independent states or feasible concatenation.

There are at most $\binom{n-1}{r-1}$ groupings and m^r index choices. Hence, uniformly in F ,

$$P_s(F) \leq e^{n\tau(H+\eta)} \sum_{r=1}^{\min(K+1, n)} \binom{n-1}{r-1} m^r e^{rC_\eta}.$$

The sum is bounded by a polynomial of degree at most K in n . Taking the maximum over windows, normalized logarithms, and then $\eta \downarrow 0$ proves the claim. Safety of s is pointwise inherited from the selected legal blocks, so the argument applies to its full infinite trajectories. $\square$

When $K = 0$, each whole name comes from one of the original feedbacks, and the analogous inequality holds for every fixed h_A . One switch suffices to invalidate that conclusion.

Example G.6 (One waiting step changes a fixed sampled rate). *There is a compact system with three distinct continuous controls, two clopen safe feedbacks $s_{\text{wait}}, s_{\text{read}}$, and a clopen safe mixture s_{mix} with at most one feedback-index switch on every trajectory, for which a single sampling sequence A satisfies*

$$h_A(s_{\text{wait}}) = h_A(s_{\text{read}}) = 0, \quad h_A(s_{\text{mix}}) = \log 2.$$

Their maximal-pattern rates are respectively 0, $\log 2$, $\log 2$.

Proof. Let $D = \{4^j : j \geq 1\}$ and

$$Y = \bigcup_{t \geq 0} \{y \in \{0, 1\}^{\mathbb{N}_0} : \text{supp } y \subset (D - t) \cap \mathbb{N}_0\}.$$

This union is closed. A positive difference between two powers of four determines those powers uniquely, by its 4-adic valuation. Thus two ones in a convergent sequence eventually

fix the same powers and the same nonnegative translate t ; every other one of the limit uses that translate. Zero and single-one sequences also belong to the union. This proves closedness, while shifting replaces t by $t + 1$, so $\sigma Y \subset Y$. Surjectivity is not assumed.

At any finite set of positions in D all binary patterns occur, so $p_Y^*(n) = 2^n$. On the other hand, take $A = (0, 4^1 + 1, 4^2 + 1, \dots)$ and its first n positions F_n . A pattern there has at most one one. A difference from zero to $4^j + 1$ is not divisible by four; a difference between two positive sampled positions uniquely forces the translate $t = -1$, which is excluded. Zero and all single-one patterns occur, whence $P_Y(F_n) = n + 1$.

Use $Q = Y \times \{0, 1\}$, $X = Q \sqcup \{\dagger\}$, and controls 0, 1, 2. At (y, r) the matching control $u = y_0$ sends the state to $(\sigma y, 0)$, the other binary control sends it to $\dagger$, and control 2 sends it to $(y, 0)$. The cemetery point is fixed. These three maps are continuous and distinct. The flag r is a coordinate of this system's state, rather than feedback memory.

Set $s_{\text{wait}} = 2$, $s_{\text{read}} = y_0$, and take $s_{\text{mix}} = 2$ when $r = 1$, $s_{\text{mix}} = y_0$ when $r = 0$. They are clopen and globally safe. The mixture waits once from flag one and then reads forever; from flag zero it reads immediately. Their actual infinite name sets are exactly $\{2^\infty\}$, Y , and $Y \cup 2Y$. Since $0 \in F_n$, the latter two pieces are distinguished by their first sampled label; the waiting piece reads all patterns on the first $n - 1$ powers of four. Consequently

$$P_{\text{wait}}(F_n) = 1, \quad P_{\text{read}}(F_n) = n + 1, \quad P_{\text{mix}}(F_n) = 2^{n-1} + n + 1.$$

These equalities prove the sampled rates. On any n -position window the mixture has at most 2^{n+1} patterns; its displayed sampled lower bound gives maximal rate $\log 2$, consistent with Lemma G.5. $\square$

The constant waiting feedback makes the full-class values $L_1 = M_1 = 0$ in this example. It refutes a fixed-sampling stability claim for finite switching, rather than the full-class minimax identity. The change comes from restarting the reading coordinates one step later on a one-sided language.

APPENDIX H. CAPACITY PEAKS AND PROBABILITY-ENTROPY BOUNDS

Actual pattern counts retain every realized name, including names carried by exceptional states. The examples below compare these counts with invariant-measure entropy, transfer norms, and collision statistics. They explain which information is lost in those estimates; the common-sampling value remains within the stated bounds.

H.1. Invariant name measures need not certify the counting rate. For $1 < \beta < 2$, put $Q = [0, M]$, $M = (\beta - 1)^{-1}$, and use $g_u(x) = \beta x - u$, $u = 0, 1$. The maps can be continuously clipped on $[-1, \beta M]$, without changing their values on Q . The safety domains are $[0, M/\beta]$ and $[1/\beta, M]$. With normalized Lebesgue measure the scalar inverse capacities are $1/\beta < 1$, and the minimum safety-cover size is two. Theorem 3.2 gives $M_1 = \log 2$. Every actual length- n fiber lies in an interval of length $M\beta^{-n}$, so consecutive observations give $\log \beta \leq L_1 \leq \log 2$.

Proposition H.1. *For every integer $r \geq 2$, take $\beta = 2^{1/r}$ in this system. The greedy feedback*

$$s_g(x) = \begin{cases} 0, & x < 1/\beta, \\ 1, & x \geq 1/\beta \end{cases}$$

is Borel and globally safe. There is an arithmetic sampling sequence $A_q = (0, q, 2q, \dots)$ with $h_{A_q}(s_g) = \log 2$. Nevertheless, for every sampling sequence A and every shift-invariant probability μ on its completed name system,

$$h_{A,\mu}(s_g; \mathcal{P}) \leq H_2\left(\frac{1}{r+1}\right) < \log 2,$$

where $\mathcal{P}$ observes the single control coordinate, $H_2(p) = -p \log p - (1-p) \log(1-p)$, and $h_{A,\mu}$ uses Shannon entropy divided by the number of samples.

Proof. The interval $[0, 1)$ is invariant under the greedy rule. After a digit one there, the next state is below $\beta - 1$. Since $\beta^r = 2$ and $2(\beta - 1) < 1$, the next r digits are zero. Let Y_r be the binary subshift whose one-positions have mutual distance at least $r + 1$. Points below M enter $[0, 1)$ after a finite initial run of ones: as long as the state is at least one, the iterate is $M - \beta^k(M - x)$. The endpoint M is fixed. Thus the completed name system is contained in the closed shift-invariant set

$$R_r = \{1^\infty\} \cup \bigcup_{k \geq 0} 1^k Y_r.$$

Closedness follows by separating bounded prefix lengths from prefix lengths tending to infinity.

An invariant probability on R_r has the form $\mu = \lambda \delta_{1^\infty} + (1 - \lambda)\nu$ with ν invariant and supported on Y_r : the increasing sets $\sigma^{-n}Y_r$ all have measure $\mu(Y_r)$ and their union is $R_r \setminus \{1^\infty\}$. For ν , invariance and $\sum_{j=0}^r \omega_j \leq 1$ give $p = \nu(\omega_0 = 1) \leq 1/(r + 1)$. For every n -element window F , subadditivity and the component indicator give

$$H_\mu(\omega|_F) \leq H_2(\lambda) + (1 - \lambda)nH_2\left(\frac{1}{r + 1}\right).$$

Division by n proves the claimed bound, including nonergodic mixtures.

Choose q so large that $\beta^{q-1}(\beta - 1) > 1$, and put $R = \beta^{q-1}/(\beta^q - 1) < 1$. For any binary sequence b , let $d_{qj} = b_j$ and set the other digits to zero. Define $x_k = \sum_{i \geq k} d_i \beta^{-(i-k+1)}$. Then $0 \leq x_k \leq R < 1$. If $d_k = 1$, then $x_k \geq 1/\beta$; if $d_k = 0$, then $x_k \leq R/\beta < 1/\beta$. The identity $x_{k+1} = \beta x_k - d_k$ therefore furnishes an actual infinite greedy trajectory. All binary patterns occur at the sampled positions, proving $h_{A_q}(s_g) = \log 2$. $\square$

The inequality is restricted to the fixed control-coordinate partition. It makes no assertion about the supremum over all measurable partitions, and it supplies no counting-entropy upper bound for any other feedback.

H.2. One feedback can saturate the transfer norm at every depth. For this construction take $Q = [0, 1]$ and

$$g_0(x) = \beta x, \quad g_1(x) = \beta x - (\beta - 1), \quad \psi_0(y) = cy, \quad \psi_1(y) = cy + 1 - c, \quad c = \beta^{-1}.$$

Here $K_0 = [0, c]$, $K_1 = [1 - c, 1]$, and again $M_1 = \log 2$. Continuous clipping on $[-2, 2]$ gives a compact ambient realization without changing the affine formulas on Q . For a legal selector s , with $S_u = s^{-1}(u)$, its actual transfer operator on $L^\infty([0, 1])$ is

$$(\mathcal{P}_s f)(y) = \sum_{u=0}^1 c \mathbf{1}_{S_u}(\psi_u(y)) f(\psi_u(y)).$$

Affine change of variables gives $\int h(T_s x) f(x) dx = \int h(y) \mathcal{P}_s f(y) dy$. In particular this is the operator of the actual closed loop, with all selector restrictions retained. Nondegenerate affine inverse maps preserve null sets, so it is well defined on L^∞ , and $\|\mathcal{P}_s^m\|_{\infty \rightarrow \infty} \leq (2c)^m$.

Proposition H.2. *For every rational $1 < \beta < 2$, there is one globally safe Borel selector s_β such that*

$$\left\| \mathcal{P}_{s_\beta}^m \right\|_{\infty \rightarrow \infty} = (2/\beta)^m \quad (m \geq 1), \quad \mathbf{P}_{s_\beta}(F) = 2^{|F|} \quad (F \subset \mathbb{N}_0 \text{ finite}).$$

At each depth the norm equality holds on a set of positive Lebesgue measure. Thus no strict exponential discount of the base $2/\beta$ holds even for this one feedback.

Proof. For a finite binary word w , let $\psi_w = \psi_{w_0} \circ \cdots \circ \psi_{w_{|w|-1}}$, including the empty word with $\psi_\emptyset = \text{id}$. If $\beta = p/q$ in lowest terms, $q \geq 2$, then different words of the same length give different affine maps. Otherwise their translation difference would produce a nonzero polynomial $P(\beta) = 0$ with coefficients in $\{-1, 0, 1\}$ and leading coefficient ± 1 . Multiplication by $q^{\deg P}$ and reduction modulo q contradict coprimality. Maps of different lengths have different rational slopes, so equality at a point can occur only at a rational point. Consequently, at any irrational y , all finite-word nodes $\psi_w(y)$ are distinct.

We use a compact nowhere dense set of positive measure. Starting with $[0, 1]$, remove at stage k a central open interval of length 4^{-k} from each remaining interval. The component length after stage k is $2^{-k-1} + 2^{-2k-1}$, by the splitting recurrence. Thus the construction is defined at every stage, the removed lengths sum to $1/2$, and the limiting compact set is nowhere dense and has measure $1/2$. Its affine copy F_J in a nondegenerate closed interval J has measure $|J|/2$.

Inductively choose intervals $J_j \subset (\beta - 1, 1)$ and set $F_j = F_{J_j}$. Let

$$D_{j-1} = \bigcup_{i < j} \bigcup_{|w| \leq i} \psi_w(F_i).$$

This finite union is compact and nowhere dense. The set

$$(\beta - 1, 1) \setminus \bigcup_{|w| \leq j} \psi_w^{-1}(D_{j-1})$$

is open dense. Choose an irrational y_j there. The finitely many nodes with $|w| \leq j$ are distinct and avoid D_{j-1} . Their pairwise distances, their distances to D_{j-1} when nonempty, and the distances of y_j to the window endpoints have a positive minimum d_j . Taking $J_j = [y_j - d_j/4, y_j + d_j/4]$ makes all the intervals $\psi_w(J_j)$, $|w| \leq j$, disjoint and disjoint from D_{j-1} , since their Lipschitz constants are at most one. This separates nodes of different lengths as well as nodes from different stages.

Define the disjoint F_σ sets

$$C_u = \bigcup_{j \geq 1} \bigcup_{\substack{1 \leq |w| \leq j \\ w_0 = u}} \psi_w(F_j), \quad u = 0, 1.$$

They lie in $\text{int } K_u$. Set

$$s_\beta(x) = \begin{cases} 1, & x \in C_1 \cup (Q \setminus K_0), \\ 0, & \text{otherwise.} \end{cases}$$

This is a single Borel state rule. It chooses zero on C_0 , one on C_1 , and is safe everywhere because $K_0 \cup K_1 = Q$. It treats every endpoint by the same formula and gives infinite safe trajectories for every initial state.

For w of length m and $y \in F_m$, the states $x_i = \psi_{w_i \dots w_{m-1}}(y)$, $0 \leq i < m$, with $x_m = y$, satisfy $s_\beta(x_i) = w_i$ and $g_{w_i}(x_i) = x_{i+1}$. Thus every word is a real prefix of this same feedback. Expansion of the actual transfer operator gives

$$(\mathcal{P}_{s_\beta}^m 1)(y) = \sum_{|w|=m} c^m \mathbf{1}_{E_w(s_\beta)}(\psi_w(y)) = (2c)^m \quad (y \in F_m).$$

Since $|F_m| > 0$, this proves the essential-norm lower bound; the general upper bound proves equality. The sets F_m need not have a common positive measure bound. Every pattern on a nonempty finite F can be completed to a word of length $1 + \max F$, which has an actual witness by the construction. The binary alphabet gives the matching count; the empty window has one pattern. $\square$

The selector has counting rate $\log 2$ along every sampling sequence. It therefore refutes a transfer-norm estimate, not the full-class equality $L_1 = M_1$. Changing the feedback can destroy the constructed inverse trees.

H.3. A conditional collision bound can miss full support. Let s be a globally safe Borel feedback on $Q = [0, 1]$ that selects among finitely many affine branches, all with nonzero slopes. Fix sampled times $a_1 < \dots < a_n$ and a terminal time $T = a_n + 1$. Let E_ω be the actual initial-state fiber of a sampled name ω . Its joint density with the terminal state is defined by

$$\int_{E_\omega} h(T_s^T x) dx = \int_Q h(y) f_\omega(y) dy.$$

Piecewise affine change of variables supplies these densities. Put $\rho = \sum_{\omega} f_{\omega}$, so $\int \rho = 1$, and set

$$R_n = \int_{\rho>0} \frac{\sum_{\omega} f_{\omega}(y)^2}{\rho(y)} dy.$$

If N_n counts all nonempty actual sampled fibers, including zero-measure fibers, Cauchy's inequality gives $1/N_n \leq R_n \leq 1$. Hence $-\log R_n$ is a lower bound for $\log N_n$, not an upper bound.

Example H.3. Use three controls on $Q = [0, 1]$,

$$g_a(x) = \frac{4}{3}x, \quad g_b(x) = 4x - 3, \quad g_d(x) = 16x - 4,$$

with safety domains $[0, 3/4]$, $[3/4, 1]$, and $[1/4, 5/16]$. Continuous clipping on $[-4, 12]$ preserves these formulas on Q . The minimum safety-cover size is two and the largest single inverse capacity is $3/4 < 1$, so $M_1 = \log 2$. The selector choosing a on $[0, 3/4)$ and b on $[3/4, 1]$ is globally safe and Borel; the third control remains available in the full feedback class.

For this selector, every a, b word has a real inverse path ending at each $y \in (0, 1)$, through the inverse maps $\psi_a(y) = 3y/4$ and $\psi_b(y) = 3/4 + y/4$. All intermediate inverse states lie in the appropriate open selector domains. The endpoints are handled by the stated rule. Summing the unsampled letters, with $p_a = 3/4$, $p_b = 1/4$, gives for every sampling sequence

$$f_{\omega}(y) = \prod_{j=1}^n p_{\omega_j} \quad \text{a.e.}, \quad \rho(y) = 1, \quad N_n = 2^n, \quad R_n = (5/8)^n.$$

Thus $h_A(s) = \log 2$ for every A , whereas $\lim_n -n^{-1} \log R_n = \log(8/5) < \log 2$. The terminal density is already uniform; the loss comes from the unequal probabilities of the labels. The calculation is the familiar nonuniform full-branch coding phenomenon, and does not establish a full-class counting-entropy gap.

For a finite atom refinement of the same physical feedback, coarse densities are sums of fine densities. Therefore the fine support count is no smaller, while its collision quantity is no larger. The equalities for the unrefined selector are not asserted for arbitrary refinements.

USE OF AI TOOLS

OpenAI ChatGPT/Codex tools assisted the argument exploration, candidate proofs, comparison of primary literature, manuscript integration, and automated source, compilation, and PDF checks. These activities included mathematical work and were not limited to language editing. Model versions for the complete earlier research history cannot be reliably reconstructed from the record.

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